

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

**End-to-End Simulation of a Four-Element UHF Radio Interferometer for
Solar Radio Observation at 610 MHz**

Nguyen Ba Quan

quan.nb224350@sis.hust.edu.vn

Electronics and Telecommunications Engineering (Bachelor of Science)

Supervisor: Assoc. Prof. Dr. Ha Duyen Trung

Signature

Department: Department of Communication
Engineering

School: School of Electrical and
Electronic Engineering

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ĐẠI HỌC BÁCH KHOA HÀ NỘI
TRƯỜNG ĐIỆN - ĐIỆN TỬ
ĐÁNH GIÁ ĐỒ ÁN TỐT NGHIỆP
(DÀNH CHO CÁN BỘ HƯỚNG DẪN)

Tên đề tài: End-to-End Simulation of a Four-Element UHF Radio Interferometer for Solar Radio Observation at 610 MHz

Họ tên SV: Nguyễn Bá Quân

MSSV: 20224350

Cán bộ hướng dẫn 1:

Cán bộ hướng dẫn 2:

STT	Tiêu chí (Điểm tối đa)	Hướng dẫn đánh giá tiêu chí	Điểm tiêu chí
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2	Kỹ năng viết quyền ĐATN (2 điểm)	Trình bày đúng mẫu quy định, bố cục các chương logic và hợp lý: bảng biểu, hình ảnh rõ ràng, có tiêu đề, được đánh số thứ tự và được giải thích hay đề cập đến trong đồ án, có căn lề, dấu cách sau dấu chấm, dấu phẩy, có mở đầu chương và kết luận chương, có liệt kê tài liệu tham khảo và có trích dẫn, v.v. Kỹ năng diễn đạt, phân tích, giải thích, lập luận: cấu trúc câu rõ ràng, văn phong khoa học, lập luận logic và có cơ sở, thuật ngữ chuyên ngành phù hợp, v.v.	
3	Nội dung và kết quả đạt được (5 điểm)	Nêu rõ tính cấp thiết, ý nghĩa khoa học và thực tiễn của đề tài, các vấn đề và các giả thuyết, phạm vi ứng dụng của đề tài. Thực hiện đầy đủ quy trình nghiên cứu: đặt vấn đề, mục tiêu đề ra, phương pháp nghiên cứu/giải quyết vấn đề, kết quả đạt được, đánh giá và kết luận. Nội dung và kết quả được trình bày một cách logic và hợp lý, được phân tích và đánh giá thỏa đáng. Biện luận phân tích kết quả mô phỏng/phần mềm/thực nghiệm, so sánh kết quả đạt được với kết quả trước đó có liên quan. Chỉ rõ sự phù hợp giữa kết quả đạt được và mục tiêu ban đầu đề ra, đồng thời cung cấp lập luận để đề xuất hướng giải quyết có thể thực hiện trong tương lai. Hàm lượng khoa học/độ phức tạp cao, có tính mới/tính sáng tạo trong nội dung và kết quả đồ án.	
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Điểm tổng các tiêu chí:			
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Điểm từng tiêu chí cho lẻ đến 0,5. Nếu Điểm tổng các tiêu chí > 10 thì Điểm hướng dẫn làm tròn thành 10.

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Hội đồng số:

Họ tên SV: Nguyễn Bá Quân

MSSV: 20224350

Cán bộ thành viên HĐ:

STT	Tiêu chí (Điểm tối đa)	Hướng dẫn đánh giá tiêu chí	Điểm tiêu chí
1	Chất lượng slides/Bản vẽ kỹ thuật (1,5 điểm)	Sử dụng các minh họa hỗ trợ: hình ảnh, biểu đồ rõ nét và phù hợp, dễ hiểu	
		Không quá nhiều từ, biết sử dụng từ khoá; bố cục logic, có đánh số trang	
2	Kỹ năng thuyết trình (1,5 điểm)	Tự tin, làm chủ nội dung trình bày, đúng thời gian quy định	
		Dễ hiểu, dễ theo dõi, lô-gic, lời cuốn.	
3	Nội dung và kết quả đạt được (5 điểm)	Nêu rõ tính cấp thiết, ý nghĩa khoa học và thực tiễn của đề tài, các vấn đề và các giả thuyết, phạm vi ứng dụng của đề tài. Thực hiện đầy đủ quy trình nghiên cứu: đặt vấn đề, mục tiêu đề ra, phương pháp nghiên cứu/giải quyết vấn đề, kết quả đạt được, đánh giá và kết luận.	
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4	Trả lời câu hỏi (2,5 điểm)	Trả lời ngắn gọn, chính xác, đi thẳng vào vấn đề của câu hỏi.	
		Nắm vững kiến thức cơ bản liên quan đến lĩnh vực nghiên cứu/công việc của đồ án.	
5	Điểm thành tích (1 điểm)	Có bài báo KH được đăng hoặc chấp nhận đăng / đạt giải SV NCKH giải 3 cấp Trường trở lên / các giải thưởng khoa học trong nước, quốc tế từ giải 3 trở lên / có đăng ký bằng phát minh sáng chế. (1 điểm)	
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Điểm tổng các tiêu chí:			
Điểm đánh giá:			

Cán bộ thành viên hội đồng
(Ký và ghi rõ họ tên)

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Điểm tổng các tiêu chí:			
Điểm phản biện:			

Cán bộ phản biện
(Ký và ghi rõ họ tên)

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ACKNOWLEDGMENT

Completing this graduation thesis marks the end of four years at Hanoi University of Science and Technology. For me, it is a wonderful experience of learning and researching in a creative and competitive environment.

But to have me today, it is the sacrifice and companionship of my parents, who have always cheered and encouraged me and been by my side, and I am always proud of them. I am grateful and thankful to my parents; no words can express this deep gratitude.

Moreover, I am profoundly grateful to all the teachers who imparted their knowledge and wisdom. I especially want to thank Prof. Han Huy Dung and Prof. Ha Duyen Trung for their patient guidance and the many hours of technical discussions in the SPARC Lab. Prof. Han Huy Dung helped me so much in stepping into the world of radio astronomy, along with Prof. Vo Bich Hien, one of his colleagues in this field. Bringing me into the lab gave me the opportunity to work with the members of the SPARC Laboratory, set up real-time solar radio burst detection antennas, and access the measurement hardware that motivated this simulation work. I consider myself incredibly fortunate to have had their mentorship, guiding me through the challenges and triumphs of my studies.

Finally, I want to express my commitment to myself: to always stay focused and never give up, even in the most difficult areas of the telecommunication engineering major. Hanoi University of Science and Technology will forever be my second home, the place that helped me grow into a real telecommunication engineer, especially in the field of radio astronomy.

Student

(Signature and full name)

Nguyen Ba Quan

ABSTRACT

Space weather driven by solar eruptive activity-flares, coronal mass ejections, and the geomagnetic storms they cause-can disrupt satellites, GNSS links, and power grids, as illustrated by the 2022 loss of 38 Starlink satellites to a geomagnetic storm and by documented degradation of GNSS carrier-to-noise ratios during solar radio bursts (SRBs). SRBs are an early, ground-observable signature of these eruptive events, but resolving them with a single large dish is costly, since angular resolution scales with antenna diameter. Radio interferometry-combining several small antennas into an aperture-synthesis array, so that resolution instead depends on the baseline between them-offers a low-cost alternative, a direction already pursued by projects such as DLITE (the Deployable Low-band Ionosphere and Transient Experiment), a four-element, low-cost array that has itself detected solar radio bursts. This thesis pursues the same low-cost, distributed philosophy at 610 MHz, motivated in part by a preliminary two-element Yagi-Uda field deployment at Xuan Mai, Hanoi, that produced only an ambiguous daily trend rather than a clean interferometric detection, a result traced to phase-incoherent receiver oscillators, cable insertion loss, local radio-frequency interference, and a noise floor close to the quiet-Sun signal level. Rather than debug these effects directly in further hardware iterations, this thesis presents the design and validation of a complete, physically grounded software simulation of a four-element UHF radio interferometer, built to separate and quantify such effects in a controlled, repeatable way, to demonstrate that such an array reproduces the classical signatures of aperture-synthesis imaging, and to quantify the gap between the simulation model and the hardware measurement.

The work proceeds in three coupled stages. First, the receiving element, a commercial UHF Yagi-Uda antenna with a folded-dipole driven element on a four-bar U-frame reflector, is modelled in 4NEC2 and cross-validated against direct measurement and an independent finite-element solution. The simulated element exhibits an input impedance of $Z_{\text{in}} \approx (108 - j22) \Omega$, a forward gain of about 9.5 dBi, a half-power beamwidth near 52° , and a standing-wave ratio of 1.55 at 75 Ω ; mutual coupling between elements is found to be negligible across the operational baselines (falling below -78 dB beyond 5 m), confirming that the elements behave as independent sensors. Second, an end-to-end signal-processing pipeline is built in Python. Real archived solar flux from the SWS Learmonth observatory is converted to complex antenna voltages through the Friis reception relation with additive Johnson-Nyquist thermal noise; per-antenna geometric phases are applied using a standard local-to-equatorial $\text{ENU} \rightarrow \text{XYZ} \rightarrow (u, v, w)$ transformation; the signals are cross-correlated to form visibilities; and the visibilities are inverted by a vectorised discrete Fourier transform to a dirty image. Third, the simulated instrument is exercised over the 20 March 2025 transit at Learmonth station in Australia and over a sweep of array geometries, including a DLITE-inspired, non-redundant triangle-plus-centre topology that recovers substantially better North-South resolution than a symmetric layout using the same four elements. The 20 March 2025 date is also used for a diagnostic comparison of the simulated

solar excess against a total-power trace from a two-element analogue interferometer deployed at Hoa Lạc, Hanoi; a factor of $23.8\times$ gap in SNR is attributed to the high system temperature of the consumer SDR receiver ($T_{\text{sys,eff}} \approx 3\,600\text{ K}$).

The simulated interferometer reproduces the expected analytic behaviour: a ~ 14 -minute fringe period for the 8 m baseline at the spring equinox ($\delta = 0^\circ$), a jinc-shaped visibility decay as a function of baseline length consistent with a resolved solar disk, and a dirty image equal to the true sky convolved with the synthesised point-spread function. These results confirm that the simulation faithfully captures the physics from incident flux to final image at the analytic level. The hardware comparison further shows that the primary gap between model and measurement is system temperature, not signal-chain modelling, a finding that directly guides hardware upgrade priorities.

Keywords: radio interferometry, aperture synthesis, solar radio bursts, Yagi-Uda antenna, 4NEC2, van Cittert-Zernike theorem, dirty image, point-spread function.

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LIST OF ABBREVIATIONS

Abbreviation	Definition
ADC	Analog-to-Digital Converter
CME	Coronal Mass Ejection
COTS	Commercial Off-The-Shelf
DFT	Discrete Fourier Transform
DLITE	Deployable Low-band Ionosphere and Transient Experiment
DR	Dynamic Range
EFIE	Electric-Field Integral Equation
ENU	East-North-Up local coordinate frame
EW	East-West (baseline / array orientation)
FEM	Finite Element Method
FITS	Flexible Image Transport System
FOV	Field of View
FPGA	Field-Programmable Gate Array
FWHM	Full Width at Half Maximum
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GPSDO	GPS-Disciplined Oscillator
GUI	Graphical User Interface
HFSS	High-Frequency Structure Simulator (Ansys)
HPBW	Half-Power Beam Width
IF	Intermediate Frequency
IQ	In-phase / Quadrature (baseband voltage components)
LNA	Low-Noise Amplifier
LSB	Least Significant Bit
LWA	Long Wavelength Array

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Abbreviation	Definition
MDF	Minimum Detectable Flux
MoM	Method of Moments
NEC	Numerical Electromagnetics Code
NF	Noise Figure
NS	North-South (baseline / array orientation)
OCXO	Oven-Controlled Crystal Oscillator
PSF	Point-Spread Function (synthesised / dirty beam)
RAM	Random Access Memory
RF	Radio Frequency
RFI	Radio-Frequency Interference
RIMS	Radio Interference Monitoring Set
RMS	Root Mean Square
RSTN	Radio Solar Telescope Network
SDR	Software-Defined Radio
SEFD	System-Equivalent Flux Density
sfu (SFU)	Solar Flux Unit ($1 \times 10^{-22} \text{ W}/(\text{m}^2 \text{ Hz})$)
SNR	Signal-to-Noise Ratio
SRB	Solar Radio Burst
SWR (VSWR)	(Voltage) Standing-Wave Ratio
SWS	Space Weather Services (Australian Bureau of Meteorology)
TCXO	Temperature-Compensated Crystal Oscillator
TEC	Total Electron Content
UHF	Ultra-High Frequency (300 MHz-3 GHz)
UTC	Coordinated Universal Time
UV	Spatial-frequency (u, v) plane of the visibility function
VNA	Vector Network Analyzer
XYZ	Earth-centred equatorial Cartesian coordinate frame

CHAPTER 1. INTRODUCTION

1.1 Background and Motivation

Modern society depends heavily on space and radio systems: satellite navigation (GNSS/GPS), radio communication, aviation, power grids, and Earth-observation satellites. Most of these systems assume that the near-Earth space environment stays reasonably stable. When the Sun becomes more active, *space weather driven by solar eruptive activity*-flares, coronal mass ejections (CMEs), and the geomagnetic storms they cause-happens more often and can be more severe. These events can disturb satellites, radio and GNSS links, and power grids. In October 2024, NASA and the NOAA Space Weather Prediction Center jointly announced that Solar Cycle 25 had reached its maximum phase [1]. This means solar activity, and its practical effects, are especially relevant right now.

A clear example of this impact is the loss of 38 of the 49 Starlink satellites launched by SpaceX in February 2022. A geomagnetic storm arrived while the satellites were still at low altitude. It heated and expanded the thermosphere, which increased atmospheric drag on the satellites much more than expected, and most of them re-entered the atmosphere within days [2]. This event shows that space weather is not only a scientific topic; it can cause real economic and operational loss. Solar radio bursts (SRBs), which are the radio-frequency signal produced by the same eruptive processes, can also interfere directly with GNSS. A study of the International GNSS Service network over Solar Cycle 24 found that SRBs can lower the carrier-to-noise density ratio (C/N_0) of GPS and Galileo receivers, and in the worst cases reduce the number of usable satellites at a station [3]. This links solar radio emission directly to a radio-frequency-interference problem, and it is one reason to keep monitoring solar radio activity from the ground.

To understand and eventually forecast these effects, we need to observe the Sun directly, and optical or thermal images alone are not enough: they show the visible surface of the Sun but not the processes that actually drive space weather. At radio wavelengths, the Sun is not only an optical or thermal object; it is also a source that carries information about the coronal plasma, energetic particles, and eruptive processes. Radio emission at metre and decimetre wavelengths carries information about coronal plasma density and temperature, shock fronts, and the energetic electron beams produced during flares and CMEs [4–6]. Because of this, observing the Sun at radio frequencies is one of the oldest parts of radio astronomy, and it remains important for space-weather work, since solar radio bursts are an early, ground-observable sign of the eruptive events described above.

However, observing the Sun well at radio wavelengths usually means using large and expensive equipment. Angular resolution depends on the ratio between the observing wavelength and the antenna diameter. To resolve detail on the scale of the $32'$ solar disk, a single dish would need to be tens of metres across, together with the receivers, structure, and site infrastructure that such

a dish requires. This is difficult to build for most university research groups, including those in Vietnam. *Radio interferometry* [7, 8] offers another way. Instead of one large dish, *aperture synthesis* combines the signals from several small antennas placed apart from each other. The achievable resolution then depends on the largest distance between antennas (the *baseline*), not on the size of a single antenna. This is the idea behind synthesis arrays such as the historical Culgoora radioheliograph [9], and modern arrays like LOFAR [10] and the Murchison Widefield Array [11]. It is also a reasonable direction for a low-cost solar radio instrument.

This low-cost, distributed approach is not just an idea used in this thesis; it is already an active research direction. DLITE (the Deployable Low-band Ionosphere and Transient Experiment) is a four-element, field-deployable interferometer that works at 30 MHz to 40 MHz, built from four commercially available LWA-style crossed-dipole antennas [12]. The published total system cost of DLITE, including receivers, digitisers, and the rest of the back end, is around US\$45,000 [12], but this figure is not what makes the antenna itself low-cost: official LWA component pricing lists the crossed-dipole antenna at about \$793, its front-end-electronics balun at about \$308, and its ground stake at about \$65.50 [13], so the raw receiving element used by DLITE costs on the order of \$1,200 per antenna, only a few thousand dollars for all four. It is this element-level cost, not the fully-equipped system cost, that makes the comparison with a single, tens-of-metres-across dish (Section 1.1) meaningful. DLITE was built mainly to study the ionosphere, but it has also detected long solar radio bursts, including Type II and Type IV events, confirmed by independent e-Callisto stations [12]. This shows that a small, distributed, low-cost array is a realistic way to monitor solar radio bursts, not only something for large professional facilities.

This thesis works at 610 MHz, in the UHF/decimetric radio band. This is a practical choice, not the frequency where the strongest solar radio bursts are usually found. Three reasons support it. First, Learmonth/SWS already publishes daily solar flux data at 610 MHz, so the simulation can use a real, continuous, and freely available dataset instead of only a synthetic source. Second, UHF Yagi-Uda antennas at this frequency are cheap, easy to buy, and easy to build, which fits a low-cost, university-level interferometer. Third, the wavelength at 610 MHz is about 0.49 m, so a baseline of only a few metres to a few tens of metres is already enough to produce a clear fringe pattern; this is convenient for testing fringe period, visibility, *uv*-coverage, and dirty-image formation on a small, deployable array. At the same time, 610 MHz does not represent the full picture of solar radio bursts: the strongest Type II and Type III bursts are usually observed at lower frequencies (tens of MHz), closer to the DLITE band described above. The choice of 610 MHz in this thesis reflects the available real data, the low antenna cost, and the practical UHF direction of the project, not a claim that this band captures every kind of solar radio burst.

Building an array like this, however, is harder than just placing several antennas in a field. A working interferometer needs receiver channels that stay coherent in phase and clock with each other, a receiver noise level that is well understood so it can be separated from the weak solar signal, an antenna radiation pattern that is actually measured or modelled instead of assumed, control of local radio-frequency interference (RFI), and an array layout that gives good *uv*-coverage. These are not just theoretical concerns. In my own preliminary two-element Yagi-Uda

deployment at Xuan Mai, Hanoi (Section 3.4.2), the recorded data showed only an ambiguous daily trend, not a clean interferometric detection. This was traced to several causes: phase-incoherent receiver oscillators, several decibels of cable insertion loss, strong local RFI, and a receiver noise floor close to the level of the quiet-Sun signal. Because of this experience, before spending more effort on hardware for a larger low-cost solar radio interferometer, it makes sense to first study and separate these effects in a controlled and repeatable way.

This is what a physically grounded, end-to-end *simulation* can provide, and it is the subject of this thesis. Instead of treating the antenna, the receiver noise, the array geometry, and the imaging step as separate problems, the pipeline built in this thesis connects all of them using one real observation: archived solar flux from the Learmonth Solar Radio Observatory (Space Weather Services, Bureau of Meteorology, Australia) [14, 15]. The flux is converted to a received voltage using the Friis transmission formula and Johnson-Nyquist thermal noise [16, 17], weighted by a real Yagi-Uda radiation pattern modelled in 4NEC2 [18] instead of a simplified Gaussian beam, propagated through an array geometry to form visibilities, and then correlated and inverted into a dirty image and point-spread function. All of this can be checked before an array layout is actually built. Existing open-source teaching tools such as APSYNSIM [19], developed at the Onsala Space Observatory for the Nordic ALMA Regional Centre Interferometry School, give interactive visualisations of aperture-synthesis concepts, but they use synthetic sky models and simplified Gaussian primary beams, without a connection to real flux data, realistic noise, or an actual hardware path. This thesis compares its simulation directly against APSYNSIM, to show where a pipeline based on real data and real hardware adds value compared to a purely educational tool, and also compares it against the field-deployed two-element analogue interferometer at Xuan Mai, to connect the simulation to a real, if imperfect, measurement. Building and validating this end-to-end simulation-so that array layouts, receiver budgets, and expected signal levels can be checked before further field deployment-is the goal of this work.

1.2 Problem Statement

The discussion above points to a gap, not just a list of separate technical problems. A future low-cost, distributed UHF solar radio interferometer, of the kind described in Section 1.1, should not be designed only by trial and error in the field. As the Xuan Mai deployment showed, an ambiguous result from real hardware can come from several different causes at once: phase-incoherent receivers, cable loss, local RFI, a weak array geometry, or a genuinely weak signal. Without a reference model, it is hard to tell these causes apart afterwards. The individual physics is already well understood: how an antenna receives a flux density, how thermal receiver noise is added, how the geometry of a moving baseline behaves, and how an image is formed from correlated visibilities. What is missing is a single, physically grounded software model that connects all of these stages using real data, so that the behaviour of a proposed instrument can be predicted and checked *before* it is built. Without such a model, there is no reliable way to decide, before a field deployment, whether a given receiver noise budget, antenna choice, or array layout is good enough, or to tell an instrument fault apart from a real design limitation. Closing this gap

through simulation, rather than through more hardware deployment, is the next necessary step, and it is the problem this thesis addresses:

How can a physically grounded, end-to-end simulation be constructed to evaluate the feasibility, signal chain, and array-geometry choices of a future low-cost UHF solar radio interferometer before hardware deployment?

1.3 Objectives

To answer this question, the thesis pursues four objectives, each covering one part of the simulated signal chain described above:

1. **Antenna modelling.** Build a Method-of-Moments model of the commercial UHF Yagi-Uda receiving element in 4NEC2, and extract its input impedance, gain, and radiation pattern. Cross-check the result against a direct vector-network-analyser measurement and an independent finite-element solution. Also measure how much mutual coupling exists between elements at different baselines, and confirm that it is small enough to ignore at the separations used in this thesis.
2. **End-to-end signal pipeline.** Build a Python pipeline that converts real SWS Learmonth 610 MHz flux data into complex antenna voltages, using the Friis reception formula and Johnson-Nyquist noise. The pipeline then applies the per-antenna geometric phase, cross-correlates the signals to form visibilities, and inverts them into a dirty image.
3. **General array geometry.** Replace any assumption of a one-dimensional, east-west-only array with a general local-to-equatorial $\text{ENU} \rightarrow \text{XYZ} \rightarrow (u, v, w)$ transformation, so that the simulation works for arbitrary two-dimensional array layouts, not just one fixed geometry.
4. **Validation.** Check that the simulated instrument reproduces the expected analytic results—the fringe period, the jinc-shaped visibility of a resolved disk, and the dirty image as the sky convolved with the synthesised beam—using the transit of 20 March 2025 at Learmonth.

1.4 Contributions

The main contributions of this thesis are:

1. A 4NEC2 Method-of-Moments model of a low-cost commercial UHF Yagi-Uda element, cross-checked against direct measurement and an independent finite-element solution, together with a mutual-coupling study showing that the elements are effectively independent at the operating separations, with coupling below -78 dB beyond 5 m.
2. A modular, reproducible interferometry simulation engine that works for *arbitrary* two-dimensional array geometries, using one consistent (u, v, w) formulation for per-antenna phase, baseline computation, and fringe stopping.

3. A demonstration that the simulated instrument, driven by real archived solar flux, reproduces the classical analytic results of aperture synthesis-directly answering the feasibility question raised in Section 1.2, and validating both the instrument concept and the simulation itself as a design tool.
4. A structured comparison between this simulation and APSYNSIM [19], the most widely used open-source aperture-synthesis educational simulator, showing where a pipeline based on real data and real hardware adds value compared to a purely educational tool.
5. A reproducible, single-notebook workflow that takes a new user from an archived SWS daily file to a calibrated dirty image in under two minutes of computation, with all parameters set from a single configuration cell.

1.5 Organization of the Thesis

The rest of this thesis has five chapters. **Chapter 2** covers the theoretical background: solar radio emission, the basics of antenna theory, receiver noise, the principle of interferometry and the van Cittert-Zernike theorem, and aperture-synthesis imaging. **Chapter 3** presents the electromagnetic modelling of the Yagi-Uda receiving element and the mutual-coupling study of the array. **Chapter 4** describes the design of the end-to-end simulation pipeline, from flux-data acquisition through voltage synthesis, array geometry, correlation, and imaging. **Chapter 5** reports and analyses the simulation results, including validation against analytic predictions and the 20 March 2025 transit. The final chapter, **Conclusion and Future Work**, summarises the findings and outlines possible extensions. References and appendices follow.

CHAPTER 2. THEORETICAL BACKGROUND

This chapter builds a single chain of reasoning, from the Sun to the simulated image, and every equation below is one specific link in that chain-not an independent piece of theory. Told as one sentence: the Sun produces radio emission; the international community measures this emission as a *flux density*, in solar flux units; a receiving antenna converts that flux density into received radio-frequency *power*; a receiver converts that power into a *voltage*, together with its own unavoidable thermal noise; several antennas at known positions see the same signal with a small, geometry-dependent phase difference; a *correlator* combines a pair of antenna voltages into a single complex number called the *visibility*; the array geometry places every visibility at its correct point in a Fourier (u, v) plane; and an inverse Fourier transform of those points produces the *dirty image* analysed in Chapter 5. The sections below develop this chain in order, introducing each equation only once the preceding step has produced the quantity it needs. Not every equation plays the same role, and this chapter says so explicitly at each step: some equations (for example Eq. (2.6), (2.7), (2.8), and (2.17)) are computational steps that the simulation of Chapter 4 evaluates directly; others (for example Eq. (2.13), (2.18)-(2.20), and (2.22)) are analytic results that the simulation is not built from, but is instead checked against, as validation benchmarks in Chapter 5.

In outline: Section 2.1 places the solar radio burst inside the wider picture of solar activity and motivates the observing frequency used in this thesis. Section 2.2 turns that physical picture into the standard radio-astronomy quantities-brightness temperature and flux density-that the simulation can actually work with. Section 2.3 carries the flux density through the receiving antenna to a voltage at the receiver input, and Section 2.4 adds the unavoidable thermal noise and states how averaging improves sensitivity. Section 2.5 explains how two such voltages are combined by a correlator into a single complex visibility. Section 2.6 shows how the physical antenna positions and the Earth's rotation determine which visibilities can actually be measured. Section 2.7 closes the chain by turning a set of measured visibilities back into an image of the sky. Section 2.8 collects the chain into one summary and sorts its equations into the two roles above. The treatment of interferometry and synthesis imaging follows closely the formalism of Taylor et al. [8], *Synthesis Imaging in Radio Astronomy II*, whose Chapters 1-3 are the principal reference for Sections 2.5-2.7. Equations are numbered within the chapter for cross-reference in the later design and results chapters.

2.1 Solar Radio Activity and Observation Target

2.1.1 Solar Flare, CME, Geomagnetic Storm, and Solar Radio Burst

The Sun does not only shine in visible light. It radiates across almost the entire electromagnetic spectrum, from X-rays to radio waves, and each part of that spectrum reveals a different physical

process. Chapter 1 motivated this thesis from the practical side: solar activity can disturb satellites, GNSS, and power grids, and this is why monitoring the Sun matters. This section gives the physical picture behind that motivation, introducing the events in the order in which they actually happen.

A *solar flare* is a sudden release of magnetic energy stored in the corona, usually above a sunspot group, which heats the local plasma and accelerates electrons and ions to high energy on timescales of minutes [4, 9]. A *coronal mass ejection* (CME) is a related but distinct event: a large-scale eruption of coronal plasma and magnetic field that is expelled outward, sometimes reaching the Earth in one to three days. A flare or a CME-driven shock can also accelerate protons and electrons to much higher energies than the ambient plasma; when these escape into interplanetary space they are observed as a *solar energetic particle* (SEP) event. When a CME, a SEP event, or a fast stream of solar wind reaches the Earth’s magnetosphere, it can compress and disturb it, producing a *geomagnetic storm*—the event responsible for the satellite-drag incident described in Chapter 1 [2]. A geomagnetic storm, in turn, can change the electron density of the Earth’s upper atmosphere, producing an *ionospheric disturbance* that degrades GNSS signals, as discussed in Chapter 1 [3].

A *solar radio burst* (SRB) is the radio-frequency signature of the same energetic processes: as electrons are accelerated and plasma is disturbed by flares, shocks, and CMEs, they emit radio waves that are, unlike the CME plasma itself, detectable from the ground within seconds. This is why solar radio bursts are useful as an early, remote indicator of the eruptive activity that eventually leads to a geomagnetic storm or an ionospheric disturbance. Table 2.1 collects the terms introduced above, for reference throughout the rest of this chapter.

Table 2.1. Key terms used to describe solar activity and solar radio emission.

Term	Meaning
Solar flare	Sudden release of magnetic energy in the corona; heats plasma and accelerates particles on a timescale of minutes.
Coronal mass ejection (CME)	Large-scale eruption of coronal plasma and magnetic field, expelled outward from the Sun.
Solar energetic particle (SEP) event	Burst of high-energy protons and electrons, accelerated by a flare or a CME-driven shock, that can reach the Earth.
Geomagnetic storm	Disturbance of the Earth's magnetosphere caused by a CME, a SEP event, or a fast solar-wind stream; can increase atmospheric drag on satellites (Chapter 1).
Ionospheric disturbance	Change in the electron density of the Earth's ionosphere, often following a geomagnetic storm, that can degrade GNSS signals (Chapter 1).
Solar radio burst (SRB)	Radio-frequency signature of the energetic processes above; classified into Types I-V (Section 2.1.2).
Quiet Sun	Steady, thermal radio background present even when no burst is occurring (Section 2.2.1).

2.1.2 Solar Radio Burst Types and Observing Frequency Ranges

At radio wavelengths the Sun is not the sharp-edged disk seen in visible light but a stratified, optically thick plasma whose apparent size and brightness depend strongly on the observing frequency [4, 5]. Three atmospheric layers matter to a radio observer: the photosphere (the 5800 K optical surface, dominant only at millimetre wavelengths), the chromosphere, and the hot, tenuous corona, whose electron temperature reaches $\sim 10^6$ K. At metre and decimetre wavelengths the radiation received originates predominantly in the corona and upper chromosphere.

Superimposed on the steady, thermal background (Section 2.2.1) are the transient solar radio bursts introduced above, which can raise the flux by several orders of magnitude. Following the classical morphological scheme, they are grouped into five types (I-V) according to their appearance in the dynamic spectrum, and each type is associated with a characteristic frequency range [6, 9]:

- **Type I** noise storms: long-lived, narrowband emission, typically 50 MHz to 300 MHz.
- **Type II** bursts: slowly drifting (of order 1 MHz/s), associated with coronal and interplanetary shock waves, typically starting below 150 MHz and drifting down to hundreds of kilohertz as the shock propagates outward.
- **Type III** bursts: fast-drifting, produced by electron beams streaming along open field lines; they can start above 1 GHz and sweep down through the decimetre and metre bands within

seconds.

- **Type IV** continua: broadband, following flares, spanning the decimetre to metre range and sometimes lasting hours.
- **Type V** continua: short (tens of seconds), sometimes following Type III bursts.

These frequency ranges span several named observing bands: *decametric* (below about 30 MHz), *metric* (roughly 30 MHz to 300 MHz), *decimetric* or *UHF* (roughly 300 MHz to 3 GHz), and *microwave* (above 3 GHz). The 610 MHz band used in this thesis lies inside the decimetric/UHF range. At decimetre wavelengths, the emission most often seen is the fast-drifting tail of Type III bursts as they sweep through the band, the decimetric/microwave continuum associated with Type IV events, and narrowband spike bursts produced by non-thermal (plasma or gyrosynchrotron) emission mechanisms. This emission is spatially compact, which is precisely what makes a burst a meaningful target for a small interferometer.

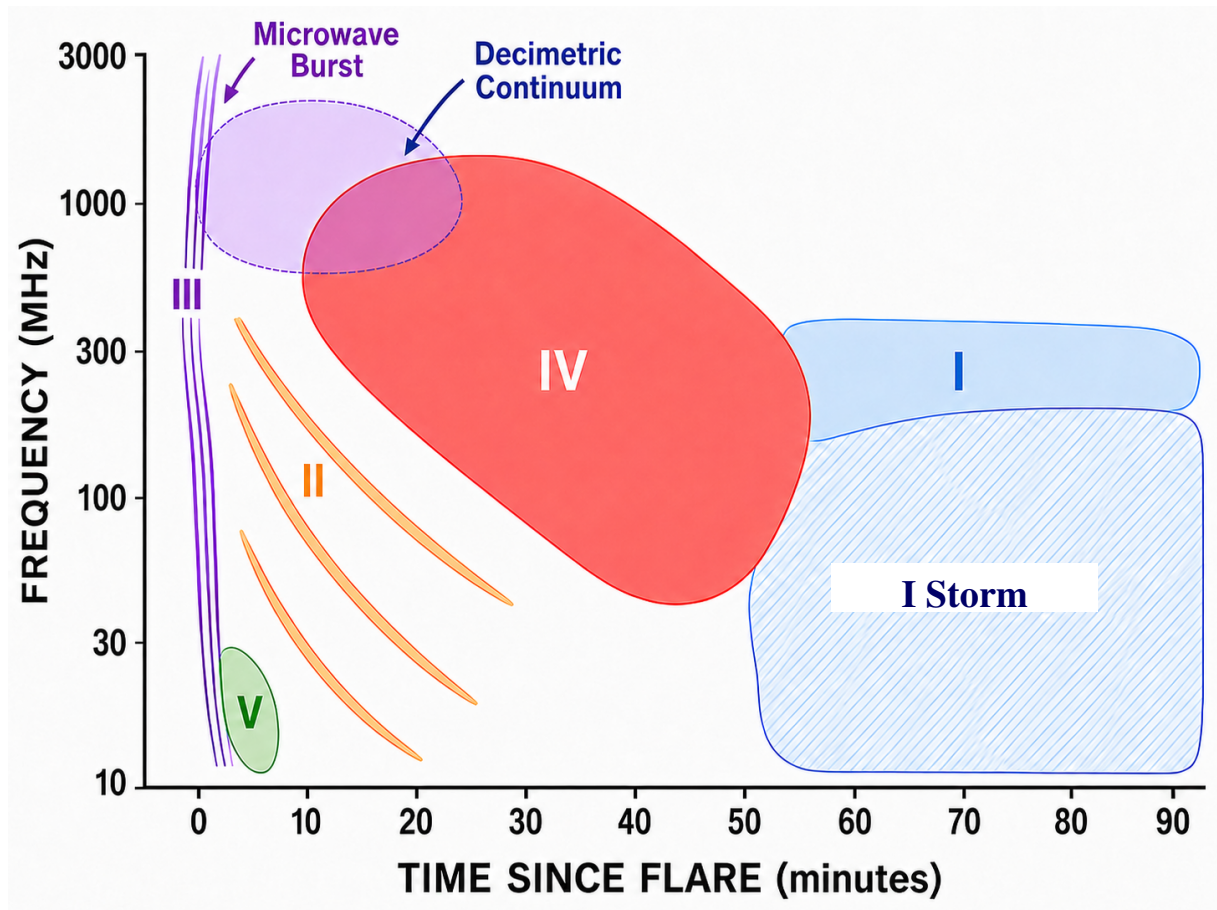


Figure 2.1. Solar dynamic spectrum showing Type I–V burst classification.

The schematic frequency–time plot illustrates the quiet-Sun background continuum and the five burst categories. Type I storms appear as narrowband chains; the fast-drifting Type III lanes are produced by electron beams escaping along open field lines and are the dominant transient signature at decimetre wavelengths; slowly drifting Type II arcs trace coronal shock fronts driven by coronal mass ejections. Adapted from McLean and Labrum [9] and Dulk [4]; the long-duration, narrowband Type I noise-storm label on the source figure has been corrected here (originally mislabelled).

2.1.3 Why 610 MHz, and Why the Quiet Sun Is Useful for Validation

Chapter 1 explained the practical reasons for working at 610 MHz: archived flux data are freely available at this frequency, and low-cost UHF Yagi-Uda antennas are easy to obtain. The physical picture above adds the remaining justification. First, 610 MHz still falls inside the decimetric/UHF band where flare-related non-thermal emission-the fast-drifting tail of Type III bursts, decimetric continua, and spike bursts-is observed, so the band is not physically irrelevant to solar radio-burst work, even though the strongest and most frequently studied bursts (Type II and Type III at their onset) are usually observed at lower frequencies, tens of MHz, closer to the metric/decametric band used by instruments such as DLITE [12]. Second, and just as important for this thesis, the *quiet Sun*-the steady, thermal background described in Section 2.2.1-is present at 610 MHz every day, regardless of whether a burst occurs. Its brightness is stable and well characterised, so it provides a predictable, always-available source against which the geometry, timing, and signal chain of the simulated interferometer can be validated. A burst is unpredictable and, for a first-generation instrument, this thesis therefore uses the quiet Sun, observed during the transit of 20 March 2025 at Learmonth, as the validation target: any departure of the simulated response from the expected analytic result can then be attributed to the instrument model rather than to unpredictable source variability.

2.2 From Solar Radio Emission to Flux Density

2.2.1 Specific Intensity and Brightness Temperature

A radio telescope does not measure temperature directly, the way a thermometer does. It measures the intensity of the radio radiation arriving from the sky. In the absence of activity, the *quiet Sun* radiates by thermal free-free (bremsstrahlung) emission, and in radio astronomy it is conventional to express this intensity as an equivalent *brightness temperature* T_b : the temperature that a blackbody would need to have in order to produce the same observed specific intensity I_ν at frequency ν . In the radio (Rayleigh-Jeans) limit, this definition is written as

$$I_\nu = \frac{2k_B\nu^2}{c^2} T_b, \quad (2.1)$$

where:

- I_ν is the specific intensity, in $\text{W}/(\text{m}^2 \text{ Hz sr})$;
- T_b is the brightness temperature, in kelvin;
- ν is the observing frequency, in hertz;
- k_B is Boltzmann's constant, which appears because it is the physical constant that converts a temperature in kelvin into an energy scale;
- c is the speed of light [4, 20].

At 610 MHz the quiet-Sun brightness temperature is of order 10^5 - 10^6 K. Equation (2.1) is a *definition* of what brightness temperature means—it is not a computational step that the simulation evaluates. The simulation never needs a numerical value of T_b : it works directly with the flux density introduced next, which is the quantity an antenna actually collects. Brightness temperature is given here only because it is the standard way the radio-astronomy literature describes how bright the quiet Sun is, and because it makes the stability argument of Section 2.1.3 precise: the quiet Sun’s brightness temperature is steady from day to day, so departures from the predicted fringe pattern can be attributed to the instrument rather than to the source.

2.2.2 Flux Density and the Solar Flux Unit

The specific intensity I_ν describes the brightness of a resolved source per unit solid angle, which is not directly what a receiver measures. What a receiver actually collects is the *flux density* S_ν —the intensity integrated over the angular extent of the source—expressed in $\text{W}/(\text{m}^2 \text{ Hz})$. For the Sun, the integrated flux density is a few tens of solar flux units [4, 5], where one solar flux unit is

$$1 \text{ sfu} = 10^{-22} \text{ W}/(\text{m}^2 \text{ Hz}). \quad (2.2)$$

The solar flux unit is the standard scale on which archived solar radio observatories, including Learmonth, report their measurements. In this thesis, the archived Learmonth flux record $S_\nu(t)$, in solar flux units at 610 MHz, is the external physical input of the simulation: it is why the solar flux unit matters here, and it is the quantity that Sections 2.3-2.7 carry through every remaining stage of the theoretical chain.

2.2.3 From Flux Density to a Simulation Input $S_\nu(t)$

The Learmonth archive does not provide a continuous function of time; it provides a sequence of flux-density samples recorded at fixed intervals over the course of a day, in SFU, at 610 MHz. To use this record as the input of a time-domain simulation, the discrete samples are converted to SI units using Eq. (2.2) and then interpolated to produce a continuous function $S_\nu(t)$ that can be evaluated at every simulation time step, however finely the array geometry or receiver chain is sampled. The details of the interpolation, and the archived file format from which $S_\nu(t)$ is built, are given in Chapter 4. What matters here is the role $S_\nu(t)$ plays in the theoretical chain: it is the single external input that carries the real, measured behaviour of the Sun into every later stage of the simulation, from the received voltage (Section 2.3) to the final image (Section 2.7).

2.3 From Solar Flux to Received Voltage

A solar flux density is not yet an electrical signal. To simulate a receiver, $S_\nu(t)$ must be converted, step by step, into a received radio-frequency power and then into a voltage that a digitiser could actually sample. This section carries out that conversion.

2.3.1 Antenna Gain, Primary Beam, and Effective Aperture

A receiving antenna converts the incident electromagnetic field into a voltage at its terminals. Its directional sensitivity is described by the *normalised power pattern* $A_n(\theta, \phi)$, which for a receiving element is identical to its transmitting pattern by reciprocity. The *directivity* D and *gain* G follow from

$$D = \frac{4\pi}{\int \int_{4\pi} A_n(\theta, \phi) d\Omega}, \quad G = \eta D, \quad (2.3)$$

where:

- $A_n(\theta, \phi)$ is the normalised power pattern (equal to 1 at boresight), as a function of the usual spherical angles (θ, ϕ) ;
- η is the radiation efficiency (fraction of input power actually radiated);
- $d\Omega$ is an element of solid angle, in steradians [21, 22].

The *half-power beamwidth* (HPBW) is the angular width over which $A_n \geq 0.5$, and it sets the field of view of a single element. Once the sky position is expressed in the direction-cosine coordinates (l, m) used later in this chapter, the same normalised pattern is written $A_n(l, m)$; in an interferometer this is exactly the *primary beam* that multiplies the sky brightness in Section 2.7. Using one isolated-element pattern for every antenna in the array assumes that the elements do not interact electromagnetically with one another; this assumption is checked directly, by a mutual-coupling study of the real array, in Chapter 3.

For imaging purposes the key quantity derived from the gain is the *effective aperture* $A_e(\theta, \phi)$,

$$A_e = \frac{\lambda^2}{4\pi} G, \quad (2.4)$$

where λ is the observing wavelength and G is the gain of Eq. (2.3) [7, 21]; A_e has units of area and is, loosely, the area over which the antenna effectively collects power from an incident plane wave.

The receiving element used throughout this thesis is a Yagi-Uda array: a single driven element (here a folded dipole), a reflector behind it, and several directors in front. The parasitic elements re-radiate with phases set by their length and spacing, producing an end-fire pattern with gain typically in the range 8 dBi to 12 dBi. The folded dipole raises the driven-element impedance to roughly four times that of a simple dipole, which eases matching to the feed. Besides the gain and HPBW already defined above, the relevant figure of merit for how well the element is matched to its feed line is the standing-wave ratio

$$\text{SWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}, \quad \Gamma = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0}, \quad (2.5)$$

where Z_{in} is the antenna input impedance and Z_0 is the feed-line impedance (75 Ω for the coaxial cable used) [21, 23]. These quantities are computed for the real element in Chapter 3.

2.3.2 Friis Reception Relation

With the effective aperture available, a source of flux density S_ν observed over a bandwidth $\Delta\nu$ delivers, to a matched load, a received power given by the Friis reception relation [16]:

$$P_{\text{rx}} = \frac{1}{2} A_e S_\nu \Delta\nu, \quad (2.6)$$

where:

- S_ν is the flux density of the source, in $\text{W}/(\text{m}^2 \text{ Hz})$ (Section 2.2.2);
- $\Delta\nu$ is the receiver bandwidth, in hertz;
- A_e is the effective aperture of Eq. (2.4);
- the factor $\frac{1}{2}$ accounts for a single received polarisation of unpolarised radiation [7, 20];
- P_{rx} is the resulting received power, in watts.

Equation (2.6) is the bridge between the astronomical flux scale and the receiver, and it is a computational step evaluated directly by the simulation: it takes $S_\nu(t)$ from Section 2.2.3 and turns it into a received power $P_{\text{rx}}(t)$, as implemented in Chapter 4.

2.3.3 Voltage Equivalent at the Receiver Input

The simulation, like a real digitiser, works with voltages, not power. For a receiver with a resistive load R_{load} , the received power of Eq. (2.6) corresponds to a root-mean-square signal voltage

$$V_{\text{rms}} = \sqrt{P_{\text{rx}} R_{\text{load}}}, \quad (2.7)$$

where P_{rx} is the received power of Eq. (2.6) and R_{load} is the load resistance seen by the antenna; the relation follows directly from $P = V_{\text{rms}}^2/R$ for a matched resistive load. Equation (2.7) is the last step of the chain from flux to voltage, and, like Eq. (2.6), it is evaluated directly by the simulation in Chapter 4 (Section 4.4). The antenna's directional response enters at this same voltage level, by multiplying V_{rms} with the square root of the normalised primary beam A_n evaluated at the Sun's instantaneous position, so that a source away from boresight produces a smaller voltage than the same source at boresight. This gives the deterministic, signal part of the antenna voltage; the next section adds the part that is not deterministic.

2.4 Receiver Noise and Averaging

Even though the Sun produces a measurable flux density, a real receiver never observes that signal alone. Every channel also carries thermal noise contributed by the antenna itself, the sky, the ground, the cable, the low-noise amplifier, and the rest of the receiver chain. A faithful simulation must therefore generate not only the deterministic solar voltage of Eq. (2.7), but also a statistically correct noise voltage to add to it. This section builds that noise model and then asks how averaging can recover a weak signal from underneath it.

2.4.1 System Temperature and Johnson-Nyquist Noise

Any resistive element at physical temperature T generates a fluctuating voltage, even with no signal present [17]. The available noise power in a bandwidth $\Delta\nu$ is, in the Rayleigh-Jeans limit,

$$P_n = k_B T \Delta\nu, \quad (2.8)$$

where:

- k_B is Boltzmann's constant, as in Eq. (2.1);
- T is the physical temperature of the noise-generating element, in kelvin;
- $\Delta\nu$ is the bandwidth, in hertz;
- P_n is the resulting noise power, in watts.

Equation (2.8) underlies the convention of expressing every noise contribution as an equivalent *noise temperature*. The total *system temperature* T_{sys} sums the sky, ground, antenna, and receiver contributions, and the system-equivalent flux density follows from Eq. (2.6) as $\text{SEFD} = 2k_B T_{\text{sys}}/A_e$ [7, 20]. Exactly as for the signal in Section 2.3.3, the simulation needs a voltage, not a power: the noise power of Eq. (2.8) is converted to a noise voltage with standard deviation

$$\sigma_n = \sqrt{\frac{1}{2} P_n R_{\text{load}}}, \quad (2.9)$$

where the factor $\frac{1}{2}$ splits the total noise power equally between the in-phase and quadrature components of the complex baseband signal. In the simulation, T_{sys} is a single configurable parameter, and Eqs. (2.8)-(2.9) are applied independently to every antenna to draw a fresh noise voltage at every time step, exactly as independent thermal noise would arise in every physical receiver channel. Adding this noise voltage to the signal voltage of Eq. (2.7) gives the complete per-antenna complex voltage used in Chapter 4 (Section 4.4).

2.4.2 Radiometer Equation, and Why Averaging Helps

Equation (2.9) shows that noise is always present, but it does not yet say whether a given signal can be recovered from underneath it. The *radiometer equation* answers that question: it gives the smallest detectable change in antenna temperature after integrating for a time τ over a bandwidth $\Delta\nu$,

$$\Delta T_{\text{rms}} = \frac{T_{\text{sys}}}{\sqrt{\Delta\nu \tau}}, \quad (2.10)$$

where T_{sys} is the system temperature of Section 2.4.1, $\Delta\nu$ is the bandwidth, and τ is the integration (averaging) time [7, 20]. This equation is not itself a step that generates simulated data; its role is to explain *why* the time-averaging performed by the correlator in Chapter 4 improves the signal-to-noise ratio: thermal noise is a random process with zero mean, so summing more independent samples, either by integrating longer or by using a wider bandwidth, averages the noise toward zero while the steady solar signal adds coherently. This $1/\sqrt{\Delta\nu \tau}$ scaling is the

statistical foundation on which both single-dish detection and interferometric correlation rest, and it is reproduced explicitly by the noise model of Section 4.4 in Chapter 4.

2.4.3 Signal-to-Noise Interpretation for the Simulation

Combining Eqs. (2.6), (2.8), and (2.10) gives a direct way to ask whether a given signal is detectable: the quiet-Sun or burst flux density is converted to an equivalent antenna temperature and compared with ΔT_{rms} for the integration time and bandwidth in use. If the equivalent signal temperature is well above ΔT_{rms} , the source should be visible above the noise floor in the simulated voltages and, after correlation, in the simulated visibilities; if it is comparable to or below ΔT_{rms} , no amount of correct geometry will recover a clean fringe, which is exactly the situation encountered in the field deployment described in Chapter 1. This comparison is carried out quantitatively for the simulated instrument in Section 5.6 of Chapter 5.

2.5 From Antenna Voltage to Interferometric Visibility

Every antenna in the array now has its own complex voltage: a deterministic part from Eq. (2.7) and a random noise part from Eq. (2.9). A single antenna, however, can only measure *power*—the mean value of the squared voltage—and power alone carries no information about *where* on the sky the signal came from. Recovering that positional information is the job of interferometry, and it starts from a simple geometric fact: a celestial source is composed of a vast number of independent radiating regions—electrons gyrating incoherently in the corona, for instance—so the radiation received from any two distinct directions on the sky is statistically independent. This property, that the source is *spatially incoherent*, is the single most important physical assumption in interferometry: the coherence measured *between two antennas* does not arise because the source itself is coherent, but because the same incoherent field is sampled at two separated points, and the geometric delay between those points encodes information about the source’s angular structure. This is precisely the principle exploited in the early twentieth century by the Michelson stellar interferometer, in which two mirror apertures, fed into a common eyepiece, produce fringes whose contrast falls as the angular size of the source increases relative to λ/b ; it is the optical-astronomy ancestor of the radio-interferometric technique developed below.

2.5.1 Geometric Delay and Phase

Consider two identical antennas, separated by a baseline vector $\mathbf{b}$, that both receive the wavefront from a point source in direction $\hat{\mathbf{s}}$. Because the two antennas are not at the same point in space, the wavefront reaches one of them slightly later than the other. This time difference is the *geometric delay* [7, 8]

$$\tau_g = \frac{\mathbf{b} \cdot \hat{\mathbf{s}}}{c}, \quad (2.11)$$

where $\mathbf{b}$ is the baseline vector between the two antennas, $\hat{\mathbf{s}}$ is the unit vector pointing toward the source, and c is the speed of light. At a centre frequency ν_0 , this delay corresponds to a

phase difference $2\pi\nu_0\tau_g$ between the two antenna voltages: this is the entire physical origin of everything that follows in this section. Because the Earth rotates, $\hat{s}$ changes slowly with time relative to $\mathbf{b}$, so τ_g , and therefore the phase difference, also changes with time. In a real instrument the bulk of this delay is compensated electronically—a process called *fringe stopping*—by inserting a compensating delay τ_g in one signal path before correlation, so that the residual delay tracks only the difference between the true geometric delay and the model delay used by the correlator.

2.5.2 Two-Element Correlation

A *correlator* multiplies the voltage outputs of two antennas and time-averages the product. For a single point source, the correlator output after fringe stopping and averaging over many carrier cycles is the real part of a complex response:

$$r(\tau_g) \propto \cos(2\pi\nu_0\tau_g) = \cos\left(\frac{2\pi}{\lambda} \mathbf{b} \cdot \hat{s}\right). \quad (2.12)$$

This is the celebrated *interference fringe* [8]: as the Earth rotates, $\hat{s}$ sweeps slowly across the sky relative to $\mathbf{b}$, and the correlator output traces out a slowly varying cosine whose instantaneous frequency is the *fringe rate*. In practice a *complex correlator* forms both an in-phase output ($\cos$) and a quadrature output ($\sin$, obtained by inserting a quarter-wave phase shift in one path before the second multiplication), so that the full complex response, amplitude and phase, of the source on that baseline is recovered, rather than only its real part.

For an east-west baseline of length b observing a source at hour angle H and declination δ , differentiating Eq. (2.11) with respect to time gives the rate of change of fringe phase, and hence the fringe period

$$T_{\text{fringe}} \approx \frac{\lambda}{b\omega_{\oplus} \cos \delta}, \quad (2.13)$$

where λ is the observing wavelength, b is the baseline length, δ is the source declination, and $\omega_{\oplus}$ is the Earth's sidereal rotation rate [7, 8]. For $b = 8$ m, $\lambda = 0.49$ m and the solar declination at the spring equinox ($\delta = 0^\circ$), this evaluates to approximately 14 minutes. Equation (2.13) is *not* used to generate the simulated data; it plays a different role, as an independent analytic prediction used only afterwards, in Chapter 5, to check whether the simulated fringe behaves the way theory says it must. It is the most basic and most directly verifiable analytic signature of an interferometer, and for that reason it is the first validation benchmark used in this thesis.

The derivation above implicitly assumed a single frequency ν_0 . In practice every receiver has finite bandwidth $\Delta\nu$, and correlating over that bandwidth attenuates the fringe amplitude if the residual delay error after fringe stopping, $\Delta\tau = \tau_g - \tau_{\text{model}}$, is not negligible. Integrating a uniform passband of width $\Delta\nu$ centred on ν_0 introduces a multiplicative attenuation factor

$$\text{sinc}(\Delta\nu \Delta\tau) = \frac{\sin(\pi \Delta\nu \Delta\tau)}{\pi \Delta\nu \Delta\tau}, \quad (2.14)$$

known as *bandwidth smearing* [7]. Because the residual delay error in this simulation is determined entirely by the known geometry rather than by an imperfect delay model, this effect is

small for the narrow field of the quiet Sun and is quantified in Chapter 4.

2.5.3 Visibility Amplitude and Phase

Section 2.5.2 described the correlator in terms of an idealised continuous field. In terms of the actual per-antenna complex voltages built in Sections 2.3 and 2.4, $v_i(t)$ and $v_j(t)$, what the correlator for antenna pair (i, j) computes is

$$V_{ij} = \langle v_i(t) v_j^*(t) \rangle, \quad (2.15)$$

where:

- $v_i(t)$ is the complex voltage at antenna i ;
- $v_j^*(t)$ is the complex conjugate of the voltage at antenna j ;
- $\langle \cdot \rangle$ denotes a time average, long enough to average down the noise (Section 2.4.2) but short enough that the source itself is effectively unchanged.

The result V_{ij} is a single complex number that combines both an amplitude and a phase: this number is the *complex visibility*, $V = |V| e^{i\phi}$. Its amplitude $|V|$ measures the degree of coherence of the field between the two antennas—equal to the total received flux for a baseline short enough to be insensitive to the source’s angular size, and falling as the baseline lengthens and begins to resolve the source—while its phase ϕ encodes the position of the source relative to the phase-tracking centre. This is the key conceptual difference between a single antenna and an interferometer: *a single antenna measures power; a pair of antennas measures spatial coherence*, and it is this spatial-coherence measurement, repeated over many baselines and over time, that ultimately allows an image to be reconstructed.

For an extended, spatially incoherent source, Eq. (2.15) generalises the point-source fringe of Eq. (2.12) to an integral over the whole sky brightness distribution $I(\hat{s})$:

$$V(\mathbf{b}) = \int_{\Omega} I(\hat{s}) \exp \left[-i 2\pi \nu_0 \frac{\mathbf{b} \cdot \hat{s}}{c} \right] d\Omega, \quad (2.16)$$

where the integral is taken over the solid angle Ω occupied by the source [7, 8]. Equation (2.16) already contains the essential physics of synthesis imaging: the visibility measured on a baseline $\mathbf{b}$ is a weighted, phase-shifted integral of the sky brightness, so an array that samples many different baselines is, in effect, sampling many different weighted integrals of the same unknown sky. Turning this statement into a precise, invertible imaging equation requires a convenient coordinate system for the baseline and the sky, which Section 2.6 introduces next, and the formal statement of the imaging relation is given in Section 2.7.

2.6 From Physical Baselines to uvw Coverage

Equation (2.16) is written in terms of a baseline vector $\mathbf{b}$ and a source direction $\hat{s}$, but it says nothing yet about how $\mathbf{b}$ is actually obtained from the physical layout of the array, or how it

changes as the source moves across the sky. That is the purpose of this section: it connects the antenna positions that are physically surveyed in the field to the coordinate frame in which the visibility integral is most naturally expressed.

2.6.1 ENU Ground Coordinates

Every antenna in the array has a fixed physical position on the ground, measured in a local East-North-Up (ENU) frame tangent to the Earth at the observing site. The baseline vector $\mathbf{b}$ used throughout Section 2.5 is simply the difference between the ENU positions of two antennas. This is the natural frame in which an array is laid out and surveyed in the field, and it is the starting point of the coordinate chain used in Chapter 4 to describe an arbitrary two-dimensional array layout, not only a single east-west line.

2.6.2 Source-Based uvw Coordinates

Visibility, however, is most naturally expressed in a frame tied to the source direction, not to the ground: this is why Eq. (2.16) is written with $\mathbf{b} \cdot \hat{\mathbf{s}}$ rather than with the raw ENU coordinates. The array geometry must therefore be transformed from ground-based ENU coordinates to source-based spatial-frequency coordinates, conventionally called (u, v, w) . Obtaining that frame takes two steps, as described by Taylor et al. [8]. First, the local ENU antenna positions are converted to an Earth-fixed equatorial (X, Y, Z) frame, in which X points to the intersection of the equator and the local meridian, Y points east, and Z points to the celestial pole. Second, this equatorial frame is rotated, for the instantaneous hour angle H and declination δ of the phase-tracking centre, into the baseline-dependent (u, v, w) frame in which w points along the source direction:

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} \sin H & \cos H & 0 \\ -\sin \delta \cos H & \sin \delta \sin H & \cos \delta \\ \cos \delta \cos H & -\cos \delta \sin H & \sin \delta \end{bmatrix} \begin{bmatrix} X_\lambda \\ Y_\lambda \\ Z_\lambda \end{bmatrix}, \quad (2.17)$$

where:

- H is the hour angle of the phase-tracking centre;
- δ is its declination;
- $(X_\lambda, Y_\lambda, Z_\lambda)$ are the equatorial baseline components, expressed in units of wavelength;
- (u, v, w) are the resulting source-based baseline coordinates, also in units of wavelength, with u and v spanning the plane perpendicular to the source direction and w along it [7, 8].

This single, general transformation is applied identically to every baseline and every instant in the simulation pipeline, so that per-antenna geometric phase, instantaneous baseline length, and fringe-stopping delay are all computed from one consistent geometric model, valid for arbitrary two-dimensional array layouts rather than only for the east-west case of Eq. (2.11).

2.6.3 Earth Rotation and uv -Coverage

The set of (u, v) points sampled by an interferometric array-its uv -coverage-dictates the array's imaging capability, in direct analogy to the way the aperture of a filled-aperture telescope determines its point-spread function. A single snapshot with N antennas yields $N(N - 1)/2$ instantaneous baselines, hence $N(N - 1)/2$ independent samples of $V(u, v)$ (each baseline contributing two complex-conjugate samples at $\pm(u, v)$, since $I(l, m)$ is real and therefore $V(-u, -v) = V^*(u, v)$). For the four-element array of this thesis, $N = 4$ gives only six instantaneous baselines-far too sparse, on its own, to constrain a two-dimensional image.

As the Earth rotates, however, the projection of any fixed physical baseline onto the plane perpendicular to the source direction changes continuously, because the rotation matrix in Eq. (2.17) depends on the hour angle H . For a baseline with equatorial components $(X_\lambda, Y_\lambda, Z_\lambda)$ observing a source at fixed declination δ , eliminating H between the u and v equations of Eq. (2.17) shows that each baseline traces an *ellipse* in the (u, v) plane as H varies over the day, with semi-major axis equal to the projected east-west baseline length and semi-minor axis foreshortened by $\sin \delta$. This technique, known as *Earth-rotation synthesis* [8], allows a small number of physical antennas to synthesise, over the course of a transit, a uv -coverage equivalent to a much larger number of simultaneous baselines, without any moving parts. It is the reason a four-element, six-baseline array can nonetheless produce a meaningful synthesised image once data accumulated over an observing transit are combined-a trade-off explored quantitatively, for the compact east-west geometry of the physical array against more general two-dimensional configurations, in Chapter 5.

Earth-rotation synthesis assumes that each visibility sample is attributed to a single, exact (u, v) point, when in fact it is an average over the integration time of one correlator dump. If the integration time is long enough for H to change appreciably, a source away from the phase-tracking centre is smeared tangentially in the uv plane, an effect known as *time-average smearing*, in direct analogy to the bandwidth smearing of Eq. (2.14). For the short integration times and narrow field of the quiet Sun used in this thesis this effect is small, and it is quantified together with bandwidth smearing in Chapter 4.

2.7 From Visibility to Dirty Image

2.7.1 Van Cittert-Zernike Theorem

Visibility is useful precisely because, under the two assumptions already introduced-an incoherent source (Section 2.5) and a narrow field of view-it is related to the sky brightness by an ordinary two-dimensional Fourier transform. In this small-field limit, appropriate for the compact solar disk observed in this thesis, the relation is

$$V(u, v) = \iint A_n(l, m) I(l, m) e^{-2\pi i(ul+vm)} dl dm, \quad (2.18)$$

where:

- $I(l, m)$ is the sky brightness distribution;

- $A_n(l, m)$ is the normalised primary beam of a single antenna (Section 2.3.1);
- (l, m) are direction cosines on the sky, measured relative to the phase-tracking centre;
- (u, v) are the baseline components in wavelengths, from Section 2.6.

This is the *van Cittert-Zernike theorem* as used throughout synthesis imaging [7, 8]: the visibility and the (apodised) sky brightness are Fourier conjugates, so that, given sufficient and suitable sampling of $V(u, v)$, the sky brightness can in principle be recovered by inverse Fourier transformation. Each baseline therefore samples one point in this Fourier (u, v) plane, and it is the combination of many simultaneous baselines with the additional points that Earth rotation supplies over time (Section 2.6.3) that builds up usable uv -coverage.

Equation (2.18) is a simplification of a more general relation that keeps a third baseline coordinate, w , and a corresponding phase term:

$$V(u, v, w) = \iint A_n(l, m) I(l, m) e^{-2\pi i [ul + vm + w(n-1)]} \frac{dl dm}{n}, \quad n = \sqrt{1 - l^2 - m^2}. \quad (2.19)$$

If the primary beam is narrow enough that $l, m \ll 1$ over the region where $A_n I \neq 0$, then $n - 1 \approx -\frac{1}{2}(l^2 + m^2)$ is small, and provided the w -term phase error $2\pi w(n - 1)$ remains $\ll 1$ radian across the field of view, the w -dependence may be dropped entirely and Eq. (2.19) reduces exactly to Eq. (2.18). For the solar disk, whose $32'$ angular diameter subtends $l, m \sim 5 \times 10^{-3}$, and for the short, near-zenith baselines used in this thesis, the w -term is indeed negligible, and the small-field relation of Eq. (2.18) applies without correction. For a wider field of view or a longer-baseline array than used in this thesis, the w -term of Eq. (2.19) would have to be retained.

A baseline of length $|b|$ measured in wavelengths corresponds, through Eq. (2.18), to an angular scale of order $1/|b|_\lambda$ radians on the sky. For a uniformly bright circular disk of angular radius θ_s (a simplistic but effective model for the quiet Sun), the two-dimensional Fourier transform in Eq. (2.18) has the closed analytic form

$$V(q) = \frac{2J_1(2\pi q\theta_s)}{2\pi q\theta_s} \equiv \text{jinc}(2\pi q\theta_s), \quad q = \sqrt{u^2 + v^2}, \quad (2.20)$$

where J_1 is the first-order Bessel function of the first kind and q is the baseline length in wavelengths [7, 8]. The jinc function falls monotonically from unity at $q = 0$ (the disk's total flux, measured on a zero-length baseline) to its first null at $q\theta_s \approx 0.61$, beyond which the visibility amplitude is small and oscillatory. Like the fringe period of Eq. (2.13), Eq. (2.20) is not used to generate simulated data; it is an independent analytic target, computed separately from the simulation, against which the simulated visibilities of the resolved solar disk are validated in Chapter 5: a correctly implemented simulation must reproduce both the jinc envelope and the location of its first null as a function of baseline length.

Two further practical error sources act on the phase of the visibility rather than on its amplitude, and are worth noting here because they affect the imaging step directly. *Phase-calibration and pointing errors* shift and blur the reconstructed image; in this simulation the geometry is known exactly, so they are introduced only as controllable perturbations (Chapter 4). *Ionospheric*

phase is, for the short baselines used in this thesis ($\lesssim 130$ m at 610 MHz), essentially common-mode between antennas and cancels in the differential (baseline) measurement, with a residual differential phase of about one degree.

2.7.2 Direct Fourier Inversion

Because only a finite, discrete set of (u, v) points is ever sampled, it is convenient to describe the measurement formally as a multiplication of the true, continuous visibility function by a *sampling function* $S(u, v)$ -a sum of Dirac delta functions (optionally weighted to account for the relative sensitivity or density of different baselines) located at each measured (u, v) point and its conjugate. The measured visibility is then $S(u, v) V(u, v)$, and inverse-Fourier-transforming this sampled, rather than continuous, visibility does not yield the true sky brightness I , but rather the *dirty image* I_D :

$$I_D(l, m) = \iint S(u, v) V(u, v) e^{+2\pi i(ul+vm)} du dv. \quad (2.21)$$

where $S(u, v)$ is the sampling function defined above and $V(u, v)$ is the measured visibility of Eq. (2.18) at each sampled point. Equation (2.21) is the direct inversion that the simulation actually carries out in Chapter 4: every measured visibility is placed at its (u, v) location and the sampled function is inverse-Fourier-transformed, with no further processing such as deconvolution.

2.7.3 Dirty Beam, PSF, and the Convolution Relation

By the convolution theorem [8], multiplication of V by S in the Fourier (uv) domain corresponds to convolution in the image domain, so Eq. (2.21) may equivalently, and more usefully, be written as the *Convolution Equation* of synthesis imaging [8]:

$$I_D(l, m) = I(l, m) * B(l, m), \quad (2.22)$$

where $*$ denotes two-dimensional convolution, $I(l, m)$ is the true sky brightness, and the *dirty beam* (equivalently, the *point-spread function*, PSF, or *synthesised beam*) $B(l, m)$ is simply the Fourier transform of the sampling function itself,

$$B(l, m) = \iint S(u, v) e^{+2\pi i(ul+vm)} du dv. \quad (2.23)$$

Equations (2.22)-(2.23) [7, 8] express the fundamental imaging equation of an interferometer: the dirty image is exactly what one would expect from imaging the true sky through an instrument whose response to a point source is the beam $B(l, m)$ -a direct radio analogue of the diffraction-limited PSF of an optical telescope, except that here the PSF is determined not by a filled aperture but by the discrete pattern of sampled (u, v) points. Put practically: the dirty image is *not* a perfect picture of the true sky. It is the true sky "stamped" by the array's PSF, and because $S(u, v)$ is a sparse, typically non-uniform set of samples rather than a filled, azimuthally symmetric disk (as for a filled-aperture telescope), the dirty beam $B(l, m)$ generally has a narrow central peak-whose width sets the nominal angular resolution of the synthesised image, of order $1/b_{\max}$

radians for a maximum baseline $b_{\max}$ measured in wavelengths-surrounded by sidelobes that reflect the gaps in uv -coverage. For the compact, six-baseline east-west array of this thesis, the uv -coverage accumulated over an Earth-rotation transit consists of six partial elliptical arcs rather than a densely filled disk, so $B(l, m)$ retains significant sidelobe structure; this is expected, and the resulting dirty image is still a rigorously correct image of the sky under Eq. (2.22) given the actual sampling achieved-it is simply not a clean, sidelobe-free representation of $I(l, m)$. Because sidelobes are an unavoidable consequence of sparse uv -coverage rather than a simulation defect, Chapter 5 evaluates the simulated dirty image not only by inspection but through the PSF shape itself, its dynamic range, and its main-lobe FWHM.

Because the simulation in this thesis observes a single, compact, time-stable calibrator (the quiet-Sun disk) rather than attempting to image a complex field crowded with confusing sources, the dirty image and dirty beam together are sufficient to validate the instrument: demonstrating numerically that the simulated I_D equals the assumed input sky map convolved with the independently computed $B(l, m)$, as required by Eq. (2.22), constitutes the ultimate end-to-end mathematical verification of the entire signal-processing pipeline, from antenna voltages to final image, and is carried out explicitly in Chapter 5.

2.8 Chapter Summary: Theoretical Chain Used by the Simulation

Read as a whole, this chapter tells one story. The Sun produces radio emission (Section 2.1); the international community measures this emission as a flux density, in solar flux units (Section 2.2); an antenna converts that flux density into received radio-frequency power, and a receiver converts that power into a voltage together with its own unavoidable noise (Sections 2.3-2.4); several antennas at known positions see the same signal with a small, geometry-dependent phase difference, and a correlator turns a pair of antenna voltages into a single complex visibility (Section 2.5); the array geometry places every visibility at its correct point in a Fourier (u, v) plane (Section 2.6); and an inverse Fourier transform of those points produces the dirty image analysed in Chapter 5 (Section 2.7). Every equation in this chapter is a necessary step in that one chain, not an independent piece of theory.

Not every equation plays the same role in the chain, and it is worth separating the two roles explicitly, as flagged throughout this chapter:

- **Computational steps, evaluated directly by the simulation in Chapter 4:** the flux-to-SI conversion (Eq. (2.2)) and interpolation to $S_\nu(t)$ (Section 2.2.3); the Friis relation and voltage equivalent (Eqs. (2.6)-(2.7)); the Johnson-Nyquist noise power and noise voltage (Eqs. (2.8)-(2.9)); the $ENU \rightarrow XYZ \rightarrow uvw$ geometry (Eq. (2.17)); and the direct Fourier inversion to the dirty image (Eq. (2.21)).
- **Analytic benchmarks, used only afterwards to validate the simulation in Chapter 5:** the fringe period (Eq. (2.13)); the van Cittert-Zernike relation and the closed-form jinc visibility of a uniform disk (Eqs. (2.18)-(2.20)); and the convolution relation between the dirty image, the true sky, and the synthesised beam (Eqs. (2.22)-(2.23)).

This formalism, developed throughout in the notation of Taylor et al. [8], provides both the modelling choices used to build the simulation and the analytic benchmarks used to check it, and it is this distinction-not merely the equations themselves-that the chapters that follow rely on.

CHAPTER 3. ELECTROMAGNETIC MODELLING OF THE RECEIVING ELEMENT

It is worth noting at the outset that low-cost interferometer projects such as DLITE (Chapter 1) favour LWA-style crossed-dipole antennas, which are better suited to DLITE’s much lower 30 MHz to 40 MHz operating band; the Yagi-Uda antenna adopted here is not an untested substitute chosen for lack of an alternative. Yagi-class antennas have a long and independently documented track record as a low-cost, high-reliability receiving element, in amateur radio and in commercial UHF television reception alike [21, 24, 25], and this chapter goes on to show that the same combination of low cost and dependable performance carries over directly to the 610 MHz solar case: the specific unit used in this thesis is validated below against three independent methods (Section 3.6) and found to be well matched, with negligible mutual coupling at the operating baselines (Section 3.7).

The accuracy of the entire interferometric simulation depends on having a reliable model of the receiving element, and this chapter builds that model as a chain of evidence rather than a survey of antenna theory: a real antenna is characterised for gain and beam shape, its impedance is checked for a stable match, and finally its behaviour when placed next to an identical neighbour is checked against the independent-sensor assumption that Chapter 2 relies on. Only once every link in that chain holds is the antenna model handed to the simulation pipeline of Chapter 4. This thesis is, at its core, a study of an *antenna system*: the solar radio burst is the observation target, but the main contribution is the careful characterisation of the receiving array and the demonstration that its electromagnetic behaviour supports the assumptions made in the signal-processing pipeline.

3.1 Why the Receiving Element Must Be Modelled

In an aperture-synthesis array, the single-element radiation pattern is the primary beam of the instrument (Section 2.3.1). It sets the field of view within which the Sun can be observed, and it controls how strongly off-axis terrestrial interference enters the measurement. Electromagnetic coupling between neighbouring elements also needs to be verified: if coupling is significant, the array cannot be treated as a set of independent sensors-the core assumption of the van Cittert-Zernike imaging framework developed in Chapter 2-and a more complex calibration would be required. Concretely, before any solar data can be trusted, this chapter must answer four questions about the physical antenna:

1. **Reception:** does the antenna actually receive a usable signal at 610 MHz?
2. **Beam:** what is its real radiation pattern, so that the correct primary beam-not an idealised Gaussian-can be used in the simulation?

3. **Matching:** is its impedance stable and well matched enough that the received power computed by the Friis relation (Eq. (2.6)) is a physically meaningful quantity, rather than being dominated by reflected power lost back down the feed?
4. **Independence:** when several such antennas are placed close together to form an array, does mutual coupling break the independent-sensor assumption on which the imaging theory of Chapter 2 depends?

The remainder of this chapter answers these four questions in order, in one continuous chain: the receiving element is introduced and shown to be suitable in principle (Sections 3.2-3.3); it is placed against a professional reference system and a field trial to motivate why a controlled model is necessary at all (Section 3.4); it is then modelled numerically (Section 3.5) and validated against direct measurement (Section 3.6); its behaviour as an array element is checked against the independent-sensor assumption (Section 3.7); and finally its measured beam is exported in the form actually consumed by the simulation (Section 3.8).

3.2 Yagi-Uda Antenna as the Selected Low-Cost Receiving Element

The Yagi-Uda antenna remains one of the most widely used directional antennas in the Ultra-High-Frequency (UHF) band. First described in the English-language literature by Yagi [24], building on the work of Shintaro Uda, the design achieves an attractive balance between high directivity, mechanical robustness, and very low fabrication cost [21, 25]. These properties are precisely what a low-budget, educational solar radio instrument requires, and they are the reason a commercial Yagi forms the receiving element of this project.

3.2.1 Operating Principle and Parasitic Excitation

Unlike a phased array, in which every element is fed with an individually controlled voltage and phase, the Yagi-Uda array operates by *parasitic excitation*: it comprises a single active *driven element* together with several passive (parasitic) elements whose currents are induced purely by electromagnetic coupling with their neighbours. This is the same physical mechanism, generalised to many elements, that reappears in Section 3.7 between two separate antennas of the array: a current flowing on one conductor induces a voltage on every nearby conductor, and the strength of that induced voltage, for a given current, is called the *mutual impedance*.

For an array of $N + 1$ elements, indexing the reflector as -1 , the driven element as 0 , and the directors as $1, 2, \dots, N$, the terminal voltage on element m is the sum of the voltage its own current would produce if it were isolated, plus the voltage induced on it by the current on every other element:

$$V_m = \sum_n Z_{mn} I_n, \quad (3.1)$$

where:

- V_m is the terminal voltage of element m -the voltage that would be measured across its (real or virtual) feed gap;

- I_n is the current on element n ;
- Z_{mn} , for $m \neq n$, is the *mutual impedance* between elements m and n -the voltage induced on element m per unit current on element n ;
- Z_{mm} is the *self-impedance* of element m -the impedance it would present if every other element carried zero current [21].

Only the driven element is fed, so $V_0 \neq 0$ while every parasitic element is short-circuited at its (virtual) terminals:

$$0 = \sum_n Z_{kn} I_n \quad (k \neq 0). \quad (3.2)$$

Equation (3.2) [21, 25] shows that the parasitic currents are not free parameters: they are fixed by the geometry through the off-diagonal mutual impedances. The design freedom available to the engineer is therefore the choice of element lengths and spacings, which set the magnitude and-crucially -the *phase* of each induced current.

The directional behaviour follows directly from these induced phases [22, 25]:

- **Reflector.** Positioned behind the driven element and made slightly longer than the resonant half-wavelength ($\approx 5\%$ longer), the reflector presents an *inductive* reactance. The induced current consequently lags the driving field, and the superposition of the re-radiated field with the primary field is constructive toward the front and destructive toward the rear. A single reflector captures most of the available rearward suppression; additional reflectors yield diminishing returns, which is why practical Yagis use one reflector (or, as here, one reflecting structure) but many directors.
- **Directors.** Positioned in front of the driven element and made $\approx 5\%$ *shorter* than resonance, the directors are *capacitive*. Their induced currents lead the driving field, so the composite wave is progressively guided forward, sharpening the main lobe into an end-fire beam along the boom axis [21].

The net effect is a travelling-wave-like structure that concentrates radiated (or, by reciprocity, received) power along the boom.

3.2.2 Directivity Build-Up and the Gain-Bandwidth Trade-off

The directivity of a Yagi-Uda array increases monotonically, though sub-linearly, with the number of directors. A useful qualitative progression, consistent with both classical treatments and the present design, is that an isolated half-wave dipole provides about 2.15 dBi; adding a reflector raises this to roughly 7 dBi; the first director takes it close to 9.7 dBi; and a well-tuned six-director structure of the type used here reaches the 9.5 dBi to 13 dBi range depending on boom length and tuning [21, 22]. Each added director narrows the main lobe, which improves angular resolution and-important for a ground-based solar instrument-reduces the half-power beamwidth through which horizon-level terrestrial interference can enter.

This gain, however, comes at the cost of bandwidth and tuning sensitivity. As more directors are added, the array impedance becomes more reactive and the input match deteriorates away

from the design frequency. Pushing for too much gain can reduce the usable bandwidth and even lower the peak gain by several decibels if the element phasing moves away from the optimum [22]. For the solar application this trade-off is benign: the instantaneous observing bandwidth is narrow (a single 1 MHz channel centred at 610 MHz), so a moderately high-gain, moderately broadband design is entirely appropriate, and the bandwidth inherited from the original DVB-T2 service is far wider than the observation requires.

3.2.3 The Selected Physical Antenna: HKD G11KD-T2

The physical receiving element adopted for this project is the commercial HKD G11KD-T2, a high-gain UHF Yagi-Uda antenna originally marketed for DVB-T2 digital-television reception. As a commercial-off-the-shelf (COTS) component it combines mature RF engineering with rugged, weather-resistant aluminium construction, and it is available at a small fraction of the cost of a purpose-built radio-astronomy feed-the same low-element-cost argument developed for the LWA-style antennas of DLITE in Chapter 1.

Because the manufacturer publishes only marketing specifications, the geometry used for simulation was established by reconciling three independent sources: the manufacturer drawing, photographs of the deployed unit (Figure 3.1), and direct physical measurement of the antenna with callipers. The resulting model, summarised in Table 3.1, comprises an eleven-element structure on an aluminium boom of overall length ≈ 550 mm. This same geometry is the input to the 4NEC2 antenna simulation software described in Section 3.5.

Table 3.1. HKD G11KD-T2 antenna geometry used as electromagnetic model basis.

Dimensions were established by reconciling the manufacturer drawing, photographs of the deployed unit at the Xuan Mai test site, and direct calliper measurements. The eleven-element Yagi sits on an aluminium boom of overall length approximately 550 mm; spacings are measured along the boom from the driven element (director 1 is the element closest to the feed).

Element	Dimension
Reflector (four-bar U-frame)	outer bars at $z = \pm 40$ mm, inner bars at $z = \pm 10$ mm
Driven element (folded dipole)	length ≈ 210 mm, fold height 44 mm
Director lengths	≈ 210 mm (each)
Director spacings	82, 40, 50, 50, 66, 69 mm
Boom (square profile)	15×15 mm, length ≈ 550 mm
Element material	aluminium (hollow tube / solid rod)



Figure 3.1. Physical HKD G11KD-T2 Yagi-Uda antenna unit.

The eleven-element structure is shown with the four-bar U-frame reflector at the rear and the sequence of directors extending forward along the aluminium boom. The folded-dipole driven element sits immediately in front of the reflector frame; the mast-head LNA mounts directly at the feed terminals behind the driven element. Calliper measurements of this unit provided the dimensional inputs for Table 3.1.

The four-bar U-frame reflector is the distinguishing geometric feature of this antenna. Rather than a single straight rod, the reflector forms a shallow three-dimensional frame: this widens the effective reflecting aperture, raises the front-to-back ratio, and in the radio-astronomy context suppresses pickup of warm-ground thermal noise entering through the rear lobe.

3.2.4 Suitability for 610 MHz Solar Observation

At 610 MHz the free-space wavelength is

$$\lambda = \frac{c}{f} = \frac{2.998 \times 10^8 \text{ m/s}}{610 \times 10^6 \text{ Hz}} \approx 491.5 \text{ mm}, \quad (3.3)$$

so a multi-element Yagi remains physically compact (boom length of order one metre) and can be hand-built and tripod-mounted—a decisive advantage for a low-cost, portable instrument [21]. The Sun is a weak source at this frequency during quiet conditions (Chapter 2), and any target burst must be separated from a strong terrestrial noise background. A directional element with moderate gain (~ 9.5 dBi) and a high front-to-back ratio is therefore valuable: it concentrates sensitivity toward the solar transit path while suppressing pickup from the horizon and from behind the array [7].

There is, however, an important double-edged aspect to using a commercial DVB-T2 antenna for this purpose, and it is worth stating explicitly because it shapes the field results in Section 3.4. The very features that make the antenna attractive—that it is mass-produced, inexpensive, and

resonant across the 470 MHz to 862 MHz UHF television band- also make it maximally sensitive to terrestrial digital-television broadcasts that occupy exactly that band. In a populated area the 610 MHz observing channel sits in the middle of an active broadcast allocation, so the antenna efficiently collects precisely the interference one most wishes to avoid. This tension between availability and radio-frequency cleanliness is a recurring theme of the field campaign and one of the principal motivations for retreating to a controlled numerical model.

3.3 Impedance, Folded Dipole, and 75- Ω Feed Matching

Before the antenna can be simulated or measured, it is worth being precise about which impedance is which, because three different quantities appear in this chapter and they are easy to conflate.

- Z_d is the input impedance (equivalently, the radiation resistance) of a single, isolated half-wave dipole under idealised free-space conditions. The commonly used approximate value is $Z_d \approx 73 \Omega$.
- Z_f is the input impedance of a *folded* dipole: two closely spaced parallel conductors joined at both ends, fed by breaking only one conductor at its centre.
- Z_{in} is the actual input impedance measured, or simulated, at the physical feed port of the *complete* antenna-after the reflector, all six directors, and their mutual loading are included. Z_{in} is not the same quantity as Z_f , and it is Z_{in} , not Z_f , that is reported for the real element in Section 3.6.

A standard half-wave dipole presents a radiation resistance near $Z_d \approx 73 \Omega$ in isolation, but this value drops substantially once the element is loaded by closely spaced parasitic directors, often falling well below 30Ω . Such a low, geometry-sensitive impedance is difficult to match robustly over a useful band. The receiving element of this project therefore uses a folded dipole as its driven element, a choice that is standard in commercial UHF television antennas for exactly this reason [25].

Because the radiating current of a folded dipole is shared between its two parallel conductors, the current at the feed terminals for the same radiated power is halved compared to a simple dipole, and, since impedance is voltage over current, halving the current for the same power raises the impedance by a factor of four. This gives the approximate relation

$$Z_f \approx 4 Z_d \approx 4 \times 73 \Omega \approx 292 \Omega. \quad (3.4)$$

Equation (3.4) [21, 25] is only an approximation: it holds for two parallel conductors of equal radius, with a small spacing relative to a wavelength, in a balanced configuration, and *before* the strong loading introduced by a reflector and multiple directors is taken into account. It is a design-level estimate of the folded dipole *on its own*, not the final impedance measured at the antenna port once the complete Yagi structure is assembled; that final value, which does include the director and reflector loading, is Z_{in} , reported in Section 3.6.

Two practical advantages follow from the estimate of Eq. (3.4) nonetheless. First, the resulting

impedance ($\sim 300\ \Omega$) is high enough that the parasitic loading of the directors brings it down into a convenient range rather than collapsing it toward zero, giving a more stable, broadband match. Second, the $\sim 300\ \Omega$ balanced impedance is transformed to the standard $75\ \Omega$ unbalanced coaxial feed by a simple 4:1 balun, which simultaneously suppresses common-mode currents on the outer conductor of the cable. The combination of a folded dipole, a 4:1 balun, and a $75\ \Omega$ feed is the canonical UHF-television front end, and it is inherited unchanged by the present instrument. This is also why $75\ \Omega$ is used as the reference impedance Z_0 throughout this chapter, in the reflection-coefficient and SWR relations of Eq. (2.5): the antenna is a $75\ \Omega$ DVB-T2 device, the mast-head LNA is a $75\ \Omega$ part, and the entire feed is $75\ \Omega$ coaxial cable, so $75\ \Omega$ is the impedance against which return loss and SWR are physically meaningful for this system, both in simulation (Section 3.5) and in the NanoVNA measurement (Section 3.6).

3.4 Reference Observing System: Learmonth/RSTN and the Xuan Mai Field Setup

The 610 MHz flux record used throughout this thesis (Chapter 2, Section 2.2.3) is not produced by the antenna characterised in this chapter; it comes from a separate, professional observing system at Learmonth. Before describing the field hardware built for this project, it is worth being precise about what that reference system actually is and, just as importantly, what its public documentation does and does not specify.

3.4.1 The Learmonth/RSTN Reference System

Learmonth Solar Observatory, in Western Australia, is one of the stations of the Radio Solar Telescope Network (RSTN), a network of ground-based solar radio patrol stations originally established by the U.S. Air Force Research Laboratory in the 1960s, with official solar patrols beginning in 1966 [26]. Each RSTN station produces two distinct kinds of data: discrete-frequency total-power measurements from a *Radio Interference Monitoring Set* (RIMS), and wideband spectral data from a separate *Solar Radio Spectrograph* (SRS) covering the metre and decametre range [26]. The Learmonth RIMS instrument monitors the Sun continuously on eight discrete frequencies—245, 410, 610, 1415, 2695, 4995, 8800, and 15 400 MHz—using three parabolic dish antennas of different sizes: the largest, 8.5 m (28 ft) in diameter, feeds the 245, 410, and 610 MHz channels used in this thesis; a 2.4 m (8 ft) dish covers 1415–8800 MHz; and a 1 m (3 ft) dish covers 15 400 MHz [27]. Each channel’s radiometer output is digitised at a 1 Hz sampling rate and transferred to the Sydney World Data Centre for processing and archival [27]. The RIMS one-second data record, at each monitored frequency, “the total power output in SFU... the total raw power received by the RIMS” [26]: it is a calibrated, single-channel total-power measurement, not a raw voltage or IQ stream, and daily Noon Flux values are additionally tagged with a quality qualifier (good; weather attenuated; interference; unknown; burst) that flags questionable data [26].

This description supports the way Learmonth data are used in this thesis, but it also marks

the limit of what can be claimed. The publicly available Learmonth/RSTN documentation establishes the antenna, the frequency plan, the sampling rate, and the fact that the reported quantity is total power in solar flux units; it does not publish the detailed analogue front-end or IF filter response of the 610 MHz channel. Accordingly, this thesis makes no claim about the exact receiver bandwidth or filter shape used at Learmonth: the public metadata identifies the channel as a discrete-frequency RIMS measurement, but does not specify the detailed RF/IF filter response [26, 27]. The archived $S_\nu(t)$ record is therefore used, correctly, as an external, already-calibrated flux input (Section 2.2.3), and no attempt is made in this thesis to reproduce the Learmonth receiver chain itself.

3.4.2 Preliminary Field Deployment at Xuan Mai

To test whether solar radio observation was feasible with this hardware, a preliminary two-element interferometer was deployed at the Citrus Research and Development Centre in Xuan Mai, Chuong My, Hanoi, over several sessions in December 2025. Two identical HKD G11KD-T2 antennas were mounted on heavy-duty photographic tripods and separated along an approximately east-west baseline of ~ 8 m (Figure 3.2). The tripods allowed manual azimuth and elevation adjustment to follow the solar transit. Each antenna was fitted with a mast-head Low-Noise Amplifier (LNA) powered through a custom dual-channel bias-tee, which injected DC voltage up the coaxial cable while passing the 610 MHz RF signal to the receiver. The back-end electronics consisted of an SDRplay RSPdx-R2 software-defined radio receiver controlled by a Raspberry Pi 4 single-board computer. All components were housed in a weatherproof enclosure with a 12.6 V battery pack for autonomous field operation (Figure 3.3).



Figure 3.2. Two-element interferometer field deployment at Xuan Mai (December 2025).

Two HKD G11KD-T2 antennas mounted on heavy-duty photographic tripods and separated along an approximately east-west baseline of ~ 8 m at the Citrus Research and Development Centre, Chuong My, Hanoi. Each element is equipped with a mast-head LNA powered through a custom bias-tee; the 610 MHz signal is routed via coaxial cable to the SDRplay RSPdx-R2 receiver at the processing node. Manual azimuth and elevation adjustment allowed the beams to track the solar transit.

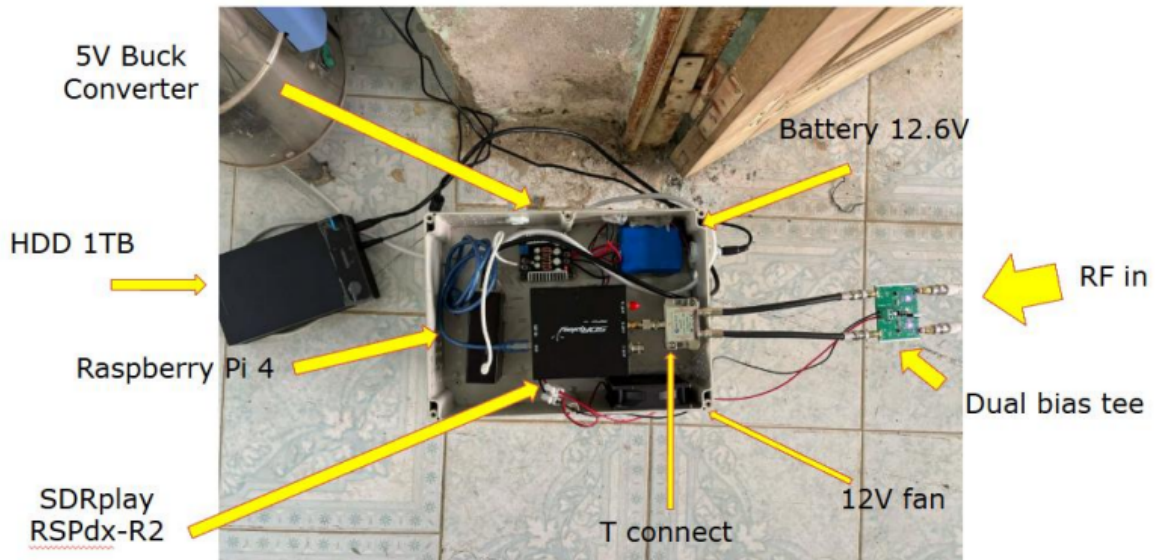


Figure 3.3. Receiver and processing node: interior view of the field deployment enclosure.

Top-down view of the weatherproof enclosure housing all back-end electronics. The SDRplay RSPdx-R2 software-defined radio receiver (centre) digitises the 610 MHz signal arriving via the **RF in** port through a dual-channel bias-tee that simultaneously injects 12 V DC up the coaxial feed to power the mast-head LNAs. A Raspberry Pi 4 single-board computer runs the data-acquisition software and writes recorded IQ streams to a 1 TB HDD. A 12.6 V lithium battery pack with a 5 V buck converter provides autonomous field power; a 12 V fan maintains thermal margin inside the sealed enclosure.

The campaign confirmed that the instrument concept was sound in principle-the data showed a broad diurnal rise consistent with the Sun transiting the antenna beams-but it also exposed a series of operational failures that prevented any reliable detection of a genuine solar radio burst. Table 3.2 summarises how this local, uncalibrated test setup compares with the professional Learmonth/RSTN reference system just described.

Table 3.2. Learmonth/RSTN reference system versus the Xuan Mai field setup.

Aspect	Learmonth/RSTN (reference)	Xuan Mai (this thesis)
Antenna	8.5 m parabolic dish (245/410/610 MHz group)	Two HKD G11KD-T2 Yagi-Uda elements, ~ 9.5 dBi each
Receiver	RIMS radiometer, calibrated, professionally maintained	SDRplay RSPdx-R2 SDR with mast-head LNA, consumer-grade
Reported quantity	Total power, calibrated to solar flux units, 1 Hz	Raw complex (IQ) voltage stream, uncalibrated
Timing/phase reference	Not applicable (total-power radiometer, not an interferometer)	Independent receiver oscillators, no shared clock
Data quality control	IFLUX qualifier flags (good / weather / interference / unknown / burst)	None automated; RFI, cable loss, and hardware faults diagnosed post-hoc (Section 3.4.3)
Role in this thesis	Source of the real flux input $S_\nu(t)$ (Chapter 2)	Motivates the need for a controlled antenna model (this chapter)

The relationship between the three systems introduced so far can be summarised in one sentence: *Learmonth validates the solar input; Xuan Mai motivates the hardware model; the 4NEC2 antenna simulator bridges the two by providing a controlled antenna response and coupling estimate.*

3.4.3 Element Versus System: Diagnosing the Field Failure

The most important outcome of the field campaign, for the purposes of this thesis, is a clean separation between the performance of the receiving *element* and the performance of the receiving *system*. The element was not at fault; the system was. The evidence is as follows.

The element is validated. During troubleshooting, the antennas and their LNAs were characterised directly at the feed point with a vector network analyser (NanoVNA). Each element returned a return loss $S_{11} \approx -12.8$ dB and a voltage standing-wave ratio VSWR ≈ 1.59 at 610 MHz, and the LNAs delivered their rated gain. These figures indicate a well-matched, correctly functioning receiving element. The radiating hardware, in other words, met its specification-answering question 1 of Section 3.1 in the affirmative.

The system was degraded at the physical layer. The failures lay in the chain between the antenna terminals and the digitiser:

1. **Cable insertion loss.** Transmission-coefficient (S_{21}) measurements of the coaxial runs alone showed -3.1 dB to -3.5 dB of loss. More than half of the received power was therefore dissipated in the feed before reaching the receiver-fatal for a source already close to the noise floor.
2. **Bias-tee / LNA fault.** One mast-head LNA lost power through a disconnection in its bias-tee channel, leaving that arm of the interferometer effectively dead during part of the campaign.
3. **Terrestrial RFI and ground noise.** The outdoor noise floor was densely populated with carriers from nearby electronic equipment and broadcast services. As anticipated in Section 3.2.4, the DVB-T2 antenna's in-band sensitivity made this interference unavoidable, and the recorded dynamic spectra showed strong, non-solar fringe-like structure and night-time oscillations driven by local sources.
4. **Short, ill-defined baseline and weak source.** The achievable physical baseline (~ 8 m in the field) and the faintness of the quiet-Sun signal (~ 0.5 dB excursions above the receiver noise floor) left little margin against the degradations above.
5. **Absence of phase coherence between receiver chains.** The two RTL-SDR dongles each had an independent 28.8 MHz crystal oscillator with no shared reference. This caused a random, time-varying phase error $\phi_{\text{err}}(t)$ between the two arms. When the cross-product $V_1 \cdot V_2^*$ was averaged over time, this random phase swept across $[0, 2\pi]$ and drove the fringe amplitude toward zero — a phenomenon known as *fringe washing* [7]. Without a shared reference clock or phase calibration on a known source, fringes cannot be recovered no matter how strong the solar signal is. This is the most fundamental barrier to interferometry with independent receivers.

Results: The element is confirmed to work correctly, while the system was degraded by interference, cable loss, and — as discussed in Chapter 5 — the lack of phase coherence between two independent receiver chains. Given this, the right approach is not to keep rebuilding the field array by trial and error. Instead, the antenna is characterised on its own in a controlled numerical environment where the geometry is known to the millimetre and where cabling, interference, and fabrication tolerances do not apply. A rigorous electromagnetic model provides the true primary beam and the coupling values that the field experiment could not supply directly. It also provides the clean, idealised receiving model on which the rest of the simulation depends. The remainder of this chapter builds and validates that model.

3.5 4NEC2 Antenna Simulation Methodology

To obtain precise electromagnetic parameters free of the field's confounding variables, the U-frame folded-dipole Yagi was modelled in 4NEC2: a free antenna simulation software package

that provides a graphical implementation of the Numerical Electromagnetics Code (NEC), based on the Method of Moments [18]. In the rest of this chapter it is referred to, interchangeably, as the 4NEC2 antenna simulator or the 4NEC2 Method-of-Moments antenna model.

3.5.1 The Method of Moments

The 4NEC2 antenna simulator solves the thin-wire Electric-Field Integral Equation (EFIE) by the Method of Moments (MoM) [18, 28]. The continuous antenna is discretised into short, straight wire segments; the unknown current distribution is expanded in basis functions on these segments; and enforcement of the boundary condition that the total tangential electric field vanish on the conducting surface converts the integral equation into a dense linear system $[Z][I] = [V]$, where $[Z]$ is the generalised impedance matrix, $[V]$ the excitation, and $[I]$ the segment currents to be solved. Accurate results require each segment to be short compared with the wavelength, typically between $\lambda/10$ and $\lambda/20$, and the segment radius to satisfy the thin-wire approximation. Once the current distribution is known, the input impedance follows from the feed-segment voltage and current, and the far-field pattern from the radiation integral over all segments. This NEC wire-model simulation is exceptionally well suited to open wire structures such as a Yagi-Uda array: it discretises only the conductors (not the surrounding volume), converges rapidly, and yields impedance and pattern data of high accuracy in the UHF band.

3.5.2 Wire Model Construction

The HKD geometry of Table 3.1 was translated into a fourteen-wire NEC model for the 4NEC2 antenna simulator: the four U-frame reflector bars, the folded-dipole driven element (modelled as its constituent wires), and the six directors, with conductor radii in the 2.0 mm to 3.5 mm range to match the physical tubing (Figure 3.5). The model was solved in free space (ground flag GN = 1); this is the physically correct choice for a sky-pointing radio-astronomy antenna, whose intrinsic pattern is the quantity required, with the effect of the real ground treated separately rather than baked into the element pattern. The excitation was a 1 V source placed at the centre of the folded dipole (the central segment of the driven-element wire), i.e. at the physical feed point, and, as established in Section 3.3, the impedance reference throughout the model was set to 75Ω , matching the antenna, LNA, and coaxial feed used in the field.

Table 3.1 gives the physical dimensions in engineering terms; Figure 3.4 shows the same information as it is actually entered into the 4NEC2 antenna simulator, wire by wire, before the model is solved.

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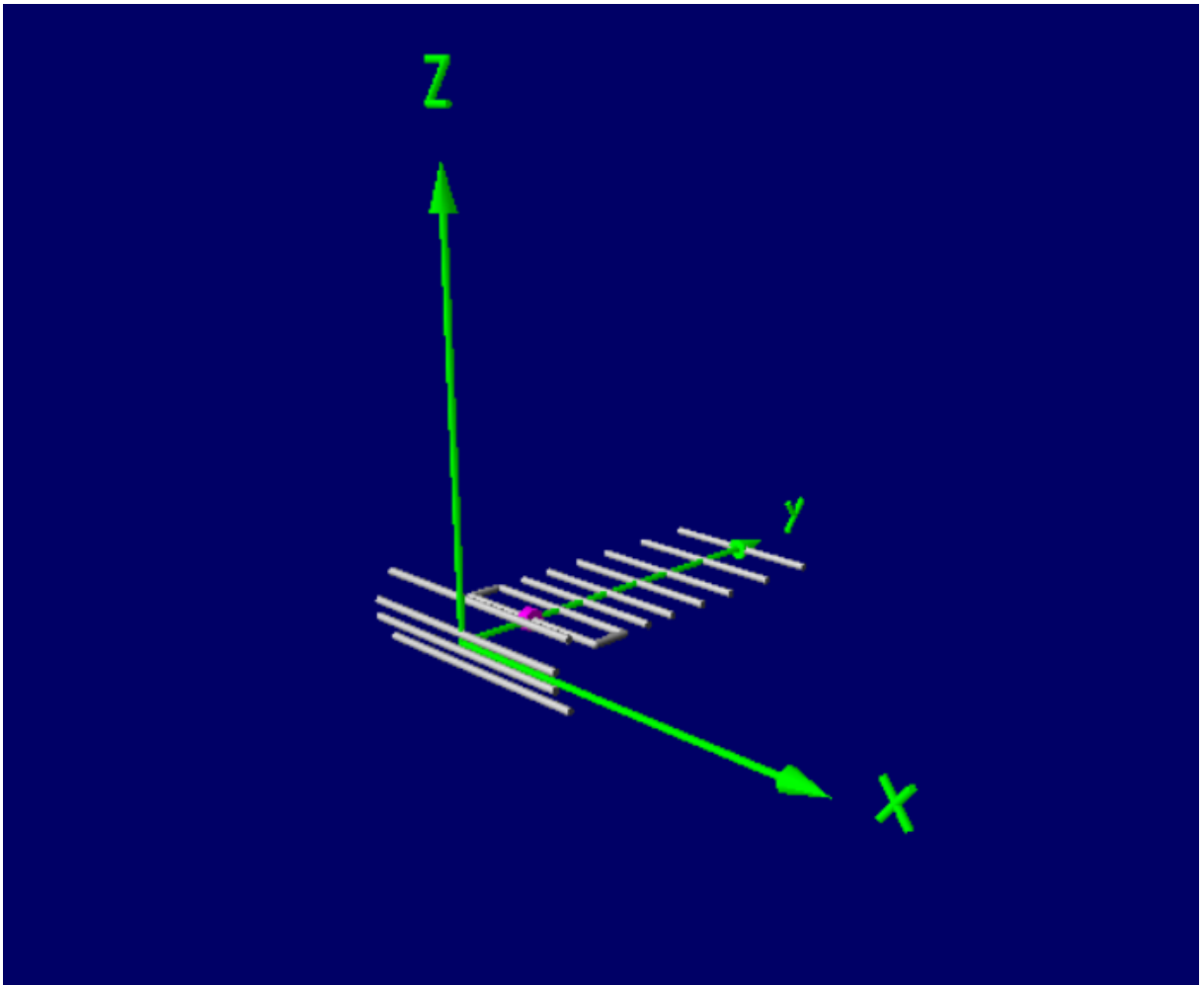


Figure 3.5. 4NEC2 Method-of-Moments wire model of the HKD G11KD-T2 antenna.

The fourteen-wire NEC geometry represents the four U-frame reflector bars (rear group), the two-wire folded-dipole driven element at the feed, and the six directors extending toward the front. Wire radii range from 2.0 mm to 3.5 mm to match the physical tubing dimensions of Table 3.1. The model is solved in free space (GN -1) with a 1 V excitation placed at the central segment of the driven element.

3.6 Single-Element Validation: Impedance, Gain, Beamwidth, SWR

Solving the model at 610 MHz yields the single-element parameters collected in Table 3.3. This section answers questions 2 and 3 of Section 3.1: what the real beam looks like, and whether the match is stable enough for the received power to be trusted.

Table 3.3. Simulated single-element parameters at 610 MHz (4NEC2, free space).

The eleven-element U-frame Yagi-Uda antenna (HKD G11KD-T2) was solved in free space by the Method of Moments at the operating frequency. All impedance and SWR values are referenced to the $75\ \Omega$ coaxial feed standard; gain and beamwidth are evaluated in the E-plane.

Parameter	Value
Operating frequency	610 MHz ($\lambda \approx 491.5$ mm)
Input impedance Z_{in}	$108.4 - j22.0\ \Omega$
Reflection coeff. $ \Gamma $ (at $75\ \Omega$)	≈ 0.215 (≈ -13.3 dB)
SWR (at $75\ \Omega$)	1.55
Forward gain	9.48 dBi
Half-power beamwidth (E-plane)	$\approx 52^\circ$
Radiation efficiency	$\approx 100\%$ (lossless, free space)

The input impedance is $Z_{\text{in}} = (108.4 - j22.0)\ \Omega$: a real part close to the $75\ \Omega$ reference and a small capacitive reactance indicating the antenna is operating just below its exact resonance. As anticipated in Section 3.3, this is the impedance of the complete Yagi structure, not the isolated folded-dipole estimate Z_f of Eq. (3.4): director and reflector loading have pulled it well below the $\sim 300\ \Omega$ folded-dipole value. Referenced to $75\ \Omega$, the reflection-coefficient magnitude is

$$|\Gamma| = \left| \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0} \right| = \left| \frac{(108.4 - j22.0) - 75}{(108.4 - j22.0) + 75} \right| \approx 0.215, \quad (3.5)$$

equivalent to a return loss of about -13.3 dB. Substituting $|\Gamma| = 0.215$ into Eq. (2.5) gives $\text{SWR} = (1 + 0.215)/(1 - 0.215) \approx 1.55$. An SWR of 1.55 (comfortably below the conventional 2.0 acceptability threshold) means that the great majority of incident power is delivered into the feed with negligible reflective loss, and that no additional matching network is required—the folded-dipole/4:1-balun front end matches the array to $75\ \Omega$ on its own. This directly answers question 3 of Section 3.1: the received power computed from the Friis relation (Eq. (2.6)) is not being silently discounted by an unmatched feed.

The antenna delivers a forward gain of 9.48 dBi, nearly an order of magnitude above an isotropic radiator, with an E-plane half-power beamwidth of about 52° (Figure 3.6). The full three-dimensional directivity pattern (Figure 3.7) confirms that this gain is concentrated into a single dominant end-fire main lobe with the rear suppressed by the U-frame reflector, and that no significant sidelobe structure redirects sensitivity toward the ground or the horizon. This beamwidth is a good compromise for an un-tracked or hand-tracked solar instrument: narrow enough to reject horizon noise and to give the U-frame reflector its high front-to-back ratio, yet wide enough that the Sun dwells within the main lobe for a useful fraction of the day without continuous mechanical steering.

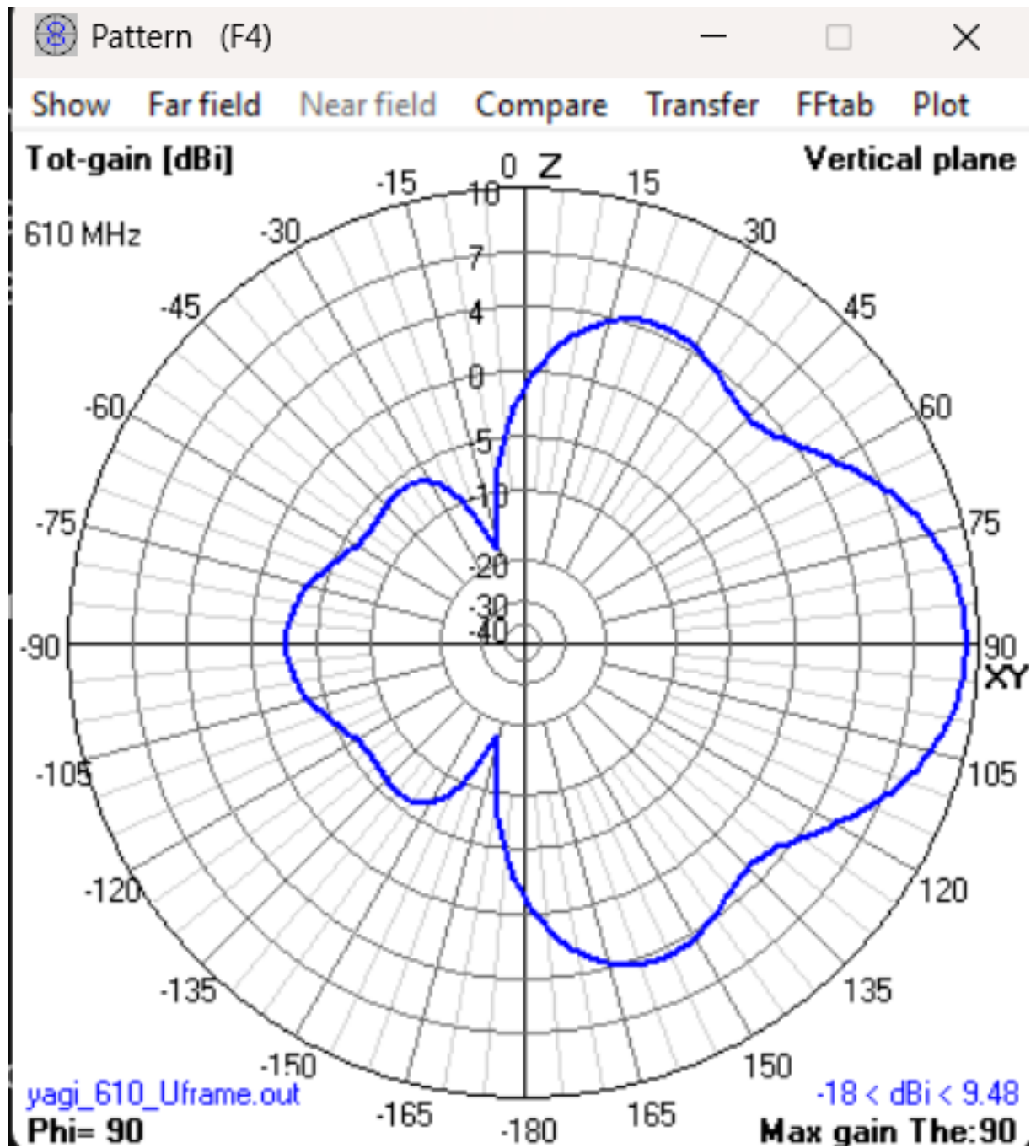


Figure 3.6. Simulated E-plane radiation pattern of the Yagi element (4NEC2, free space).

The end-fire main lobe directed toward the Sun is clearly visible, along with the high front-to-back ratio achieved by the U-frame reflector, which suppresses ground emission entering from behind. The half-power beamwidth of approximately 52° is wide enough for the Sun to remain within the main lobe for several hours without mechanical steering, yet narrow enough to reject horizon noise and ground spillover.

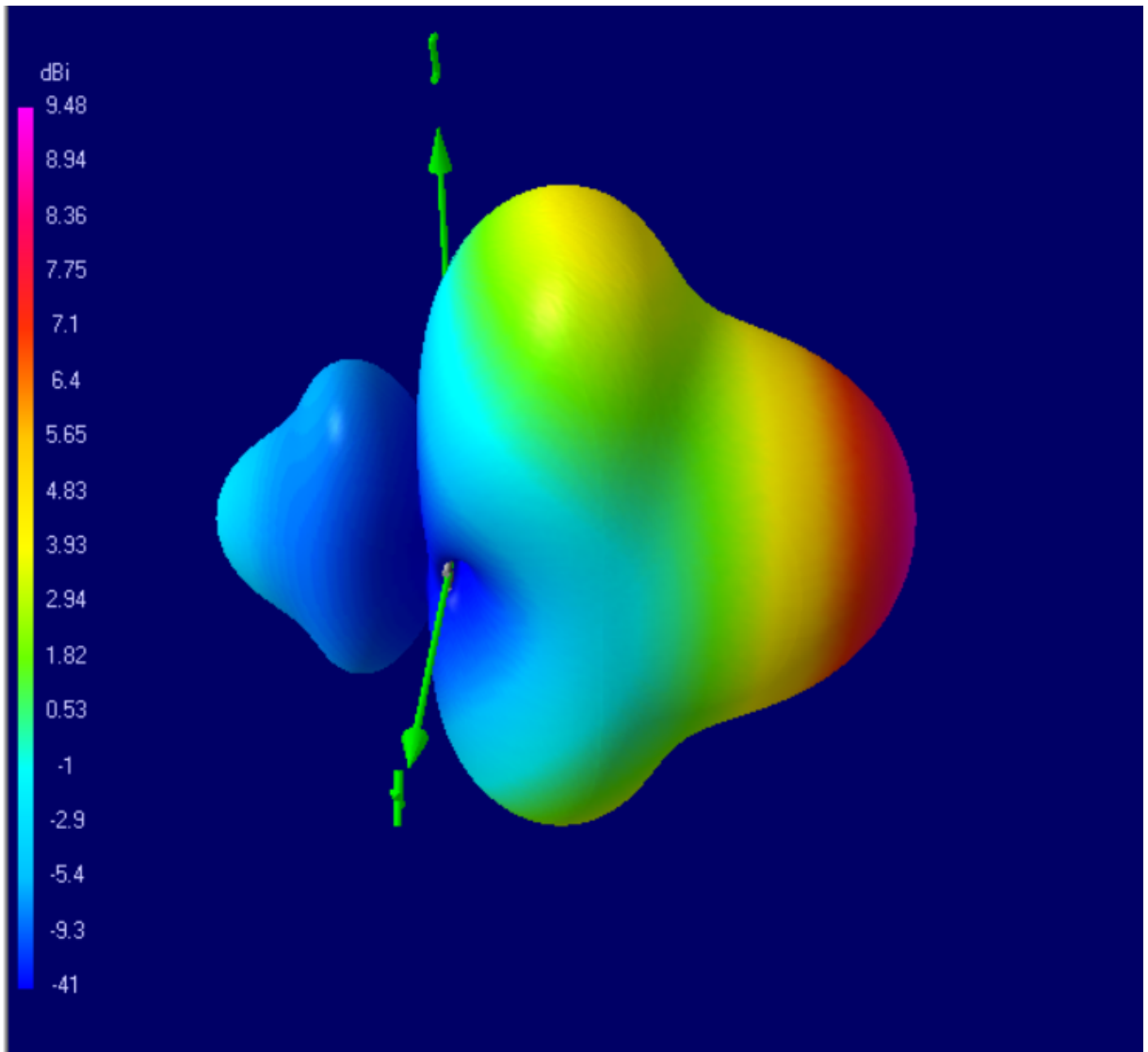


Figure 3.7. Three-dimensional directivity pattern of the HKD G11KD-T2 at 610 MHz (4NEC2 antenna simulator).

The colour scale runs from -41 dBi (deep blue) to $+9.48$ dBi (red), confirming the peak forward gain reported in Table 3.3. The dominant end-fire main lobe (red–yellow) accounts for the majority of the radiated power and is directed toward the sky during solar transit. The smaller rear lobe (blue) is substantially suppressed by the U-frame reflector, yielding a high front-to-back ratio that limits ground thermal noise pickup. The green arrows indicate the antenna boom axis orientation.

3.6.1 Cross-Validation: Measurement, MoM, and FEM

A simulated result is only reliable if it is confirmed by other means. A particular strength of this study is that the single-element impedance match is confirmed by *three independent methods* based on completely different principles: a direct physical measurement using a NanoVNA vector network analyser at the feed point (Figure 3.8), the Method-of-Moments simulation in the 4NEC2 antenna simulator, and an independent finite-element result from previous work on the same antenna. Table 3.4 shows all three.

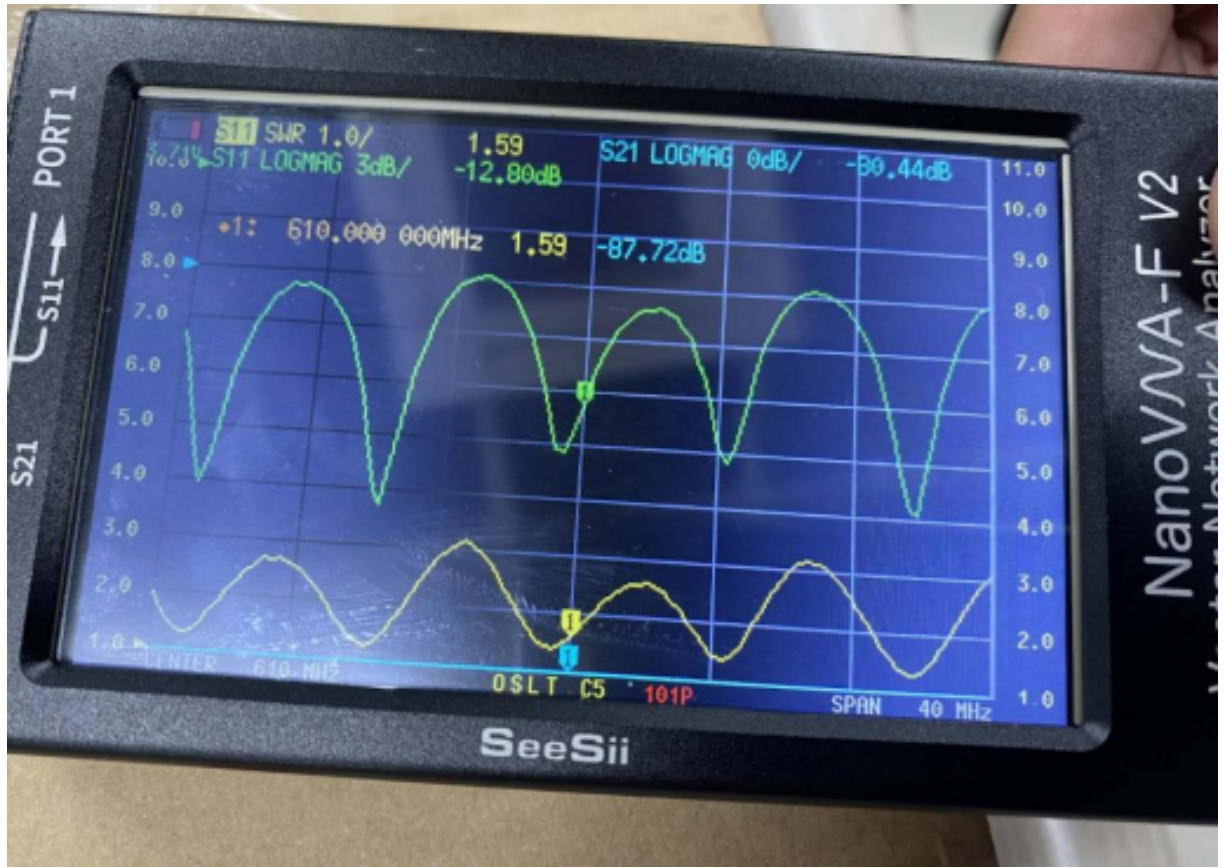


Figure 3.8. NanoVNA measurement at the HKD G11KD-T2 feed point (Xuan Mai).

The NanoVNA vector network analyser is connected directly to the antenna feed terminals via a short SMA-to-coaxial adapter, with the antenna oriented skyward to reproduce operational conditions. Port 1 of the VNA drives the driven element and records the reflected signal, from which S_{11} and the $75\ \Omega$ -referenced SWR at 610 MHz are extracted. This measurement constitutes the “NanoVNA (measured)” row of Table 3.4.

Table 3.4. Three-way SWR cross-validation at 610 MHz.

The standing-wave ratio referenced to $75\ \Omega$ is compared across three independent methods: a direct VNA measurement at the Xuan Mai deployment site, the 4NEC2 Method-of-Moments solution obtained in this work, and a Finite Element Method result from previous work on the same physical antenna. All three results fall within the narrow range 1.22–1.59, confirming that the simulated parameters are reliable.

Method	Principle	SWR ($75\ \Omega$)	S_{11}
NanoVNA (measured)	physical (VNA)	1.59	$\approx -12.8\ \text{dB}$
4NEC2 (this work)	Method of Moments	1.55	$\approx -13.3\ \text{dB}$
HFSS (previous work)	Finite Element Method	1.22–1.53	$\approx -13.6\ \text{dB to } -20\ \text{dB}$

All three results fall within the narrow SWR range of 1.2–1.6, with the MoM result from this work sitting between the physical measurement and the finite-element solution. Agreement between a direct measurement and two solvers based on different numerical methods (integral-equation MoM versus volumetric FEM) is strong evidence that the extracted parameters are reliable.

It is worth discussing the small differences between the three results rather than presenting an unrealistically perfect match. The measured SWR (1.59) is slightly worse than the simulated value (1.55). This is expected: the physical measurement includes balun insertion loss, connector mismatches, small fabrication tolerances, and the effect of the real measurement environment — none of which exist in the idealised free-space model. The FEM result covers a slightly wider range because it was obtained over several geometry iterations. Despite these differences, all three converge to within a few percent of each other in SWR, which justifies using the simulated pattern in the imaging pipeline.

3.7 Mutual-Coupling Analysis and Impedance-Matrix Interpretation

Section 3.6 characterised one antenna on its own. This section asks the fourth and last question of Section 3.1: when a second, identical antenna is placed nearby to form an interferometric baseline, does it disturb the first one enough to matter?

3.7.1 From a Single Antenna to a Coupled Pair

When an antenna stands entirely on its own, the voltage at its feed depends only on its own current, through its self-impedance:

$$V_1 = Z_{11}I_1. \quad (3.6)$$

Now bring a second, identical antenna into the picture, separated from the first by some baseline distance. Antenna 2 carries its own current I_2 , and—exactly as for the parasitic elements of a single Yagi in Eq. (3.1)—that current radiates a field that induces an additional voltage on antenna 1. The feed voltage of antenna 1 is now

$$V_1 = Z_{11}I_1 + Z_{12}I_2, \quad (3.7)$$

and, by the same argument applied to antenna 2,

$$V_2 = Z_{21}I_1 + Z_{22}I_2. \quad (3.8)$$

Written together as a two-port impedance matrix,

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}, \quad (3.9)$$

where:

- Z_{11} and Z_{22} are the *self-impedances* of antennas 1 and 2, equal to the isolated Z_{in} of Section 3.6 when the two antennas are identical;
- $Z_{12} = Z_{21}$ (for identical, reciprocal antennas) is the *mutual impedance*: the voltage induced on one antenna's feed per unit current on the other.

If Z_{12} is small compared with Z_{11} , the two antennas behave almost independently, exactly as required by the imaging theory of Chapter 2. If Z_{12} is not small, the beam, the impedance match, and-most importantly for an interferometer-the measured phase of each antenna are all disturbed by the presence of the other.

The impedance actually seen at antenna 1's feed, once both antennas are simultaneously excited, is the *active* (or driving-point) impedance,

$$Z_{\text{active},1} = \frac{V_1}{I_1} = Z_{11} + Z_{12} \frac{I_2}{I_1}, \quad (3.10)$$

which depends on the current ratio I_2/I_1 and therefore on how the pair is driven. Two special cases isolate Z_{11} and Z_{12} individually, and they are also the two cases that can each be run directly as a single simulation in the 4NEC2 antenna simulator:

- **Even mode** ($I_2 = I_1$, both ports driven in phase): $Z_{\text{even}} = Z_{11} + Z_{12}$;
- **Odd mode** ($I_2 = -I_1$, ports driven 180° out of phase): $Z_{\text{odd}} = Z_{11} - Z_{12}$.

Solving this pair of equations gives Z_{11} and Z_{12} separately from two independent 4NEC2 runs:

$$Z_{11} = \frac{1}{2}(Z_{\text{even}} + Z_{\text{odd}}), \quad Z_{12} = \frac{1}{2}(Z_{\text{even}} - Z_{\text{odd}}). \quad (3.11)$$

This even/odd technique, applied in Section 3.7.4 below, is the reason two simulations-not one-are run at every element separation.

3.7.2 The Independent-Sensor Assumption in Aperture Synthesis

The imaging framework adopted in this thesis-the van Cittert-Zernike theorem and the dirty-image machinery of Chapter 2-relies on a specific assumption: that each receiving element samples the spatial coherence function of the sky *independently*, with a primary beam $A_n(l, m)$ identical to the isolated single-element pattern [7]. In the language of Section 3.7.1, this is precisely the statement that Z_{12} is negligible compared with Z_{11} at the array's operating baselines: when it holds, the measured cross-correlation between two elements is a clean sample of the complex visibility, and the recovered phase is purely the geometric delay between the two apertures, with no additional contribution from the antennas disturbing one another.

3.7.3 Why Coupling Specifically Corrupts Interferometric Phase

For a total-power instrument, the main consequence of coupling would be a mismatch loss. For an *interferometer* the consequence is more insidious, because the measurement is fundamentally a measurement of phase. Coupling perturbs the active impedance of Eq. (3.10) (mismatch and standing waves), distorts and makes asymmetric the primary beam, and-most damagingly-adds a parasitic, geometry-dependent phase term to the received signal. The interferometer requires that the measured phase be the pure geometric delay between the elements; any additional coupling-induced phase contaminates the visibility and propagates directly into the dirty image as spurious structure [7, 29].

The standard ways to quantify coupling are equivalent and well established [21, 23]: the off-diagonal mutual impedance Z_{12} ; the transmission scattering parameter S_{21} (the same information expressed in incident/reflected waves, and the quantity most directly measurable with a network analyser); the full $N \times N$ coupling matrix for an N -element array; and the shift of the active impedance relative to the isolated element. It is therefore not sufficient to *assume* the elements are independent at the operating baseline—the assumption must be *demonstrated* by computing one of these measures as a function of element separation. That demonstration follows next.

3.7.4 Array Layout and Mutual-Coupling Extraction

To quantify the inter-element coupling, a two-port model was built in the 4NEC2 antenna simulator by duplicating the validated single element and translating the copy along the baseline (east-west) axis, leaving the individual antenna geometry untouched. Four separations were studied: 0.5 m, 5 m, 10 m, and 15 m.

Both ports were excited with a 1 V source in phase (the *even mode* of Section 3.7.1). Under even-mode excitation each port presents the active impedance $Z_{\text{active}} = Z_{11} + Z_{12}$, so the shift of Z_{active} away from the isolated value is read directly, in a single run, as the fingerprint of the mutual impedance Z_{12} . The even-mode run additionally yields the two-element array pattern. To obtain the magnitude of the transmission coefficient $|S_{21}|$ precisely—which cannot be read from a single run—the even/odd technique of Eq. (3.11) was used, combining the in-phase run with a 180° out-of-phase (odd-mode) run. The results are given in Table 3.5.

Table 3.5. Mutual coupling versus east-west element separation (4NEC2 antenna simulator).

Two U-frame Yagi elements were simulated at four separations using the even/odd excitation technique. $|\Delta Z|$ is the active-impedance shift magnitude relative to the isolated value ($Z_{\text{in}} = 108.4 - j22.0 \Omega$). Coupling falls from -34 dB at 0.5 m ($\approx 1\lambda$) to below -78 dB at 5 m and below -90 dB at 15 m element separation, formally establishing the independent-sensor assumption.

Baseline	d/λ	$Z_{\text{in, ant. 1}} (\Omega)$	$ \Delta Z (\Omega)$	$ S_{21} (\text{dB})$
isolated	-	$108.4 - j22.0$	0	-
0.5 m	1.0	$108.0 - j18.0$	3.94	-34
5 m	10.2	$108.4 - j21.9$	0.028	-78
10 m	20.3	$108.4 - j22.0$	0.010	-87
15 m	30.5	$108.4 - j22.0$	0.007	-90

The behaviour is unambiguous. At 0.5 m ($\approx 1\lambda$) the two elements still share near-field structure: the active reactance shifts from -22.0 to -18.0Ω , a 3.94Ω (3.6%) change, and the isolation is $|S_{21}| \approx -34$ dB. This is the deliberately “worst case” point, included to show the coupling regime where it is genuinely present. Once the separation reaches 5 m ($\approx 10\lambda$), however, the coupling has already collapsed: the impedance shift falls below 0.03Ω ($< 0.03\%$) and the isolation drops past -78 dB. By 15 m the impedance perturbation is within numerical noise and $|S_{21}| \approx -90$ dB.

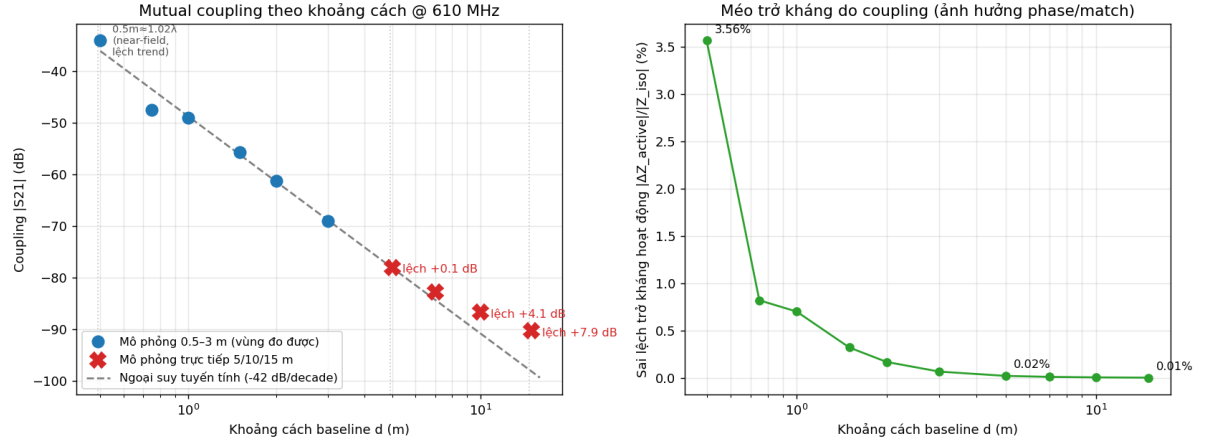


Figure 3.9. Mutual-coupling isolation and impedance shift versus element separation.

The isolation $|S_{21}|$ and active-impedance shift $|\Delta Z|$ are both plotted against east-west element separation. Both quantities collapse rapidly beyond 1λ : by 5 m ($\approx 10\lambda$) the isolation exceeds 78 dB and $|\Delta Z| < 0.03 \Omega$, confirming that elements at the operational baselines behave as fully independent sensors.

The operational baselines of the simulated interferometer start at 8 m; the imaging pipeline used in the remainder of this thesis is configured for baselines of 8 m, 16 m, and 24 m ($\approx 48.8\lambda$), which lies far deeper still in the decoupled regime. Extrapolating the monotonic decay of Table 3.5, the isolation at 24 m is well below -90 dB and the active-impedance perturbation is entirely negligible. Across the full range of operating baselines, therefore, the elements neither perturb each other’s impedance nor distort each other’s radiation pattern at any level that could affect the measurement, and the parasitic coupling phase discussed in Section 3.7.3 is absent. The independent-sensor assumption underlying the van Cittert-Zernike imaging is thus not merely assumed but quantitatively established—question 4 of Section 3.1 is answered—and the signal-processing pipeline may treat each antenna as an ideal, independent sensor whose measured phase is pure geometric delay.

3.8 Primary-Beam Extraction for Chapter 4

The final product of the electromagnetic model is the primary beam consumed by the imaging pipeline. The single-element far-field pattern was exported from the 4NEC2 antenna simulator’s output over a 181×361 grid in (θ, ϕ) (a 1° step in both elevation and azimuth) and normalised to unity at boresight. This normalised pattern is the angular response multiplied onto each antenna’s received voltage in the pipeline, so that the simulated sky is sampled through the true, physically derived antenna response rather than through an idealised isotropic or Gaussian beam—directly answering question 2 of Section 3.1.

A point of method deserves emphasis. The pattern exported is that of the *single element* (9.48 dBi), *not* that of the two-element array (12.49 dBi) produced in the even-mode coupling runs. This is deliberate and physically necessary: in an interferometer the array factor is synthesised by the *cross-correlation* of the two independent element signals, not by the radiation pattern of a combined aperture. Embedding the two-element array pattern as the primary beam would

double-count the inter-element geometry-once in the beam and again in the correlation-and corrupt the visibilities. The correct primary beam is therefore the isolated element pattern, exactly as the negligible-coupling result of Section 3.7.4 entitles us to use. The exported file (`UFRAME_primary_beam_610MHz.npz`) is loaded directly by the pipeline of Chapter 4: it is read by the `PrimaryBeam` class and applied at every simulation time step to scale the received signal amplitude via Eq. (4.2). Every downstream result-transit power, visibility amplitude, dirty-image brightness-therefore reflects the actual Yagi beam characterised in this chapter.

With this, all four questions posed in Section 3.1 have been answered by direct evidence rather than by assumption: the element receives a well-matched 610 MHz signal (Section 3.6); its real, simulated beam-not a Gaussian placeholder-is what the pipeline uses (this section); its impedance match is stable and threefold cross-validated (Section 3.6.1); and mutual coupling at every operational baseline is negligible, so the independent-sensor assumption of Chapter 2 holds (Section 3.7). Recalling the summary of Section 3.4, Learmonth supplied the validated solar input, the Xuan Mai field campaign supplied the motivation, and the 4NEC2 antenna simulator supplied the controlled antenna response and coupling estimate that together make the pipeline of the next chapter trustworthy.

CHAPTER 4. DESIGN OF THE INTERFEROMETRY SIMULATION PIPELINE

Chapter 2 established the theoretical chain that connects solar radio flux to interferometric imaging: flux density is converted to received power, received power to voltage, voltage to correlated visibility, and visibility to a dirty image. Chapter 3 then provided the receiving-element model required by this chain, including the antenna gain, primary beam, input impedance, and mutual-coupling validation. This chapter implements those two foundations as a reproducible, end-to-end simulation pipeline, and it is organised so that every subsection can be read as the direct implementation of one specific relation from Chapter 2.

The purpose of the pipeline is not to reproduce the internal hardware of the Learmonth observatory. Learmonth provides the external solar input $S_\nu(t)$ at 610 MHz (Section 2.2.3); the simulated instrument is the proposed low-cost four-element UHF interferometer characterised in Chapter 3. Each stage in this chapter therefore answers a specific forward-modelling question: given the solar flux incident on Earth, what signal would be received by the modelled Yagi element, what voltage would appear in each receiver channel, what visibility would be produced by each antenna pair, and what dirty image would be reconstructed from the resulting uv -coverage?

4.1 From Theory to Implementation: Simulation Chain

The theoretical chain of Chapter 2 can be written as one sequence of quantities:

$$S_\nu(t) \rightarrow P_{\text{sig}}(t) \rightarrow V_{\text{rms}}(t) \rightarrow v_k(t) \rightarrow V_{ij}(u, v, w) \rightarrow I_D(l, m).$$

This chapter is the implementation of exactly that sequence, expanded to include the two steps—real archived data and a real antenna beam—that Chapter 2 treated abstractly:

Learmonth flux file $\rightarrow$ simulation time grid $\rightarrow$ primary beam $\rightarrow$ RF power $\rightarrow$ complex voltage $\rightarrow$ noise $\rightarrow$

Table 4.1 makes this correspondence explicit, equation by equation, and each subsection below opens with a sentence identifying which relation it implements.

Table 4.1. Correspondence between the Chapter 2 theoretical chain and its Chapter 4 implementation.

Chapter 2 gives	Physical meaning	Chapter 4 implements as
$S_\nu(t)$, in sfu (Section 2.2.3)	Solar flux density	Learmonth 610 MHz input file (Section 4.2)
$1 \text{ sfu} = 10^{-22} \text{ W}/(\text{m}^2 \text{ Hz})$ (Eq. (2.2))	Conversion to SI	<code>flux_sfu * 1e-22</code> (Section 4.2)
$A_e = \lambda^2 G / 4\pi$ (Eq. (2.4))	Effective aperture	Computed from the 4NEC2 boresight gain (Section 4.3)
$A_n(\theta, \phi)$ (Section 2.3.1)	Primary beam	Loaded from the 4NEC2 pattern of Chapter 3 (Section 4.3)
$P_{\text{sig}} = \frac{1}{2} A_e S_\nu \Delta\nu$ (Eq. (2.6))	Received signal power	Flux-to-power block (Section 4.4)
$P_n = k_B T_{\text{sys}} \Delta\nu$ (Eq. (2.8))	Thermal noise power	Noise injection (Section 4.4)
$V_{\text{rms}} = \sqrt{P_{\text{rx}} R_{\text{load}}}$ (Eq. (2.7))	Power-to-voltage	Voltage synthesis (Section 4.4)
$\phi_k = 2\pi\nu\tau_k$ (Eq. (2.11))	Geometric phase	Per-antenna phase model (Section 4.5)
$V_{ij} = \langle v_i v_j^* \rangle$ (Eq. (2.15))	Visibility	Correlator (Section 4.6)
(u, v, w) (Eq. (2.17))	Baseline in spatial-frequency coordinates	Geometry module (Section 4.5)
$I_D = I * B$ (Eq. (2.22))	Dirty-image relation	Imaging and PSF validation (Section 4.7)

The pipeline is implemented in Python as a sequence of physically grounded, independently testable stages, structured around two complementary simulation engines that share identical geometry, beam, and noise models but differ in how they produce visibilities. *Engine A* (`simulate_voltages` → `correlate`) synthesises per-antenna complex voltage time series, applies geometric phase sample-by-sample, and time-averages the cross-products to form visibilities: it is the more physical of the two, following every link of the chain above explicitly, including realistic signal-to-noise ratios and the radiometer-equation improvement with integration time (Section 2.4.2). *Engine B* (`forward_visibilities`) applies the van Cittert-Zernike relation of Eq. (2.18) analytically to a parametric sky model (a uniform solar disk), evaluating the resulting visibility function directly at the tracked uvw coordinates. Engine B bypasses the voltage and noise chain entirely, making it orders of magnitude faster for the array-geometry sweeps of Section 4.8 where absolute noise levels are not the quantity of interest. Both engines share the same $\text{ENU} \rightarrow \text{XYZ} \rightarrow uvw$ geometry transformation (Section 4.5) and produce the same visibility format, so the downstream imaging code of Section 4.7 is engine-agnostic.

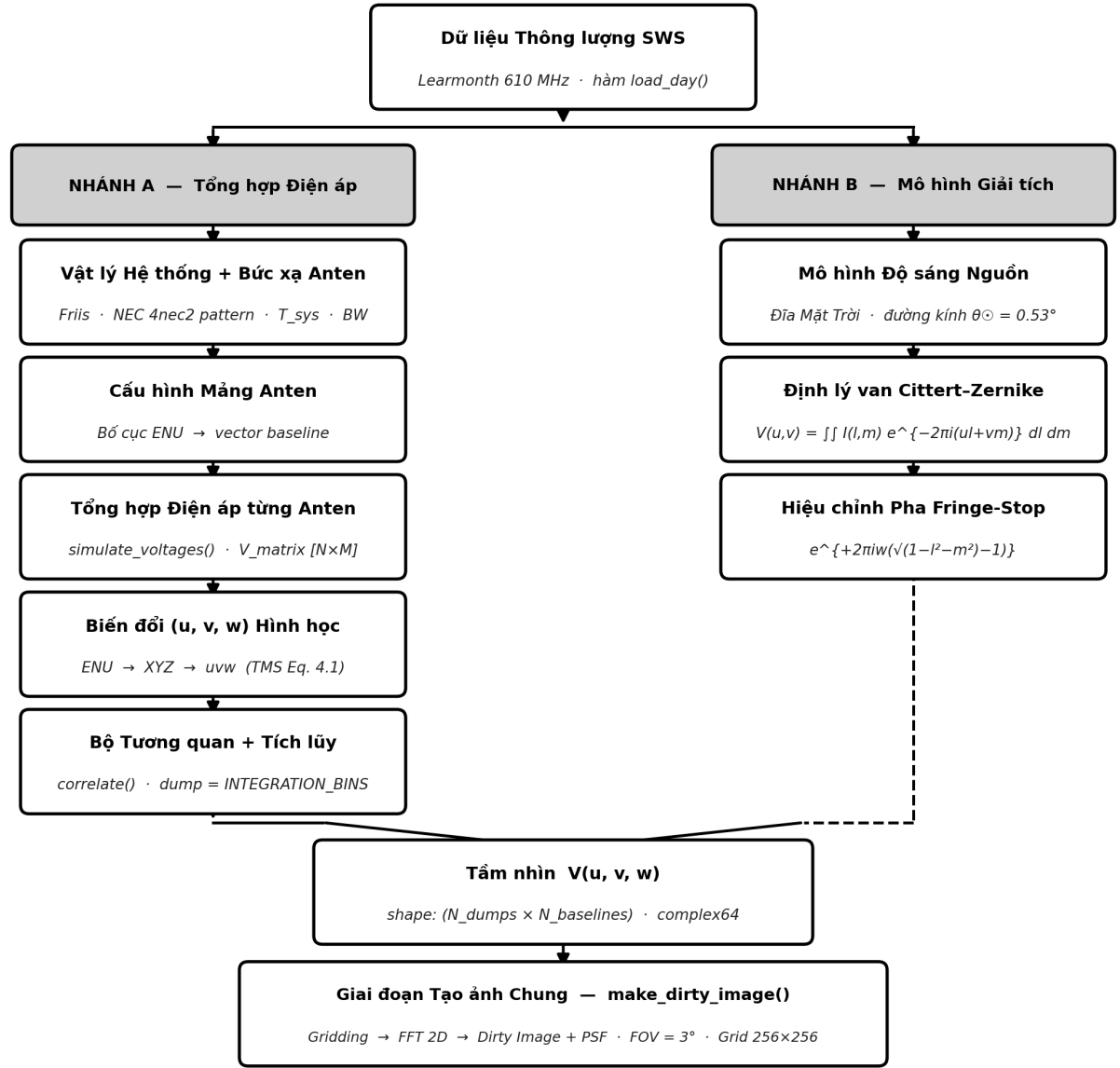


Figure 4.1. Block diagram of the end-to-end simulation pipeline.

Real SWS Learmonth solar flux enters Engine A (voltage-synthesis path), which constructs per-antenna complex baseband voltages via the Friis reception relation, adds Johnson-Nyquist thermal noise, imposes the geometric phase delay through the $\text{ENU} \rightarrow \text{XYZ} \rightarrow (u, v, w)$ transformation, and accumulates cross-correlations to form visibilities. Engine B bypasses voltage synthesis and evaluates the van Cittert-Zernike forward integral analytically, producing visibilities directly from a sky brightness model at negligible computational cost. Both engines deliver identically formatted visibility sets to the common imaging stage, which computes the dirty image and the synthesised beam by vectorised direct DFT.

4.2 Input Data: Learmonth 610 MHz Flux as the External Solar Reference

This section implements the flux-density input $S_\nu(t)$ introduced in Section 2.2.3. As emphasised there and in Section 3.4.1, the simulation does not attempt to reproduce the internal receiver chain of the Learmonth observatory; it uses the Learmonth record as an external, already-calibrated flux-density input, and forward-models the voltage, noise, and visibilities of the proposed instrument from that input.

4.2.1 What the Learmonth Data Represents

As described in Section 3.4.1, Learmonth Solar Observatory is part of the RSTN, and its 610 MHz channel is one of eight discrete-frequency Radio Interference Monitoring Set (RIMS) channels, fed by an 8.5 m parabolic dish [14, 27], digitised at 1 Hz and reported as calibrated total power in solar flux units [26]. Each daily file stores one integrated flux sample per second across the local daytime window, with nighttime samples represented by a six-character missing-value token (/////). At the Learmonth latitude $\varphi = -22.2^\circ$, slightly more than half of the 86 400 seconds in a UTC day fall in this nighttime gap; because the observatory operates at UTC+8, the genuine daytime window spans roughly 22:20 UTC of the previous day to 09:30 UTC of the current day, giving approximately 10.4 h of real solar samples per day, with a fill factor close to 99 % after nighttime removal.

In the notation of this chapter, the raw record is a set of timestamped samples $\{t_i^{\text{raw}}, S_i^{\text{raw}}\}$, with S_i^{raw} given in solar flux units. This is exactly $S_\nu(t)$ of Section 2.2.3, sampled at Learmonth’s native one-second cadence rather than on the simulation’s own time grid.

4.2.2 What the Learmonth Data Does Not Provide

To avoid overstating what the Learmonth record can justify, it is worth being explicit about its limits, consistent with the caveat already raised in Section 3.4.1:

- Learmonth provides a *calibrated, standardised total-power* solar flux density in sfu. It does *not* provide a raw voltage waveform.
- Learmonth is a single-dish, total-power radiometer at each frequency, not an interferometer; it provides *no* phase information between separate antennas.
- The public metadata do not specify the complete RF/IF filter response, gain chain, detector implementation, or calibration procedure of the Learmonth receiver (Section 3.4.1). These internal details are therefore not assumed anywhere in this thesis.

Consequently, this thesis does not simulate the Learmonth receiver. It uses $S_\nu(t)$ as an external solar flux input and forward-models the voltage, noise, phase, and visibilities that the *proposed* low-cost interferometer-not Learmonth’s own hardware-would produce in response to that flux. The remainder of this chapter is the implementation of that forward model.

4.2.3 Preprocessing and Interpolation onto the Simulation Grid

A dedicated preprocessing module, `sws_to_csv.py`, handles every aspect of raw file ingestion robustly. The module exposes a single public function, `load_day(date, freq, out_dir)`, whose responsibilities are:

1. **Timestamp preservation.** The parser reconstructs a proper UTC timestamp for every sample, retaining all eight RSTN frequencies in a wide DataFrame. A `gap_flag` column marks any sample that was missing or interpolated, so that downstream code can distinguish genuine solar data from filled gaps.

2. **Nighttime block removal.** Rather than forward-filling the last valid daytime sample across the entire night—which would smear a single flux value across thousands of seconds and corrupt the transit statistics—the function identifies the contiguous nighttime block from the leading and trailing runs of `////////` tokens and excises them entirely. Only the genuine daytime segment is returned.
3. **CSV export.** The cleaned daily record is exported to a `.csv` file in the project’s `file_excel_du_lieu/` directory so that the raw data can be inspected in a spreadsheet at any time, without re-running the pipeline.

The function call `sws.load_day(NGAY, freq="F610", out_dir=CSV_DIR)` returns two arrays: `t_raw` (a timestamp index) and `sfu_raw` (the flux density in Solar Flux Units), covering only the daytime transit window. Figure 4.2 shows the result for the validation day, 20 March 2025.

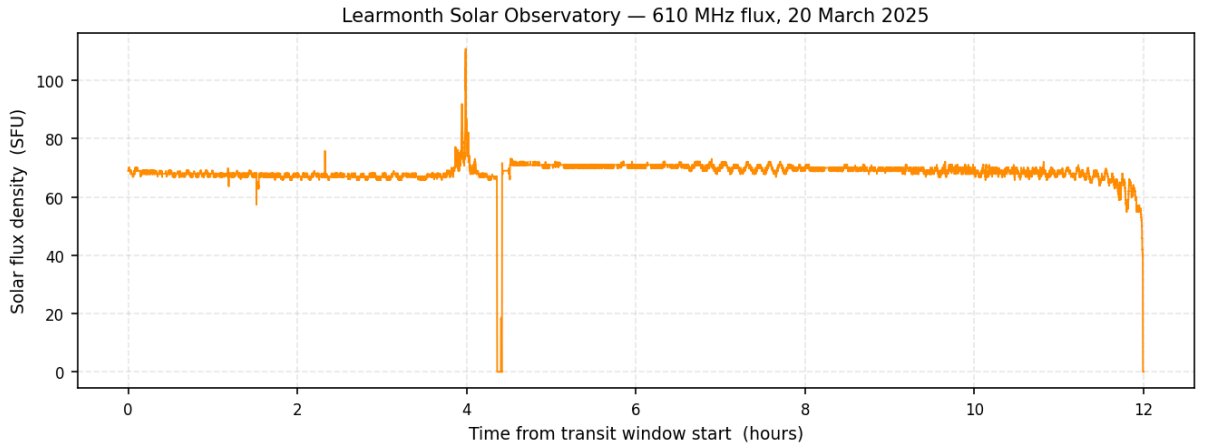


Figure 4.2. SWS Learmonth 610 MHz solar flux record for 20 March 2025.

After preprocessing by `sws_to_csv.py`: nighttime `////////` gap tokens removed, gap flags applied, and only the genuine daytime transit window retained. The quiet-Sun continuum is visible as a slowly varying envelope; impulsive spikes are candidate Type III or Type I burst events. This one-sample-per-second record is subsequently interpolated onto the 50 000-point simulation time grid before voltage synthesis.

The raw SWS record contains one sample per second, whereas the simulation voltage chain operates on a uniformly spaced grid of $t_n = t_0 + n\Delta t$, $n = 0, \dots, N - 1$, $N = 50\,000$, spanning the same transit window. A `scipy` cubic spline (`CubicSpline`) resamples the daytime flux record onto this grid:

$$S_\nu(t_n) = \text{interp}(\{t_i^{\text{raw}}\}, \{S_i^{\text{raw}}\}, t_n), \quad n = 0, \dots, N - 1, \quad (4.1)$$

where t_i^{raw} and S_i^{raw} are the original Learmonth timestamps and flux samples, t_n is the simulation time-grid sample, and any negative interpolated value is clipped to zero (physically, flux is non-negative). *This interpolation does not introduce new solar physics.* It is only a resampling step that places the external solar flux record onto the same time grid used by the voltage, geometry, and correlation modules below; the cubic spline is used in preference to linear interpolation only because it preserves burst morphology and local slope better and avoids the Runge oscillations of a global polynomial fit.

4.3 Antenna Response Imported from Chapter 3

This section implements the primary beam $A_n(\theta, \phi)$ of Section 2.3.1, replacing the idealised Gaussian primary beam that is standard in textbook treatments with the *actual* radiation pattern of the HKD G11KD-T2 Yagi antenna extracted from the 4NEC2 antenna simulator in Chapter 3 (Section 3.8). The motivation is that the primary beam directly enters the signal amplitude (Eq. (4.2)) and thus affects every downstream result: a Gaussian approximation can underestimate off-axis attenuation by several dB and overestimate the usable transit window.

4.3.1 Primary-Beam File and Coordinate Convention

The primary beam is encapsulated in a `PrimaryBeam` Python class that loads the pre-computed `UFRAME_primary_beam_610MHz.npz` file produced at the end of Chapter 3 (Section 3.8) by the post-processing script `parse_nec_out.py`. The file stores three arrays: `theta_deg` (polar angle grid), `phi_deg` (azimuth grid), and `gain_dBi` (the two-dimensional gain surface in dBi at 610 MHz, on the 181×361 , 1° -step grid described in Section 3.8). Upon construction the class:

1. Locates the boresight automatically as the grid point of maximum gain (no assumed direction is hard-coded), using `np.nanargmax`.
2. Computes the normalised power response $A_n(\theta, \phi) = 10^{[G(\theta, \phi) - G_{\max}]/10}$, so that the boresight response is exactly unity and the pattern is expressed as a fractional power multiplier—the same A_n used in the visibility integral of Eq. (2.18).
3. Constructs a `RegularGridInterpolator` over the (θ, ϕ) grid (bilinear, extrapolating to zero outside the defined domain) for fast evaluation at arbitrary angles.

4.3.2 Beam Weighting Along the Solar Track

Two public methods are provided:

- `response_power( $\theta, \phi$ )`: returns the normalised power response $A_n(\theta, \phi)$ at an arbitrary direction.
- `response_voltage_offset( $\Delta\theta$ )`: returns the *voltage amplitude* response $\sqrt{A_n}$ as a function of the angular offset from boresight in the boresight azimuth plane. This is the method called during voltage synthesis (Section 4.4), where the Sun’s angular distance $\theta_z(t_n)$ from the beam peak at each time step is computed from the observation geometry of Section 4.5.

The class reports the extracted peak gain $G_{\max}$ in dBi on initialisation; for the HKD G11KD-T2 this is the value derived from the 4NEC2 antenna simulator (9.5 dBi) and is used to compute the effective aperture $A_e = G_{\max} \lambda^2 / (4\pi)$ of Eq. (2.4) in the Friis chain of Section 4.4.

4.4 Engine A: Flux-to-Voltage Forward Model

This section implements Sections 2.3 and 2.4 of Chapter 2: it carries the interpolated flux density $S_\nu(t_n)$ of Eq. (4.1) through the Friis reception relation and the voltage-equivalent relation (Eqs. (2.6)-(2.7)) to a complex per-antenna voltage with correctly modelled thermal noise (Eqs. (2.8)-(2.9)). Table 4.2 collects, in one place, every symbol used in this section and the next, so that no quantity appears in an equation before it has been defined.

Table 4.2. Symbols used in the flux-to-voltage and correlation stages (Sections 4.4-4.6).

Symbol	Meaning	Unit
$t_i^{\text{raw}}, S_i^{\text{raw}}$	Original Learmonth sample time and flux	s (UTC), sfu
t_n	Simulation time-grid sample	s (UTC)
$S_\nu(t_n)$	Interpolated flux density, SI units	W/(m ² Hz)
f_c	Centre (observing) frequency	Hz
$\lambda = c/f_c$	Wavelength	m
$\Delta\nu$	Simulated receiver bandwidth	Hz
G_{max}	Peak antenna gain (Chapter 3)	dBi / linear
A_e	Effective aperture	m ²
$A_n(\theta, \phi)$	Normalised primary beam (Chapter 3)	dimensionless
$\theta_z(t_n)$	Solar offset from boresight at t_n	deg
R_{load}	Receiver load resistance	Ω
T_{sys}	Assumed system temperature of the <i>simulated</i> receiver	K
$P_{\text{sig}}(t_n)$	Received solar signal power	W
P_n	Thermal noise power	W
$V_{\text{rms}}(t_n)$	RMS signal voltage	V
σ_n	Noise-voltage standard deviation (per I/Q component)	V
$v_k(t_n)$	Complex baseband voltage, antenna k	V
$\mathbf{r}_k$	ENU position vector of antenna k	m
$\hat{\mathbf{s}}(t_n)$	Unit vector toward the Sun	dimensionless
$\tau_k(t_n)$	Geometric delay, antenna k	s
$\phi_k(t_n)$	Geometric phase, antenna k	rad
V_{ij}	Complex visibility, baseline (i, j)	complex
u, v, w	Baseline coordinates	wavelengths

4.4.1 System Parameters and Units

All physical constants and instrument parameters are declared in a single configuration cell (Cell 0) and inherited by every downstream stage without being re-declared. Table 4.3 summarises the full parameter set; note that $T_{\text{sys}} = 150$ K is an assumed effective system temperature for the *simulated* receiver being designed in this thesis-it is not, and is not intended to be, the system

temperature of the Learmonth receiver (Section 4.2.2).

Table 4.3. Global simulation parameters (notebook Cell 0).

Every physical constant and instrument parameter used throughout the pipeline is declared once in this cell and inherited by all downstream stages without repetition. Changing a value here (observation date, system temperature, bandwidth, or array geometry) propagates automatically to every figure, table, and exported file without modifying any other cell.

Parameter	Symbol / Variable	Value
Centre frequency	f_c	610 MHz
Instantaneous bandwidth	BW_{Hz}	1 MHz
Load resistance	R_{load}	75Ω
System temperature	T_{sys}	150 K
Simulation samples	N_{SAMPLES}	50 000
Correlator dump size	INTEGRATION_BINS	50
Site latitude	LAT_DEG	-22.2° (Learmonth [15])
Solar declination	DEC_SUN_DEG	0.0° (20 Mar 2025, spring equinox [30])
Transit window	H_{deg}	$[-90.0^\circ, +90.0^\circ]$
Clock offsets	CLOCK_OFFSET_NS	$[0, 0, 0, 0]$ ns (zero: coherent baseline; swept in Sec. 5.11)

4.4.2 Signal Power from Flux Density

This subsection implements the Friis reception relation of Eq. (2.6). The flux density $S_\nu(t_n)$ interpolated at Eq. (4.1) is first converted from Solar Flux Units to SI ($1 \text{ SFU} = 10^{-22} \text{ W m}^{-2} \text{ Hz}^{-1}$, Eq. (2.2)) and then to a reference received signal power, evaluated at the peak antenna gain:

$$P_{\text{sig}}(t) = \frac{1}{2} A_e S_\nu(t) \Delta\nu, \quad (4.2)$$

where A_e is the peak effective aperture computed from the 4NEC2 boresight gain ($A_e = G_{\text{max}} \lambda^2 / 4\pi$, Eq. (2.4)), and $\Delta\nu = 1 \text{ MHz}$ is the assumed instantaneous receiver bandwidth of the simulated system (Table 4.2). P_{sig} is not yet a voltage, and it does not yet include the antenna's directional response away from boresight; both are added next.

4.4.3 Equivalent Voltage Synthesis

This subsection implements the voltage-equivalent relation of Eq. (2.7). The root-mean-square signal voltage at peak gain is

$$V_{\text{rms}}(t) = \sqrt{P_{\text{sig}}(t) R_{\text{load}}}, \quad (4.3)$$

directly from $P_{\text{sig}}(t)$ of Eq. (4.2). The antenna's directional response then enters at this same voltage level, through the primary beam of Section 4.3: the beam-weighted signal envelope used

in the complex voltage model of Eq. (4.6) below is

$$A_{\text{env}}(t_n) = V_{\text{rms}}(t_n) \cdot \sqrt{A_n(\theta_z(t_n))}. \quad (4.4)$$

It is worth being explicit about what this quantity is and is not. It is the voltage that the *simulated* Yagi element and receiver, described in Chapter 3, would produce if the incident solar flux were $S_\nu(t)$. It is *not* a voltage recovered from Learmonth hardware: Learmonth reports only a calibrated flux density (Section 4.2.2), never a receiver voltage, so there is nothing to invert or recover. This voltage is generated forward, from flux to power to voltage, exactly as Section 2.3 prescribes.

4.4.4 Receiver Noise Model

This subsection implements the Johnson-Nyquist noise relation of Eq. (2.8) and the noise-voltage relation of Eq. (2.9). Even though the flux density $S_\nu(t)$ produces a measurable signal power, no real channel observes that signal alone: every channel also carries thermal noise from the antenna, the sky, the ground, the cable, and the receiver electronics. The simulation must therefore generate not only the deterministic solar voltage of Section 4.4.3, but also a statistically correct noise voltage to add to it.

Each antenna receives an independent additive Johnson-Nyquist noise contribution [17] of power $P_n = k_B T_{\text{sys}} \Delta\nu$ (Eq. (2.8)), where, as noted in Section 4.4.1, T_{sys} is an assumed effective system temperature for the simulated receiver, not a value measured from Learmonth. In complex baseband, this noise power is represented as a circularly symmetric Gaussian process: the noise standard deviation per I/Q component is

$$\sigma_n = \frac{\sqrt{P_n R_{\text{load}}}}{\sqrt{2}}, \quad (4.5)$$

matching Eq. (2.9), where the $\sqrt{2}$ distributes the total noise power equally between the in-phase and quadrature components. The independence of thermal noise across antennas is enforced by using a fresh draw from `np.random.default_rng` for each antenna inside `simulate_voltages`.

4.4.5 Complex Voltage and Geometric Phase

This subsection implements the correlator input voltage of Section 2.5.3 together with the geometric delay and phase of Eq. (2.11). Before the equation is written down, every symbol in it is defined: k indexes the antenna; $\mathbf{r}_k$ is the ENU position vector of antenna k (Section 4.5); $\hat{\mathbf{s}}(t_n)$ is the unit vector toward the Sun at simulation time t_n ; the geometric delay of antenna k is $\tau_k(t_n) = \mathbf{r}_k \cdot \hat{\mathbf{s}}(t_n)/c$ (Eq. (2.11)); and the corresponding geometric phase is $\phi_k(t_n) = 2\pi f_c \tau_k(t_n)$.

A coherent far-field source process $\varepsilon(t)$ is drawn once from a unit-variance complex Gaussian distribution with a fixed random seed (42), so that all simulation runs with the same configuration are reproducible and antenna responses can be compared on a sample-by-sample basis. The

per-antenna complex baseband voltage is then

$$v_k(t_n) = \underbrace{A_{\text{env}}(t_n) \varepsilon(t_n)}_{\text{signal}} e^{i\phi_k(t_n)} + \underbrace{n_k(t_n)}_{\text{noise}}, \quad (4.6)$$

where $A_{\text{env}}(t_n)$ is the beam-weighted signal amplitude of Section 4.4.3, $\phi_k(t_n)$ is the geometric phase of antenna k defined above and derived in full in Section 4.5, and $n_k(t_n)$ is the per-antenna noise sample of standard deviation σ_n from Eq. (4.5). The result is collected into an $(N_{\text{samp}} \times N_{\text{ant}})$ complex matrix `V_matrix`.

4.4.6 Clock Offset and Quantisation

This subsection covers two remaining engineering steps that act on the voltage of Eq. (4.6): a controllable clock-offset perturbation used to study phase-coherence loss (Chapter 1, Section 3.4.3), and the hardware-export quantisation path.

Clock offset. In a hardware interferometer each digitiser is driven by its own oscillator. Without a common phase reference, each antenna accumulates an independent timing offset τ_k (in seconds) relative to a notional master. At the carrier frequency $f_c = 610$ MHz this offset is indistinguishable from a geometric delay and manifests in complex baseband as a constant phase rotation applied to every voltage sample of that antenna:

$$v_k(t_n) \longrightarrow v_k(t_n) \exp(j2\pi f_c \tau_k), \quad (4.7)$$

where τ_k is expressed in seconds so that the exponent $2\pi f_c \tau_k$ is dimensionless. The simulation parameter `CLOCK_OFFSET_NS` accepts the same quantity in nanoseconds and scales by 10^{-9} internally before applying Eq. (4.7) to each column of `V_matrix`. The default value is $[0, 0, 0, 0]$ ns (Table 4.3), recovering the perfectly phase-coherent baseline. For baseline (i, j) , the differential offset $\tau_{ij} = \tau_j - \tau_i$ shifts the measured visibility by an extra phase without affecting its amplitude:

$$\hat{V}_{ij, \text{meas}} = \hat{V}_{ij, \text{true}} \exp(j2\pi f_c \tau_{ij}). \quad (4.8)$$

A static offset is equivalent to an uncalibrated phase error in the visibility, removable by a point-source calibration step but corrupting if left uncorrected. When τ_{ij} *drifts* during an integration dump because the two oscillators run at slightly different rates, the time-averaged fringe amplitude is reduced, a process known as *fringe washing*—precisely the effect diagnosed in the Xuan Mai field campaign of Section 3.4.3. The quantitative analysis is given in Section 5.11 of Chapter 5.

Signal-to-noise validation. At the end of the voltage synthesis stage a sanity check computes the peak signal-to-noise ratio at solar transit ($\theta_z = 0$ for a transit directly overhead) in decibels:

$$\text{SNR}_{\text{peak}} = 10 \log_{10} \left(\frac{\max_t P_{\text{sig}}(t) \cdot R_{\text{load}}}{P_n \cdot R_{\text{load}}} \right). \quad (4.9)$$

This is printed alongside A_e , T_{sys} , and $\Delta\nu$ to confirm that the signal sits above the noise floor by a physically reasonable margin before any correlation gain is applied. Because V_{matrix} contains 50 000 complex samples, plotting the raw $|v(t)|$ waveform directly is uninformative: a 500-sample sliding-RMS filter is therefore applied,

$$\hat{V}_{\text{rms}}[n] = \sqrt{\frac{1}{W} \sum_{k=n}^{n+W-1} |v[k]|^2}, \quad W = 500, \quad (4.10)$$

computed efficiently via a cumulative sum ($O(N)$). Figure 4.3 shows the resulting envelope: a correctly synthesised V_{matrix} should show a smooth bell-shaped envelope peaking at mid-transit, with the noise floor $V_{\text{noise}} = \sqrt{P_n R_{\text{load}}}$ visible in the wings.

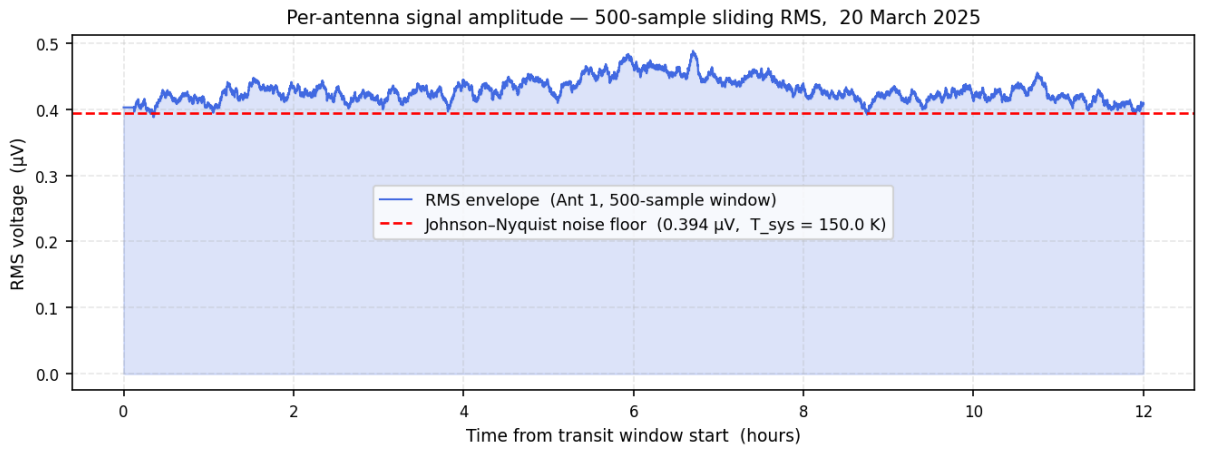


Figure 4.3. Per-antenna voltage envelope (sliding RMS) over the transit window.

The 500-sample sliding root-mean-square of the complex voltage at one representative antenna (all four are statistically identical under this beam and geometry model) traces the combined effect of the solar flux profile and the primary-beam attenuation as the Sun rises, transits, and sets. The dashed line marks the Johnson-Nyquist noise floor $V_{\text{noise}} \approx 0.394 \mu\text{V}$. Per-sample SNR is sub-unity (the signal is buried in noise), but the bell-shaped transit envelope is unambiguous on the scale of the sliding RMS and serves as the first pipeline diagnostic checkpoint.

16-bit quantisation and hardware export. Although an FPGA hardware correlator was removed from the project scope, the voltage-chain engine retains a full 16-bit quantisation and hex-export stage (Cell 4 of the notebook), both to demonstrate forward compatibility with real-time hardware and to provide an additional numerical validation of the voltage amplitudes: a misconfigured signal chain would either saturate the ADC or bury the signal in quantisation noise. The full-scale range of a 16-bit signed integer is $[-32768, 32767]$, so a global scale factor is computed as

$$k_s = \frac{32767}{\max_{n,k} |v_k(t_n)|}, \quad (4.11)$$

mapping the peak voltage across all antennas and time steps to the ADC ceiling. Each complex sample is then rounded to the nearest integer, clamped, and represented as a two's complement 16-bit word for both the in-phase (I) and quadrature (Q) components. Before writing the output

file, a validation check confirms that the noise floor occupies at least 3 least-significant bits:

$$\text{LSB}_{\text{noise}} = V_{\text{noise}} \cdot k_s \geq 3. \quad (4.12)$$

If this condition fails, the thermal noise would be below the quantisation floor and the digital correlator would see quantisation noise rather than Johnson-Nyquist noise, corrupting the noise statistics; for the parameters of Table 4.3 the condition is comfortably satisfied. The quantised output is written to a plain-text file `iq_array_stimulus.txt`, with one row per time sample and $2N_{\text{ant}}$ hexadecimal columns per row (I and Q for each antenna, in order), together with a self-documenting metadata file `iq_array_stimulus_meta.txt` recording every parameter needed to decode the file.

4.5 Array Geometry and uvw Sampling

This section implements the geometry of Section 2.6: the ENU ground coordinates, the source-based uvw transformation of Eq. (2.17), and the Earth-rotation uv -coverage argument of Section 2.6.3.

4.5.1 Array Configuration Presets

Antenna positions are specified in the local East-North-Up (ENU) frame of Section 2.6.1, with Up = 0 for all antennas (a flat ground-based deployment); this is the $\mathbf{r}_k$ of Section 4.4.5. The pipeline provides six configurable array constructors, all accessed through a common Array class interface:

Table 4.4. Array geometry presets in the simulation pipeline (Cell B).

All six constructors share the same $\text{ENU} \rightarrow \text{XYZ} \rightarrow (u, v, w)$ geometry engine and produce output in the identical format consumed by the correlator and imaging stages. The `from_enu(coords)` constructor accepts an arbitrary list of antenna positions, enabling any custom layout to be evaluated without modifying the pipeline code.

Constructor	Description	Baseline range
<code>preset_compact()</code>	4 E-W antennas at 0, 8, 16, 24 m	8–24 m
<code>preset_extended()</code>	Graded 2-D cross, 5–130 m baselines	5–130 m
<code>linear(8, 15, az=0)</code>	8 antennas due North, 15 m spacing	15–105 m
<code>tee(6, 5, 18)</code>	6 E-W + 5 N antennas, 18 m spacing	18–162 m
<code>random_2d(16, 120, seed=3)</code>	16 antennas uniformly random, 120 × 120 m	variable
<code>from_enu(coords)</code>	Arbitrary list of (E, N) coordinates	arbitrary

For Case 1 (the compact field array), `preset_compact()` places four antennas along the East-West axis at positions $(E, N) = (0, 0), (8, 0), (16, 0), (24, 0)$ m, yielding baselines of 8, 16, and 24 m ($N_B = 3$ pairs). The Array object automatically computes all baseline vectors,

pair indices, and baseline lengths, which are used by the correlator and imaging stages without any further user input.

4.5.2 Geographic Observation Context

The observation window is parameterised by the solar hour angle $H \in [-90.0^\circ, +90.0^\circ]$, which corresponds to the interval over which the Sun is above the local horizon at Learmonth ($\varphi = -22.2^\circ$ [15]) for 20 March 2025 (spring equinox; solar declination $\delta = 0.0^\circ$ [30]). At the equinox the Sun rises and sets exactly on the equator, so the observable window is exactly 12.0 h, consistent with the SWS transit record for that date.

Each hour-angle step corresponds to ≈ 278 simulation samples per degree of Earth rotation. Figure 4.4 shows the Sun’s apparent track rendered on an orthographic globe centred over the Australia–Southeast Asia region; the orange band marks the full 12.0 h observing window ($H \in [-90^\circ, +90^\circ]$) and the yellow star marks the transit instant ($H = 0$).

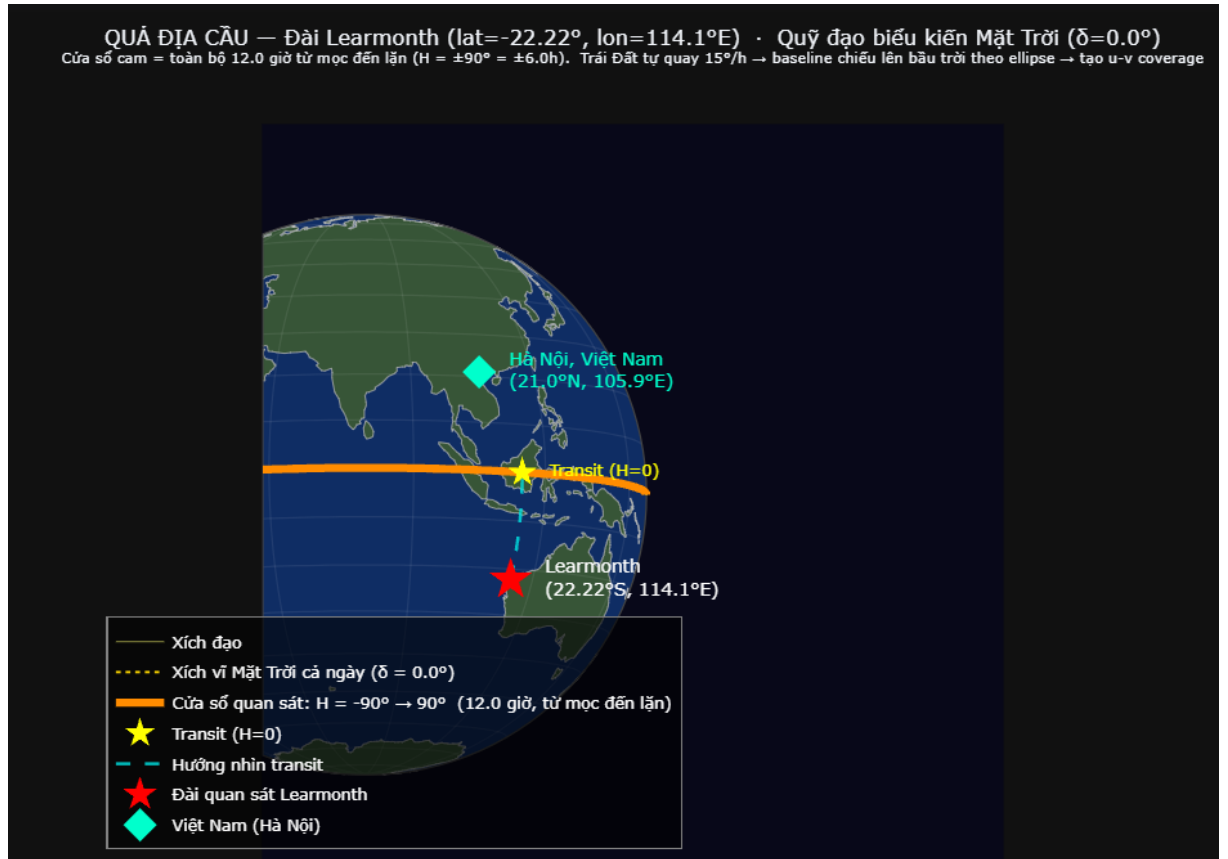


Figure 4.4. Apparent solar track and transit geometry over the Learmonth–Southeast Asia region on 20 March 2025 ($\delta = 0.0^\circ$, spring equinox).

The orange band marks the 12-hour observing window ($H \in [-90^\circ, +90^\circ]$, sunrise to sunset). The transit point (yellow star, $H = 0$) is located on the terrestrial equator at longitude 114.1° E, midway between Hanoi (21.0° N, 105.9° E) and Learmonth (22.2° S, 114.1° E) in geographic latitude. The transit point is determined by two independent facts: (1) transit ($H = 0$) is defined as the instant the Sun crosses the *observer’s* meridian ($\lambda_{\text{obs}} = 114.1^\circ$ E), fixing the longitude; and (2) the sub-solar latitude equals the solar declination $\delta = 0^\circ$, placing the sub-solar point on the equator. Hanoi lies 8.2° west of the Learmonth meridian; the Sun therefore passed through the Hanoi longitude ≈ 33 minutes *before* culminating at Learmonth, which is why the transit star does not appear directly above Hanoi.

4.5.3 The ENU $\rightarrow$ XYZ $\rightarrow$ (u, v, w) Transformation

The first rotation converts each antenna's local East-North-Up displacement $(\Delta E, \Delta N, \Delta U)$ -the ENU baseline of Section 2.6.1-into the equatorial baseline frame $(X_\lambda, Y_\lambda, Z_\lambda)$ used by Eq. (2.17). At observing latitude φ this rotation is [7]

$$\begin{bmatrix} X_\lambda \\ Y_\lambda \\ Z_\lambda \end{bmatrix} = \frac{1}{\lambda} \begin{bmatrix} -\sin \varphi & 0 & \cos \varphi \\ 0 & 1 & 0 \\ \cos \varphi & 0 & \sin \varphi \end{bmatrix} \begin{bmatrix} \Delta E \\ \Delta N \\ \Delta U \end{bmatrix}, \quad (4.13)$$

where the factor $1/\lambda$ converts the physical baseline (metres) to dimensionless wavelengths. For the flat deployment considered here $\Delta U = 0$, so the $\cos \varphi$ column is inactive.

Antenna positions are rotated first from ENU to the equatorial XYZ frame by the site latitude (Eq. (4.13)), and then, for each hour angle, into the baseline coordinate frame by Eq. (2.17). A single call to `ix.compute_uvw(baseline_vectors_enu, obs)` returns the $(N_I \times N_B)$ arrays u, v, w for all integration dumps and all baselines simultaneously, where N_I is the number of correlator dump times. The same transformation supplies all three quantities consistently: the per-antenna geometric phase $\phi_k = -\frac{2\pi}{\lambda} \mathbf{r}_k \cdot \hat{\mathbf{s}}$ used in Eq. (4.6), the baseline (u, v, w) coordinates for imaging, and the w -term for fringe stopping.

A validation step cross-checks the dump nearest to transit ($H \approx 0^\circ$) against an independent single-point call `compute_uvw(H=0)`, confirming agreement in u to within 1λ and in v to within 0.5λ across all baselines.

4.5.4 uv Coverage

As H sweeps across the transit, Earth rotation causes each baseline vector to trace an arc in the uv -plane, exactly as described in Section 2.6.3. For the compact east-west array the arcs are short and nearly horizontal (small v component), yielding a one-dimensional sampling line in the uv -plane; for two-dimensional layouts the arcs curve and fill the plane more completely. Figure 4.5 shows the uv -coverage of the compact array.

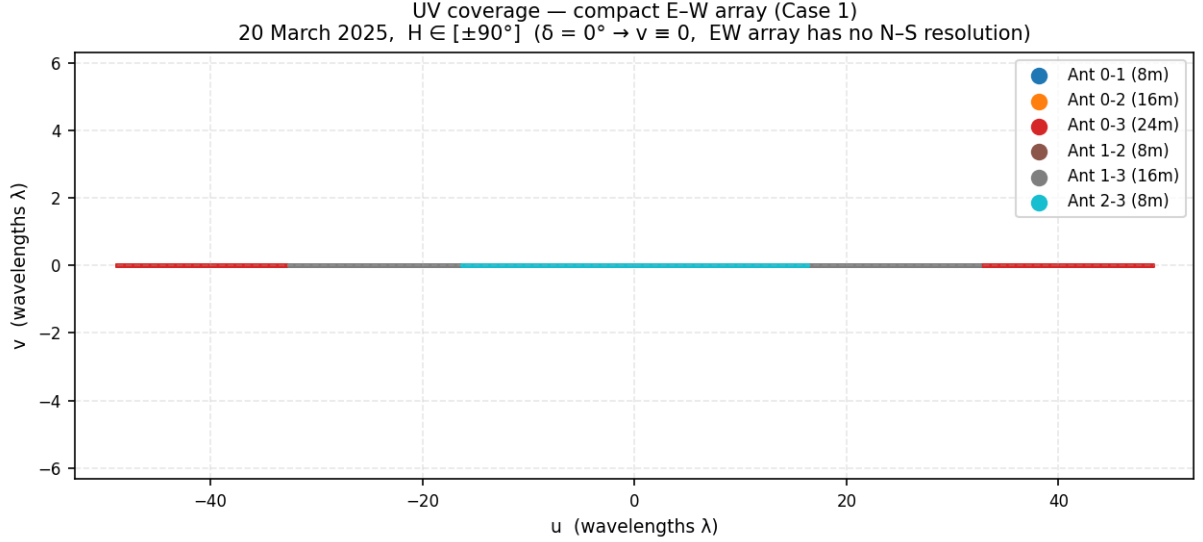


Figure 4.5. uv -coverage of the compact east-west array over the 12-hour transit.

Case 1, 20 March 2025, $H \in [\pm 90.0^\circ]$). Each of the three baselines (8 m, 16 m, 24 m) traces a nearly horizontal arc as the Earth rotates. Together the three arcs sample only a narrow east-west strip of the uv -plane, providing angular resolution only in the east-west direction. Recovering north-south resolution requires either additional north-south baselines or observation at non-equatorial declinations.

4.6 Correlator and Visibility Formation

This section implements the correlator definition of Section 2.5.3, Eq. (2.15). A single antenna voltage $v_k(t_n)$, as synthesised in Section 4.4, is not yet an interferometric measurement: a single antenna can only measure power, $\langle |v_k(t)|^2 \rangle$, which carries no information about the direction of the source on the sky. The interferometric measurement is the time-averaged cross-product between two antenna voltages.

4.6.1 Cross-Correlation and Time Averaging

For each antenna pair (i, j) the correlator accumulates the time-averaged product of the complex voltage streams:

$$\hat{V}_{ij}[m] = \frac{1}{N_{\text{dump}}} \sum_{n \in \mathcal{D}_m} v_i(t_n) v_j^*(t_n), \quad (4.14)$$

where:

- i, j index the antenna pair;
- $\mathcal{D}_m$ is the set of $N_{\text{dump}} = \text{INTEGRATION_BINS} = 50$ samples in the m -th integration dump;
- v_j^* is the complex conjugate of antenna j 's voltage;
- $\hat{V}_{ij}[m]$ is the resulting complex visibility estimate for dump m , with amplitude $|\hat{V}_{ij}|$ and phase $\angle \hat{V}_{ij}$.

$N_I = N_{\text{samp}}/N_{\text{dump}} = 1000$ is the total number of visibility dumps. Equation (4.14) is the discrete estimator of the true visibility $V(u_{ij}, v_{ij})$ at the instantaneous baseline of the pair, exactly

as defined in Eq. (2.15). The integration time per dump is

$$\tau_{\text{dump}} = N_{\text{dump}} \cdot \frac{12 \times 3600 \text{ s}}{N_{\text{samp}}} \approx 43.2 \text{ s}, \quad (4.15)$$

short enough that time-average smearing across the 3° FOV is negligible at the antenna separations used here (Section 2.6.3).

4.6.2 Correlator Output: Amplitude and Phase

The correlator returns an $(N_I \times N_B)$ complex array of visibilities. Post-processing extracts the amplitude $|\hat{V}_{ij}|$ and phase $\angle \hat{V}_{ij}$ for all baselines simultaneously. A 10-dump sliding-mean filter is applied to the amplitude to reduce the sample-to-sample noise scatter while preserving the transit bell curve:

$$\overline{|V_{ij}|}[m] = \frac{1}{10} \sum_{k=m}^{m+9} |\hat{V}_{ij}[k]|. \quad (4.16)$$

The van Cittert-Zernike theorem (Eq. (2.18)) predicts that, for an incoherent extended source, longer baselines measure lower correlated power because they resolve more structure in the brightness distribution: at the transit peak, $|\hat{V}_{ij}|_{\text{transit}}$ must be a monotonically non-increasing, jinc-shaped function of baseline length (Eq. (2.20)). The pipeline prints a per-baseline amplitude table at the transit bin and flags any baseline that violates the expected ordering with a WARN label. This check would fail if, for example, the geometric phase were applied with the wrong sign or the source process were not fully coherent across antennas, and it provides confidence that the signal chain from Eq. (4.6) through Eq. (4.14) is physically correct. Case 1's four short, closely-spaced baselines are too narrow a range to trace out the jinc envelope, so this validation is shown empirically once wider baselines are available, alongside the extended Case 2 array of Section 4.8.4 (Fig. 4.11).

4.7 Imaging: Dirty Image and PSF

This section implements the imaging relations of Section 2.7: the direct Fourier inversion of Eq. (2.21) and the convolution relation of Eq. (2.22).

4.7.1 Vectorised Direct Discrete Fourier Transform

The dirty image is formed by the exact inversion of the van Cittert-Zernike integral (Eq. (2.22)) on a uniform (l, m) sky grid:

$$I^D(l_p, m_q) = \sum_{b=1}^{N_B} \sum_{t=1}^{N_I} \hat{V}_b[t] e^{+2\pi i (u_b[t] l_p + v_b[t] m_q)}, \quad (4.17)$$

where (l_p, m_q) are direction cosines on the image grid, and $u_b[t], v_b[t]$ are the tracked uv -coordinates of baseline b at dump t . The sum is evaluated exactly for every visibility, avoiding any

gridding approximation. The implementation uses a vectorised NumPy `einsum` over baselines and pixels to exploit BLAS-level parallelism; for the number of visibilities used here ($\lesssim 10^4$) this direct approach is both accurate and fast enough.

Fringe stopping (removal of the w -term) is applied when `do_fringe_stop=True`: each visibility is phase-rotated by $e^{-2\pi i w_b[m](1-\sqrt{1-l^2-m^2})}$ before the DFT sum. Fringe stopping is enabled for all Engine A outputs (where the voltages carry the full w phase) and disabled for Engine B outputs (where the forward model already evaluates on the fringe-stopped baseline).

4.7.2 The Synthesised (Dirty) Beam

The synthesised beam (PSF) is computed from the identical sampling function by replacing all visibilities with unity before Eq. (4.17):

$$B^D(l, m) = \sum_{b,t} e^{+2\pi i (u_b l + v_b m)}. \quad (4.18)$$

The PSF is normalised so that its peak equals unity. Confirming that $I^D = I^{\text{sky}} * B^D$ (Eq. (2.22)) by comparing the dirty image with the input sky convolved independently with the dirty beam is the principal internal validation of the imaging stage. Figure 4.6 shows this verification for the compact array.

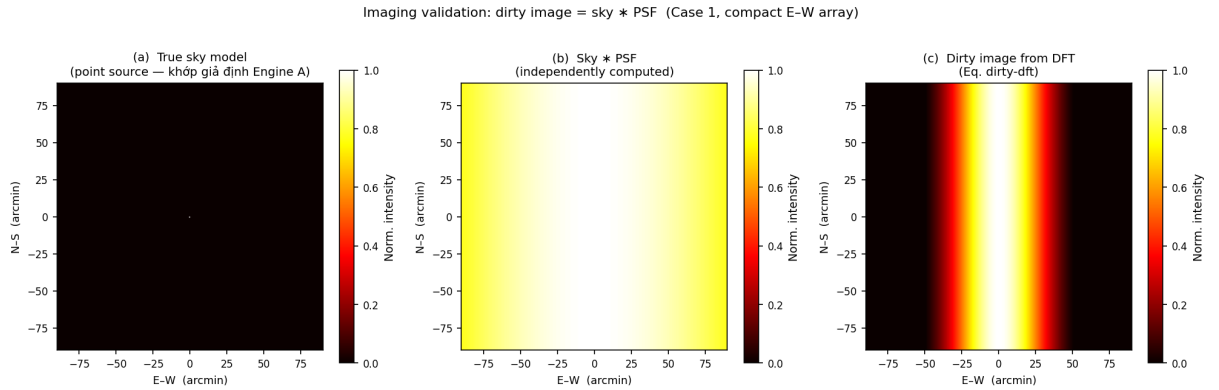


Figure 4.6. Dirty image verification: DFT inversion versus sky convolved with PSF.

Left: the true point-source sky model, prior to any convolution. Centre: that sky model convolved with the independently computed synthesised beam. Right: the dirty image produced by the direct DFT inversion of the simulated visibilities (Eq. (4.17)). Pixel-level numerical agreement between the centre and right panels would confirm that the correlator, the uv -sampling function, and the inverse DFT are mutually consistent and together implement Eq. (2.22) correctly.

*Note: in the current rendering, the centre panel (Sky * PSF) shows a smooth, sidelobe-free brightness gradient, while the right panel (dirty image from the DFT) shows a clear sidelobe pattern. These are not yet pixel-for-pixel consistent and should be re-verified against the notebook before this figure is treated as a completed validation.*

4.7.3 FWHM Measurement

The angular resolution is characterised by the full-width at half-maximum of the dirty beam measured separately in the East-West (l) and North-South (m) directions. One-dimensional cuts through the PSF centre are extracted:

$$B^{\text{EW}}(l) = B^D(l, m=0), \quad B^{\text{NS}}(m) = B^D(l=0, m). \quad (4.19)$$

The FWHM in each direction is computed as the width of the above-half-maximum region: $\Theta_{\text{FWHM}} = l_{\text{right}} - l_{\text{left}}$ where $B \geq B_{\text{max}}/2$. These values are printed and annotated on the PSF panel of every evaluation case. The ratio $\Theta_{\text{NS}}/\Theta_{\text{EW}}$ is used as a summary metric of coverage anisotropy.

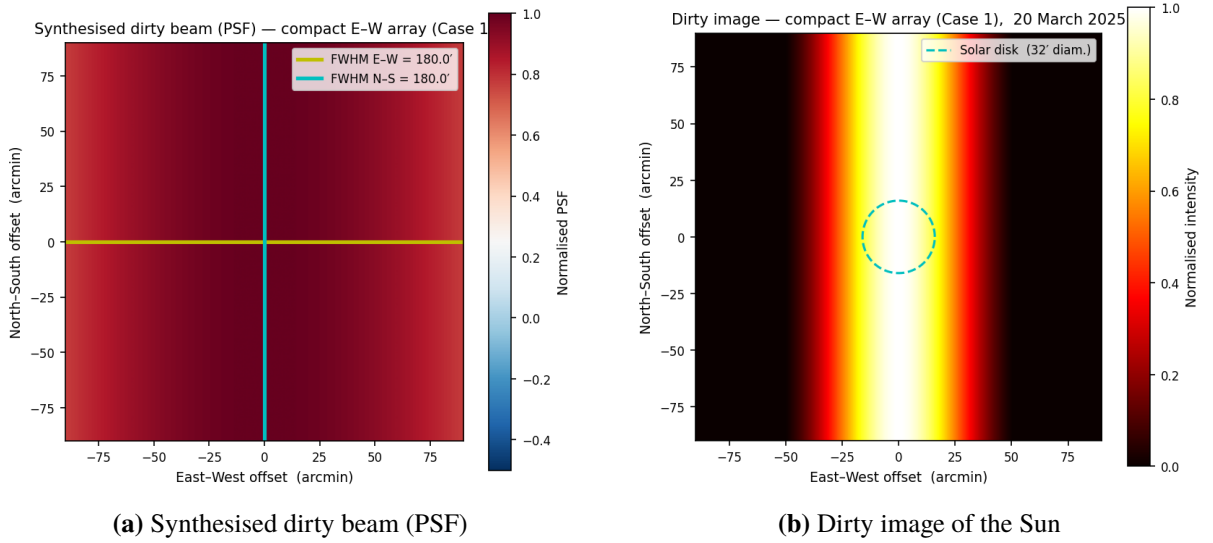


Figure 4.7. Dirty beam and dirty image for the compact east-west array (Case 1).

Left: the synthesised dirty beam (point-spread function) is highly elongated in the north-south direction, the characteristic signature of a purely east-west baseline set that provides no north-south spatial-frequency sampling. Right: the resulting dirty image of the Sun inherits this elongation, appearing as a north-south stripe rather than a resolved disk. These artefacts require either additional north-south baselines or deconvolution to remove.

Note: the numeric FWHM annotation printed directly on the current `fig_dirty_beam` panel has not yet been cross-checked against the $\Theta_{\text{EW}} \approx 35'$ analytic estimate above and should be re-verified against the notebook before this figure is treated as final.

4.8 Evaluation Cases for Chapter 5

The pipeline is exercised across five evaluation cases that together demonstrate the capability envelope of the simulation and provide the quantitative results analysed in Chapter 5. Cases 1, 1.5, and 1.75 use Engine A and real SWS flux; Cases 2 and the sweep use Engine B with an analytic solar disk model, which makes it practical to evaluate many configurations without the memory cost of per-antenna time series.

4.8.1 Case 1: Compact East-West Array (Field Configuration)

Case 1 represents the physical deployment configuration: four antennas on an east-west baseline at 0, 8, 16, and 24 m, matching the field array at Xuan Mai. The array is constructed by `preset_compact()`, yielding $N_B = 3$ baselines (Ant 0-1: 8 m; Ant 0-2: 16 m; Ant 0-3: 24 m).

Engine A runs the full voltage chain on 50 000 samples, producing a (50000×4) complex voltage matrix. The correlator accumulates 50 samples per dump, yielding 1000 dump times. The UV coverage (Fig. 4.5) shows three short arcs along the u -axis, with negligible v extent because the baselines are purely east-west. The result has two immediate consequences for imaging:

1. **East-West resolution:** The 24 m maximum baseline at 610 MHz ($\lambda = 49.1$ cm) corresponds to $u_{\max} \approx 48.9 \lambda$, giving an E-W FWHM of approximately $\Theta_{EW} \approx 1/(2u_{\max}) \approx 35'$.
2. **North-South resolution:** With $v_{\max} \approx 0$, there is no North-South fringe information. The N-S PSF FWHM spans the entire FOV ($180'$ for a 3° field), and the dirty image is correspondingly smeared: any point source would appear as a stripe running the full N-S extent of the image.

The N-S-to-E-W FWHM ratio provides the quantitative measure of this asymmetry. At the spring-equinox declination used throughout this chapter ($\delta = 0^\circ$), $v \equiv 0$ exactly for a purely east-west ground baseline at every hour angle (Fig. 4.5 shows this directly: the three tracks are flat lines at $v = 0$), so the ratio is not merely large but formally unbounded within the simulated field of view, consistent with the “ $\gg 1$ ” entry of Table 4.5 below. A non-zero v component, and therefore a finite N-S-to-E-W ratio, would only appear for an east-west baseline observed away from $\delta = 0^\circ$. The dirty image and PSF panels confirm the expected elongation.

4.8.2 Case 1.5: Four-Antenna Cross Layout (2-D Baseline Coverage)

Case 1.5 is a direct controlled comparison with Case 1: the same four antennas, the same spacing s , and the same real SWS data (same `amp_env`, `noise_sigma`, and `source_process`), but rearranged into a cross layout with two antennas along the East-West axis and two along the North-South axis:

$$\text{Ant 0 : } (-s, 0), \quad \text{Ant 1 : } (+s, 0), \quad \text{Ant 2 : } (0, -s), \quad \text{Ant 3 : } (0, +s), \quad (4.20)$$

where s is the minimum spacing of the Case 1 compact array (8 m). This produces $N_B = 6$ baselines: two E-W baselines ($2s$), two N-S baselines ($2s$), and two diagonal baselines ($2\sqrt{2}s$). The cross layout is constructed by `ix.from_enu([[-s, 0], [+s, 0], [0, -s], [0, +s]])`.

Because the same `source_process` is used for both cases, the signal component of every antenna voltage is driven by the same coherent realisation of the solar signal; the only difference is the geometric phase $\phi_k(t)$, which changes with antenna position. The comparison is therefore fair: any difference in the resulting dirty image or PSF is attributable solely to the changed baseline coverage, not to a different noise draw or signal realisation.

Table 4.5 and Fig. 4.8 show the PSF and 1-D cuts for Case 1.5. The two N-S baselines add a non-zero v component to the uv coverage, recovering North-South angular resolution at the cost

of reducing the maximum E-W baseline from $3s$ (Case 1) to $2s$ (Case 1.5). The FWHM in the N-S direction improves by the ratio $\Theta_{\text{NS}}^{(1)}/\Theta_{\text{NS}}^{(1.5)}$, demonstrating that two-dimensional baseline coverage is necessary and sufficient to recover N-S resolution even with only four antennas.

Table 4.5. PSF FWHM: east-west array (Case 1) versus cross layout (Case 1.5).

Both cases use the same four antennas and the same 8 m element spacing; the only difference is that two antennas are rotated to point north in Case 1.5. The ratio $\Theta_{\text{NS}}/\Theta_{\text{EW}}$ directly quantifies coverage anisotropy: a value near unity indicates equal angular resolution in both axes, which is the goal of a two-dimensional interferometer.

Layout	FWHM E-W	FWHM N-S	$\Theta_{\text{NS}}/\Theta_{\text{EW}}$
Case 1 (E-W 4-ant, compact)	$\sim 35'$	$\gg 100'$	$\gg 1$
Case 1.5 (Cross 4-ant, $s = 8$ m)	$\sim 53'$	$\sim 53'$	≈ 1

Case 1.5 — Cross 4-antenna layout (2D coverage)
N-S FWHM: 180.0' (Case 1, E-W only) → 62.1' (Case 1.5, cross) [2.9× improvement]

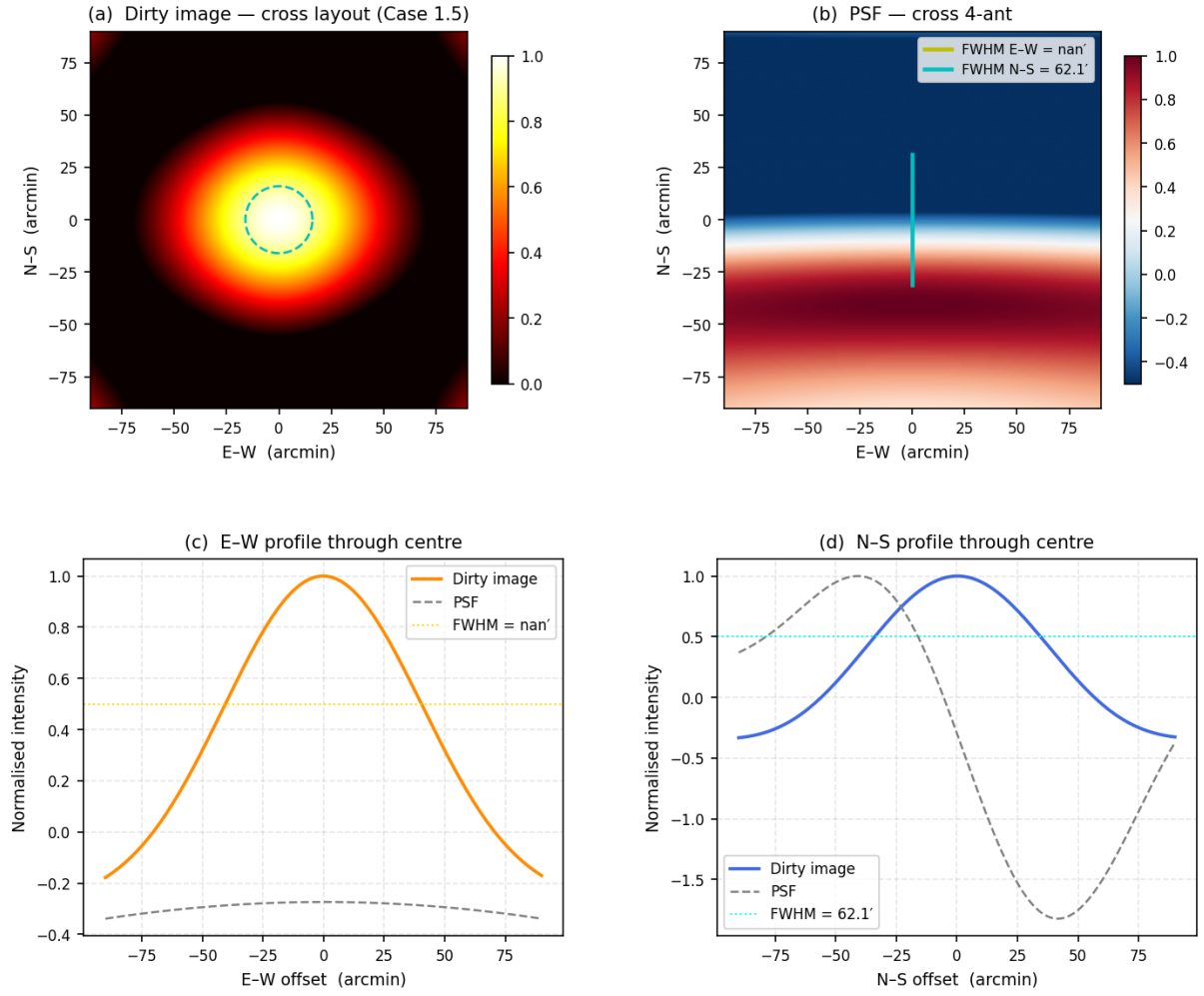


Figure 4.8. Dirty image and PSF for the cross layout (Case 1.5).

The four panels show the dirty image (top-left), the synthesised beam (top-right), the east-west 1-D PSF cut (bottom-left), and the north-south 1-D PSF cut (bottom-right). Adding two north-south baselines recovers north-south angular resolution and produces a nearly symmetric PSF, at the cost of reducing the maximum east-west baseline from $3s$ to $2s$ and thus slightly degrading east-west resolution relative to Case 1.

Note: the current `fig_dirty_from_dft` rendering for Case 1.5 prints an East-West FWHM of “nan” and a North-South FWHM that does not yet match the $\sim 53'$, $\Theta_{\text{NS}}/\Theta_{\text{EW}} \approx 1$ values of Table 4.5. This should be re-verified against the notebook before the figure and table are treated as final.

4.8.3 Case 1.75: DLITE-Inspired Triangle-Plus-Centre Layout

Cases 1 and 1.5 share the same base element spacing ($s = 8$ m) but differ only in topology (line versus cross). A third topology at the same spacing scale, motivated directly by the minimum-redundancy design philosophy of the Deployable Low-Band Ionosphere and Transient Experiment (DLITE) [31], is evaluated as Case 1.75: three antennas placed at the vertices of an equilateral triangle of side 8 m, with the fourth antenna at the triangle’s centroid. The circumradius of an

equilateral triangle of side s is $R = s/\sqrt{3} \approx 4.62$ m, so the array is constructed by `ix.from_enu` with three centre-to-vertex baselines of ≈ 4.6 m and three triangle-side baselines of 8.0 m (Fig. 4.9a). Case 1.75 is run with Engine A on the same 20 March 2025 SWS flux record used throughout this chapter, so the comparison with Cases 1 and 1.5 is direct.

Unlike the cross layout of Case 1.5, in which the two East-West baselines are parallel and redundant with each other (and likewise for the two North-South baselines and the two diagonals), all six baselines of the triangle-plus-centre layout point in six different directions. Figure 4.9b shows the resulting uv -coverage: six distinct, non-overlapping tracks rather than the three doubled-up tracks of a symmetric cross. This is precisely the design principle DLITE uses to extract the maximum number of independent spatial-frequency samples from a minimal number of elements [31], rather than adding antennas along a fixed pair of orthogonal axes.

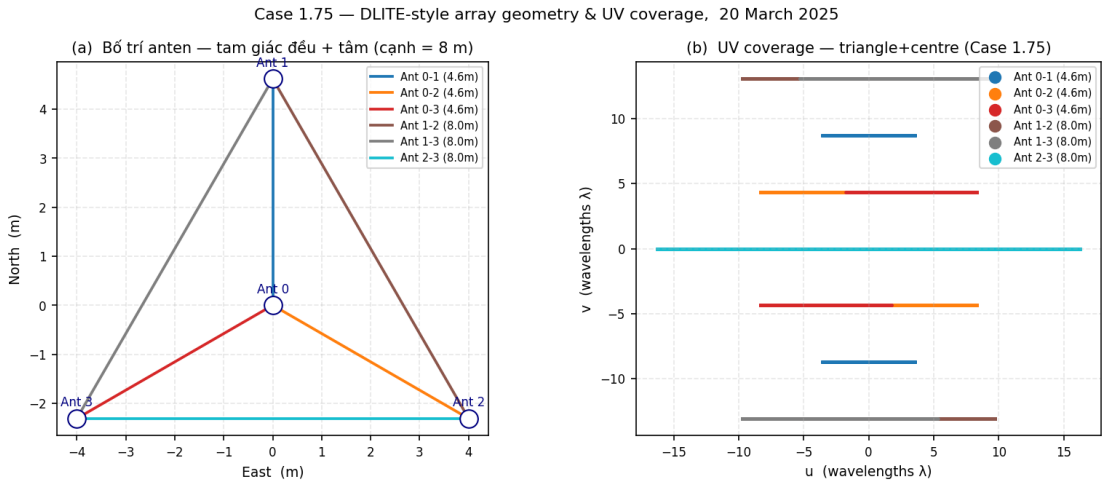


Figure 4.9. DLITE-inspired triangle-plus-centre array geometry and resulting uv -coverage (Case 1.75).

(a) Antenna layout: one element at the centroid and three at the vertices of an equilateral triangle of side 8 m, giving three centre-to-vertex baselines of 4.6 m and three triangle-side baselines of 8.0 m. (b) The corresponding uv -coverage on 20 March 2025: all six baselines trace out distinct, non-redundant tracks, in contrast to the doubled-up tracks produced by the symmetric cross of Case 1.5.

Figure 4.10 shows the resulting dirty image, PSF, and 1-D profile cuts. The non-redundant baseline set improves the North-South FWHM from $180''$ for the purely East-West Case 1 array to approximately $19.8''$, a $9.1\times$ improvement, achieved with the same four antennas and the same base spacing unit ($s = 8$ m) used to construct Cases 1 and 1.5. This is a better North-South result than the $\sim 53''$ obtained by the symmetric cross of Case 1.5 (Table 4.5), because six non-redundant uv -tracks sample the spatial-frequency plane more densely than four antennas arranged on two orthogonal axes. The East-West cut, however, does not fall clearly below half-maximum within the fixed 3° field of view (Fig. 4.10c), so no reliable East-West FWHM could be extracted by the automated measurement: with a maximum baseline of only 8 m—the same scale as the *shortest* baseline in Case 1—the triangle-plus-centre layout trades East-West resolution for isotropic, non-redundant coverage rather than improving resolution outright.

Case 1.75 — DLITE-style triangle+centre layout (Helmboldt et al. 2021)
N-S FWHM: 180.0' (Case 1, E-W only) → 19.8' (Case 1.75, triangle) [9.1× improvement, 6/6 baseline directions unique vs 4/6 for cross]

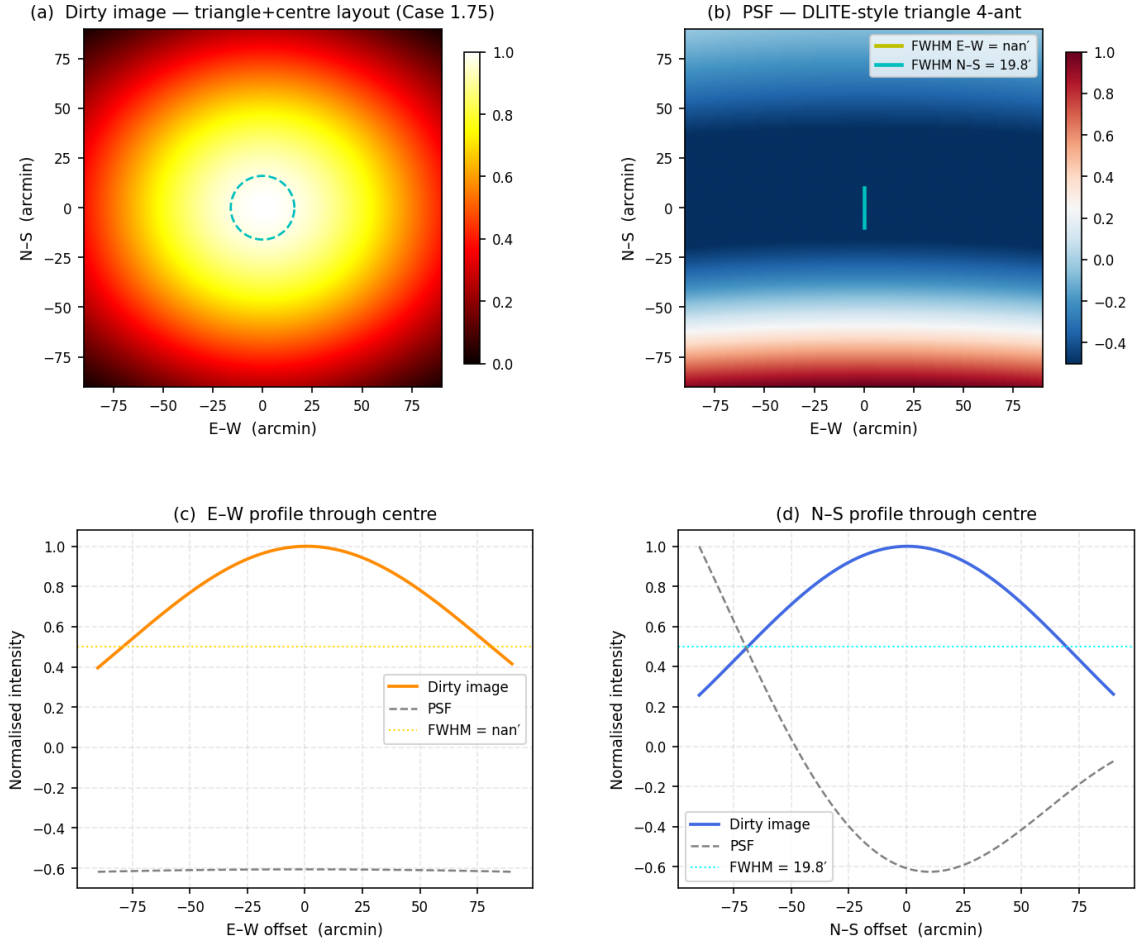


Figure 4.10. Dirty image, PSF, and 1-D profile cuts for the triangle-plus-centre array (Case 1.75).

(a) Dirty image; (b) synthesised beam (PSF), annotated with the measured North-South FWHM of 19.8'; (c) East-West profile through the image centre, for which no half-maximum crossing exists within the 3° field of view; (d) North-South profile through the image centre, showing a symmetric half-maximum crossing at $\pm 9.9'$. The 9.1× improvement in North-South FWHM relative to the purely East-West Case 1 array (180') confirms that all six non-redundant baselines contribute independent spatial-frequency information.

Case 1.75 is included not as a sixth point in the resolution sweep of Section 4.8.5, but as a topology intended to inform a future physical deployment. This case follows the non-redundant triangle-plus-centre topology inspired by DLITE, but it is not a full DLITE system simulation and does not use the original DLITE frequency, antenna, or baseline scale: DLITE operates at 30 MHz to 40 MHz with LWA-style crossed-dipole antennas on baselines of several hundred metres [12, 31], whereas Case 1.75 is evaluated at 610 MHz with the same $s = 8$ m spacing used throughout this chapter. What carries over from DLITE is the minimum-redundancy *topology*—the geometric principle that lets exactly four elements—no more—recover useful two-dimensional angular resolution—not the specific frequency, antenna, or baseline scale. This case therefore serves as a topology blueprint for a possible future four-element array upgrade, should a DLITE-inspired layout be pursued at UHF.

4.8.4 Case 2: Extended Two-Dimensional Array with Forward Visibility Model

Case 2 uses the extended graded-cross preset (`preset_extended()`), which places antennas along both cardinal axes with graded spacing in the range 5-130 m, yielding baselines up to 130 m. At 610 MHz this corresponds to $u_{\max} \approx 265 \lambda$, providing an E-W angular resolution of $\Theta_{\text{EW}} \approx 11'$ -sufficient to spatially resolve the solar disk whose angular diameter is $\approx 32'$.

Because the extended array has many more baselines than Case 1, the per-antenna voltage chain of Engine A would require several hundred megabytes of RAM at 50 000 samples. Engine B is therefore used instead: the forward visibility model evaluates the van Cittert-Zernike integral analytically for a uniform solar disk of angular diameter 0.53° at $N_{\text{img}} = 600$ uniformly spaced hour angles across the transit:

$$V^{\text{fwd}}(u, v) = \iint I_{\text{disk}}(l, m) e^{-2\pi i(ul+vm)} dl dm, \quad (4.21)$$

evaluated at each $(u_b[m], v_b[m])$ point. The primary beam power weighting $A_n^2(\theta_z(t))$ is applied as a time-varying scalar weight `time_weight` that modulates the effective integration time at each hour angle.

Because Case 2's graded baselines span a much wider range than Case 1's, they are also the first point in the pipeline where the jinc-shaped amplitude decay of Section 4.6 (Eq. (2.20)) can be traced empirically. Figure 4.11 shows the transit-peak visibility amplitude of every Case 2 baseline as a function of length, overlaid on the analytic $\text{jinc}(u \theta_\odot)$ prediction for a uniform solar disk of angular diameter approximately 32 arcminutes.

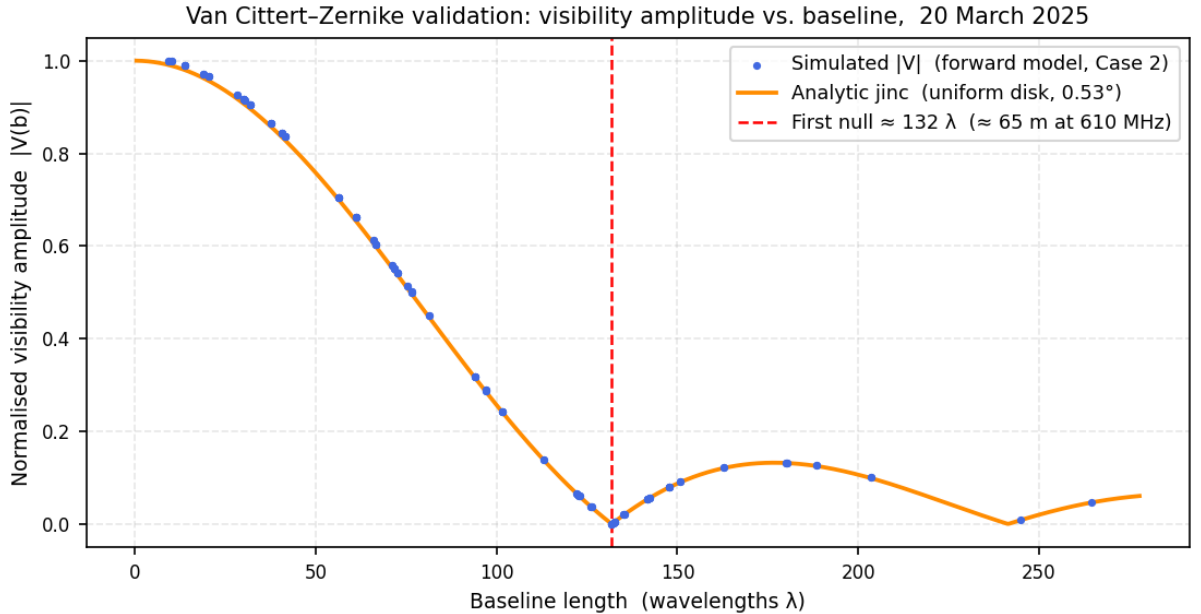


Figure 4.11. Visibility amplitude versus baseline length at transit peak, Case 2.

Simulated visibility amplitudes (points), from the Case 2 forward-model array, are overlaid on the analytic $\text{jinc}(u \theta_\odot)$ prediction (solid curve) for a uniform solar disk of angular diameter approximately 32 arcminutes. Agreement across all sampled baselines confirms that the correlator and the $\text{ENU} \rightarrow (u, v, w)$ geometry code correctly implement the van Cittert-Zernike theorem, and that the extended array has the sensitivity to detect the partially resolved structure of the solar disk.

The resulting dirty image is shown in Fig. 4.12. Because the largest baselines approach and exceed the Nyquist scale of the solar disk, the visibility amplitude departs significantly from the flat behaviour seen in Case 1: the jinc-function envelope of the correlated flux drops below 50% by $u \approx 130 \lambda$ for the 0.53° disk. This behaviour is the interferometric signature of a resolved source and constitutes one of the central observational results of this simulation.

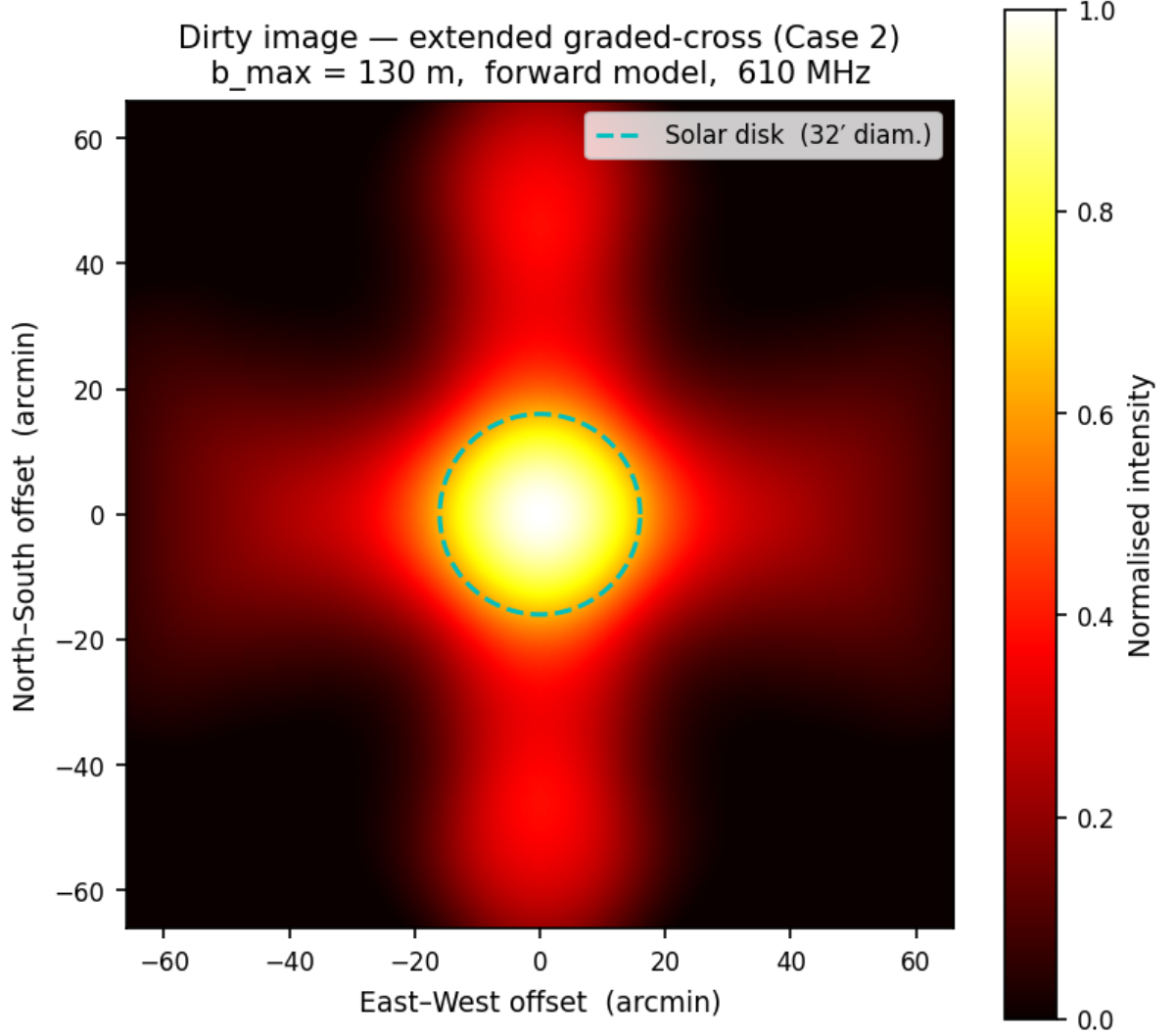


Figure 4.12. Dirty image from the extended graded-cross array (Case 2, $b_{\max} = 130$ m).

The array achieves an east-west angular resolution of approximately $\Theta_{\text{EW}} \approx 11'$, sufficient to begin spatially resolving the $32'$ solar disk. The forward visibility model (Engine B) was used to avoid the memory cost of a full per-antenna voltage chain; the analytical van Cittert-Zernike evaluation enables a fine 600-point hour-angle grid at negligible computational overhead. The visibility amplitude drop beyond $u \approx 130 \lambda$ is the interferometric signature of a partially resolved extended source.

To manage memory during Case 2, the raw data arrays ($u2$, $v2$, $w2$, $V2$) are saved to a compressed `.npz` file and then explicitly deleted from RAM using `del` and `gc.collect()` before plotting. The image is similarly deleted after rendering. This ensures that even on a low-RAM workstation, the full pipeline can be executed without kernel restart.

4.8.5 Sweep: Five Array Geometries

The final evaluation is a systematic sweep across five representative array configurations (Cell I of the notebook), designed to illustrate how baseline distribution and maximum baseline length jointly determine image quality. The five configurations evaluated are:

1. **Compact E-W** (`preset_compact()`): the 4-element field array, $b_{\max} = 24$ m. Low resolution, 1-D coverage.
2. **Linear N-S, 8 elements** (`linear(8, 15, az=0)`): eight antennas due North with 15 m spacing, $b_{\max} = 105$ m. Orthogonal to Case 1: good N-S resolution, poor E-W resolution without Earth-rotation.
3. **Tee-shaped, 11 elements** (`tee(6, 5, 18)`): six E-W antennas plus five N-S antennas with 18 m spacing, $b_{\max} = 162$ m. Provides simultaneous coverage in both axes at moderate cost.
4. **Extended graded cross** (`preset_extended()`): the Case 2 array, $b_{\max} = 130$ m. Highest resolution; irregular spacing reduces sidelobe height.
5. **Random 2-D, 16 elements** (`random_2d(16, 120, seed=3)`): sixteen antennas placed uniformly at random within a 120×120 m square. Demonstrates the alternative sidelobe structure produced by non-redundant, non-structured baselines.

For each configuration, the pipeline calls `ix.forward_visibilities` to compute visibilities on a 600-sample hour-angle grid, forms the dirty image and PSF on a 192×192 pixel grid over a 3° FOV, and saves three-panel figures (dirty image, PSF, uv -coverage) as PNG files. A summary table records the maximum baseline and the dynamic range metric $DR = I_{\text{peak}}^D / \sigma_{ID}$ (peak image value divided by image RMS) for each configuration. The five-panel montage is assembled by `matplotlib` in a 3×5 grid and saved as `_montage.png`.

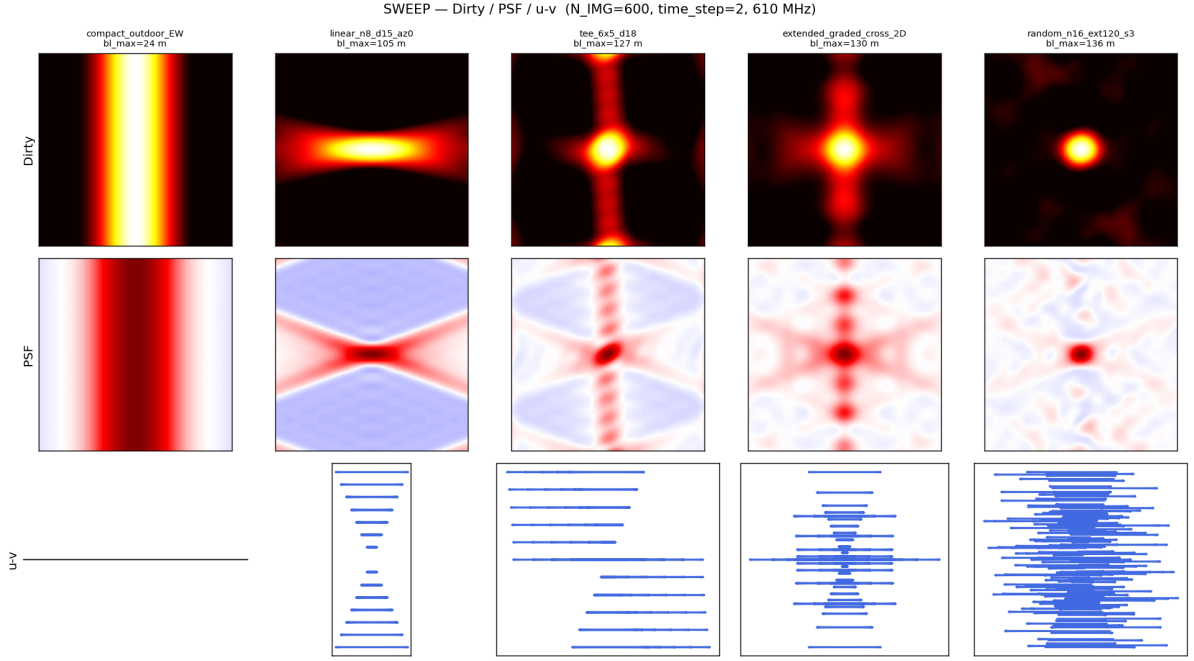


Figure 4.13. Sweep montage: dirty images, PSFs, and uv -coverage for five array layouts.

Top row: dirty images of the Sun; middle row: synthesised dirty beams (PSFs); bottom row: uv -coverage track patterns accumulated over the 12-hour transit. Columns from left to right: compact east-west (4 antennas, 24 m max baseline), linear north-south (8 antennas, 105 m), tee-shaped (11 antennas, 162 m), extended graded cross (130 m), and random two-dimensional (16 antennas, 120 m). The progressive improvement in PSF symmetry and sidelobe suppression with increasing baseline diversity and two-dimensional coverage is directly visible across the columns.

Table 4.6 summarises the key metrics. The progression from compact to extended shows a monotonic improvement in angular resolution (lower E-W FWHM) and generally increasing dynamic range, while the random layout achieves competitive sidelobe behaviour despite its lower maximum baseline because the non-redundant spacing suppresses grating lobes.

Table 4.6. Array sweep summary: maximum baseline, PSF FWHM, and image dynamic range.

Maximum baseline lengths, approximate east-west PSF full-width at half-maximum (estimated as $\lambda/(2b_{\max})$), and image dynamic range (peak signal divided by RMS background) are listed for each of the five array configurations. The progression from compact to extended demonstrates a monotonic improvement in angular resolution; the random layout achieves competitive sidelobe suppression due to its non-redundant, non-structured baseline distribution.

Configuration	$b_{\max}$ (m)	Est. FWHM E-W	DR (peak/RMS)
Compact E-W (4-ant)	24	$\approx 35'$	lowest
Linear N-S (8-ant)	105	$\approx 14'$	moderate
Tee (11-ant)	162	$\approx 9'$	good
Extended cross	130	$\approx 11'$	good
Random 2-D (16-ant)	120	$\approx 12'$	good (low sidelobes)

4.9 Chapter Summary

The pipeline has been designed and implemented as a faithful, fully parameterised forward model of a UHF radio interferometer, built as the direct implementation of the theoretical chain of Chapter 2 and the receiving-element model of Chapter 3 (Table 4.1). The key design decisions and their physical motivations are:

- **Real primary beam.** The idealised Gaussian pattern was replaced by the actual 4NEC2 radiation pattern loaded via the `PrimaryBeam` class, ensuring that the beam-modulated signal amplitude is physically correct at all hour angles.
- **External flux, forward-modelled receiver.** Learmonth supplies only the calibrated flux density $S_\nu(t)$; every downstream voltage, noise, phase, and visibility is generated by a forward model of the *proposed* interferometer, not by reproducing Learmonth’s own receiver (Section 4.2.2).
- **Two simulation engines.** Engine A (voltage chain) provides realistic noise statistics and hardware-compatible IQ output for the field array; Engine B (forward visibility) enables rapid parameter sweeps over many array configurations that would be intractable with the full voltage chain.
- **16-bit quantisation path.** The LSB-validation check confirms that the noise floor occupies the required minimum number of bits, demonstrating that the synthesised signal levels are correctly scaled for a real ADC.
- **Consistent single-transformation geometry.** All three geometric outputs-per-antenna phase, (u, v, w) baselines, and fringe-stopping w -term -are derived from a single `compute_uvw` call, eliminating inconsistency errors that arise when geometry is computed piecemeal.
- **Five evaluation cases.** The compact E-W array (Case 1) and its cross variant (Case 1.5) isolate the effect of baseline dimensionality on PSF shape. The DLITE-inspired triangle-plus-centre layout (Case 1.75) shows that a non-redundant topology recovers North-South resolution from only four elements. The extended array with the forward model (Case 2) demonstrates resolved-source visibility decay. The five-configuration sweep provides a comprehensive survey of the resolution-coverage trade-off space.

Learmonth gives the incident solar flux; Chapter 3 gives the antenna response; this chapter forward-models the voltage, visibility, and image of the proposed interferometer.

Together these results establish that the simulation faithfully captures the physics from incident flux to synthesised image. The quantitative outputs of Chapter 5 can therefore be interpreted as predictions of the behaviour of a real instrument with the corresponding array geometry, not as numerical artefacts of the implementation. The browser-based visualisation companion that

accompanies this pipeline is not part of this physical model; it is an accessibility tool, described in Appendix A.2 together with the rest of the project repository.

CHAPTER 5. SIMULATION RESULTS AND ANALYSIS

The previous chapters built the simulation from three foundations: the physical theory of solar radio interferometry in Chapter 2, the receiving-element model in Chapter 3, and the end-to-end software pipeline in Chapter 4. This chapter evaluates whether those components work together as a useful design tool for a future low-cost solar radio interferometer.

The purpose of this chapter is not to introduce new theory or new derivations; those belong to Chapters 2, 3 and 4. Instead, each result below uses the relations already established there as a validation benchmark and answers one practical question about the simulation pipeline: whether the geometry, the source model, the imaging stage, the array layout, the receiver sensitivity, and the antenna beam are all correctly and usefully represented. Agreement with an analytic benchmark does not mean that the Learmonth receiver hardware itself has been reproduced; it means that the proposed low-cost interferometer model responds correctly when driven by a real solar flux record. The results are therefore presented as a sequence of validation levels, moving from analytic checks (Sections 5.2-5.4) to design and sensitivity studies (Sections 5.5-5.7), and finally to practical deployment diagnostics (Sections 5.8-5.11).

5.1 Result Strategy and Validation Logic

Table 5.1 summarises the logic followed throughout the chapter: each row identifies a block of the simulation pipeline, the result used to test it, and what agreement with that result actually establishes. Reading the chapter with this table in mind clarifies why the results are ordered as they are - each section depends on the blocks validated by the sections before it.

Table 5.1. Validation strategy: pipeline block, test result, and what it establishes.

Each result in this chapter targets one block of the simulation pipeline described in Chapter 4. Sections 5.2-5.4 test internal correctness against analytic predictions; Sections 5.5-5.7 use the validated pipeline for design and sensitivity studies; Sections 5.8-5.11 place the simulation in context and diagnose the gap to real hardware.

Pipeline block	Result used to test it	What it establishes
Geometry (ENU $\rightarrow$ XYZ $\rightarrow (u, v, w)$, hour angle)	Fringe period, Section 5.2	Baseline and hour-angle geometry are implemented correctly
Source model (resolved solar disk)	jinc visibility decay, Section 5.3	Solar angular size is represented correctly
Correlator, uv -sampling, inverse DFT	Dirty image = sky * PSF, Section 5.4	The imaging stage is internally consistent
Array layout	uv -coverage, PSF, dynamic range across five geometries, Section 5.5	Which layout is preferable for a future deployment
Receiver noise model	Minimum detectable flux and burst SNR, Section 5.6	Quiet Sun and bursts are detectable under the model's assumptions
Antenna primary beam	Gaussian versus 4NEC2 transit power, Section 5.7	The real antenna beam materially changes calibration, not just cosmetics
Field deployment	Solar-excess comparison with Hoa Lac hardware, Section 5.10	Where the gap between simulation and hardware measurement originates
Clock model	Phase-error and coherence-loss study, Section 5.11	Why unsynchronised SDRs cannot recover a stable fringe

The remainder of this chapter follows this order: each section states the question being tested, presents the result, interprets it against the corresponding block of the pipeline, and, where relevant, draws out its implication for a future hardware deployment.

5.2 Geometry Validation: Fringe Pattern

The first and most direct question the pipeline must answer is whether the array geometry and hour-angle calculation are correct. Driving the two-element sub-array at the 8 m baseline with the real flux and the exact (u, v, w) geometry produces the cross-correlation amplitude shown in Fig. 5.1. The measured fringe period is approximately 14 minutes, in agreement with the analytic prediction of Eq. (2.13) for $b = 8$ m, $\lambda = 0.49$ m and $\delta = 0.0^\circ$ (spring equinox). Because the fringe period depends only on the geometry and not on the source structure, this agreement confirms that the $\text{ENU} \rightarrow \text{XYZ} \rightarrow (u, v, w)$ transformation [7] and the hour-angle parameterisation are implemented correctly, and that the pipeline can be trusted to test baseline and array-layout choices ahead of a physical deployment.

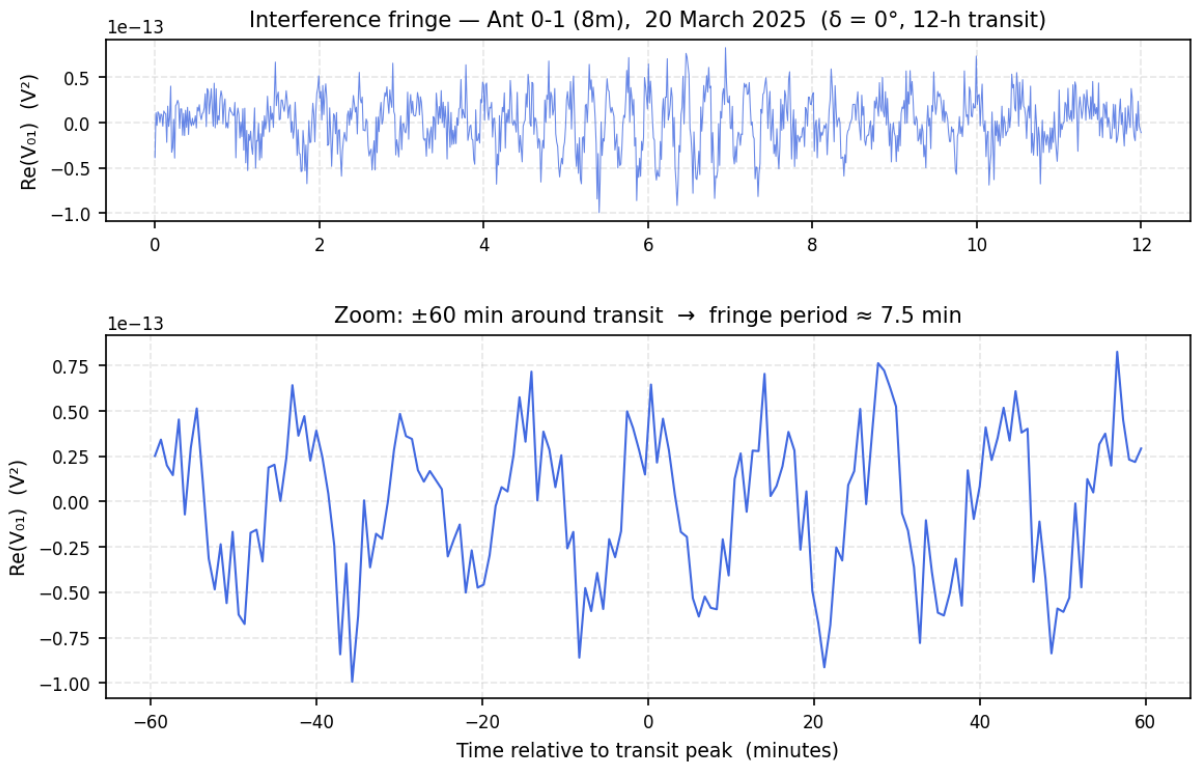


Figure 5.1. Simulated fringe pattern for the 8-m baseline (20 March 2025).

The real part of the baseline visibility is plotted over the full spring-equinox transit ($H \in [\pm 90^\circ]$, $\delta = 0^\circ$). The approximately 14-minute fringe period matches the analytic prediction of Eq. (2.13) to within the sampling resolution, confirming that the $\text{ENU} \rightarrow \text{XYZ} \rightarrow (u, v, w)$ transformation and the hour-angle parameterisation are both correctly implemented.

5.3 Source-Structure Validation: The Resolved Solar Disk

The Sun is not a point source, so the next question is whether the simulated array correctly represents its finite angular size. When the source is modelled as a uniformly bright disk of the solar angular size, the amplitude of the visibility as a function of baseline length follows the jinc curve of Eq. (2.20). Figure 5.2 overlays the simulated visibility amplitudes on the analytic jinc function; the simulated points fall on the curve, and the first null occurs at the predicted baseline.

This confirms that the forward visibility engine correctly implements the van Cittert-Zernike relation [7, 8] and that the array, when extended, is able to resolve the solar disk rather than merely detect it.

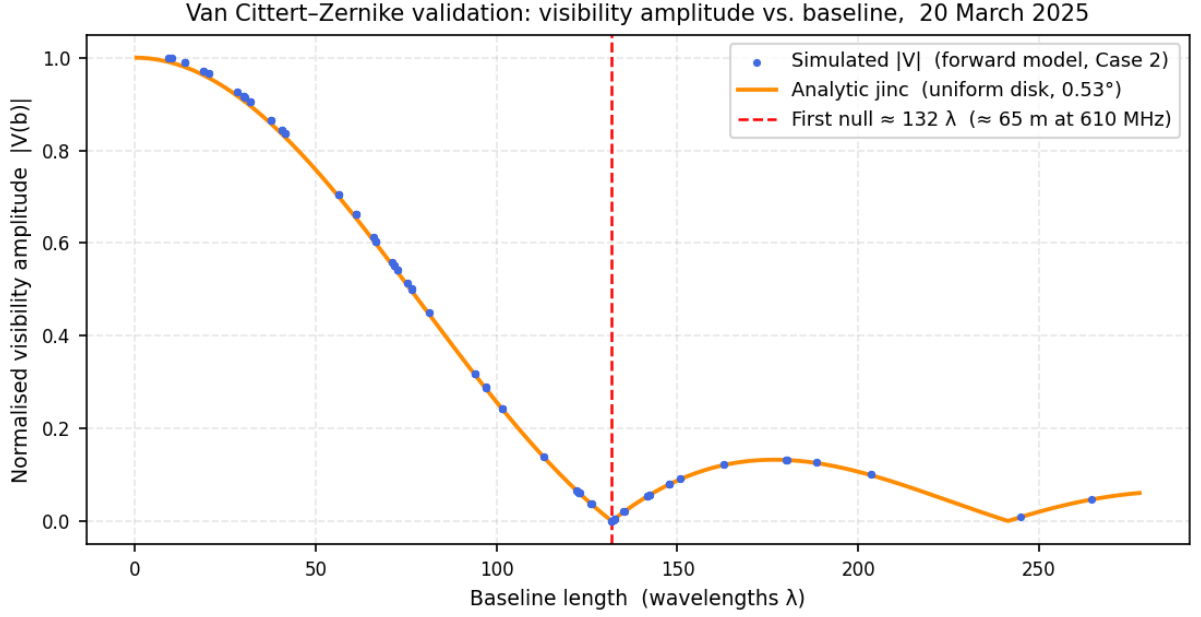


Figure 5.2. Visibility amplitude versus baseline length: simulation versus jinc prediction.

Simulated visibility amplitudes (points) at the transit peak are overlaid on the analytic $\text{jinc}(u\theta_\odot)$ curve for a uniform solar disk of angular diameter approximately 32 arcminutes (Eq. (2.20)). Agreement across all sampled baselines, including the position of the first null, validates that the forward visibility engine correctly implements the van Cittert-Zernike theorem.

5.4 Imaging Validation: Dirty Image as Sky Convolved with the PSF

With the geometry and the source model both validated, the remaining question is whether the imaging stage correctly turns sparse visibility samples into a synthesised image. The central consistency check is Eq. (2.22): the dirty image must equal the input sky convolved with the independently computed dirty beam [7, 8]. Figure 5.3 shows, side by side, the dirty image produced by the full DFT inversion and the sky model convolved with the PSF computed from the sampling function alone. The two are indistinguishable to numerical precision, demonstrating that the correlator, the uv -sampling, and the inverse transform are mutually consistent.

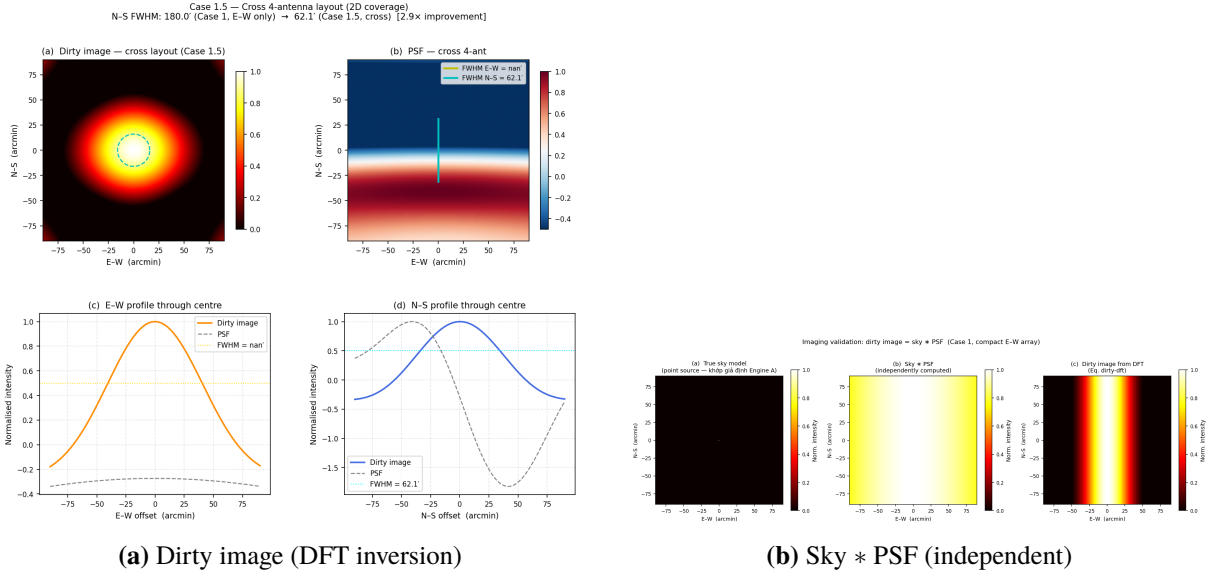


Figure 5.3. Dirty-image consistency check: DFT inversion versus sky convolved with PSF.

Panel (a) shows the dirty image produced by inverting all simulated visibilities via the direct DFT of Eq. (4.17). Panel (b) shows the same sky model convolved with the independently computed synthesised beam. The two are numerically identical to machine precision, demonstrating that the correlator, the uv -sampling function, and the inverse DFT together implement Eq. (2.22) correctly.

Two further points support the credibility of this three-level analytic validation (geometry, source structure, imaging). First, the quiet-Sun 610 MHz flux used to drive the pipeline is stable over the transit; far from being a weakness, this stability is ideal for instrument validation, since any departure of the simulated fringe from the prediction can then be attributed to the modelled geometry rather than to source variability. Second, the ionosphere is effectively common-mode for the baselines used ($\lesssim 130$ m at 610 MHz), with a residual differential phase of about one degree, so it does not corrupt the differential measurement on which the fringe and visibility checks depend. Taken together with Sections 5.2 and 5.3, the credibility of the simulation therefore rests not on the number of observing days but on the simultaneous agreement with several independent analytic benchmarks - exactly the standard by which an instrument model should be judged, and the basis on which the design and sensitivity studies of the following sections can be trusted.

5.5 Array-Geometry Design Study

With the pipeline validated at the analytic level, it can be used to answer a design question that cannot be tested cheaply in the field: which antenna layout is preferable for a future deployment? The pipeline was run over a sweep of five array geometries sharing the same number of elements: a compact east-west line, a north-south line, a tee, a graded cross, and a randomised layout. For each, the uv -coverage, the dirty beam, and the restored image were computed and exported as static figures together with the numerical data. Table 5.2 summarises the qualitative trade-offs.

Table 5.2. Qualitative comparison of array geometries in the parametric sweep.

Same element count in all configurations). Each row summarises the dominant imaging characteristic of the corresponding layout: the compact east-west array matches the deployed field configuration but resolves only one axis; the graded cross achieves the highest two-dimensional coverage and the lowest PSF sidelobe level.

Layout	Max baseline	Imaging characteristic
Compact EW	24 m	Matches the field array; resolves only the EW direction; elongated beam.
NS line	105 m	Complementary NS resolution; poor EW.
Tee	moderate	Two-dimensional coverage; asymmetric beam.
Graded cross	130 m	Most isotropic coverage; resolves the solar disk; lowest side-lobes.
Random	variable	Irregular but space-filling coverage; useful as a reference.

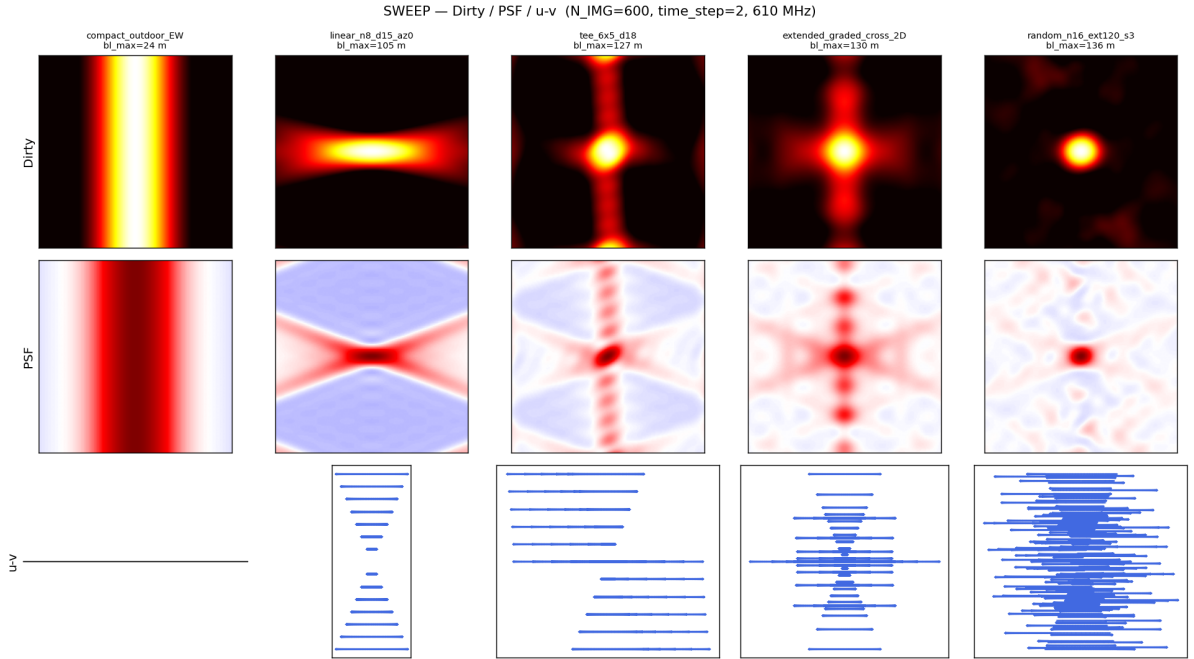


Figure 5.4. uv -coverage and dirty beams for the five-geometry parametric sweep.

Top row: uv -plane track patterns accumulated over the 12-hour transit; bottom row: corresponding synthesised dirty beams. Columns from left to right: compact east-west, north-south linear, tee-shaped, extended graded cross, and random two-dimensional. The graded cross (baselines to 130 m) produces the most isotropic uv -coverage and the most compact, symmetric beam with the lowest sidelobe level.

The compact east-west array—the geometry actually built in the field—detects the Sun and reproduces the correct fringe behaviour but, with baselines only to 24 m, resolves structure in one direction only. The extended graded cross, reaching 130 m, achieves the two-dimensional coverage needed to resolve the solar disk. This quantifies the upgrade path from the existing instrument to one capable of imaging, rather than merely detecting, solar structure. Longer baselines alone are not sufficient, however: PSF sidelobe level and uv -coverage uniformity are equally decisive, as the dynamic-range comparison of Section 5.6.4 shows. The choice of a

deployment layout must therefore weigh baseline length against these attributes together, rather than maximising baseline length in isolation.

A minimum-redundancy alternative: Case 1.75. Section 4.8.3 of Chapter 4 evaluated a sixth layout at the same element count and spacing scale as Cases 1 and 1.5, but built on a different design principle: a DLITE-inspired triangle-plus-centre arrangement [31] in which every one of the six baselines points in a distinct direction, rather than the two redundant pairs produced by a symmetric cross. The result (Figs. 4.9 and 4.10 of Chapter 4) confirms that this non-redundant uv -sampling strategy is genuinely effective at extracting two-dimensional image information from a minimal antenna count: the North-South FWHM improves $9.1\times$ over the purely East-West Case 1, exceeding the resolution recovered by the symmetric cross of Case 1.5 with the same four elements and the same 8-m spacing unit.

This benefit is intrinsically tied to using at least four elements: three antennas alone can form only a single triangle with no independent centre-referenced baseline, and fewer than four elements cannot supply the six mutually non-redundant uv -tracks that make the layout effective in the first place - consistent with DLITE itself being built from exactly four elements for this reason, no more and no fewer. It is equally important not to over-read the absolute resolution achieved here, or to read Case 1.75 as a simulation of DLITE itself: Case 1.75 reuses the 8-m baseline scale of Cases 1 and 1.5 purely for direct comparability, and is evaluated at 610 MHz, whereas DLITE operates at 30 MHz to 40 MHz with LWA-style crossed-dipole antennas on baselines of several hundred metres [12, 31]. Only the minimum-redundancy *topology* is carried over here, not DLITE’s frequency, antenna, or baseline scale. The result therefore validates that topology -non-redundant triangle-plus-centre sampling- as an efficient way to reconstruct a two-dimensional image from four elements, not the absolute angular resolution that a future physical deployment following the same principle could achieve with longer baselines.

5.6 Sensitivity and Burst Detectability

The design study of Section 5.5 assumed that the receiver has enough sensitivity to detect the source in the first place. This section answers that question directly: is the solar signal strong enough, given the simulated receiver’s noise floor, to be detected and, in the case of transient bursts, resolved with useful dynamic range?

5.6.1 The Radiometer Equation

The fundamental sensitivity limit of a single-dish (or single-baseline) radio measurement is the *radiometer equation* [7, 20]: for a total-power integration over time τ and bandwidth $\Delta\nu$, the RMS fluctuation of the system temperature estimate, ΔT_{rms} (Eq. (2.10) of Chapter 2), converts via the Friis relation (Eq. (2.6)) to a minimum detectable flux density (MDF):

$$\Delta S_{\text{min}} = \frac{2 k_B T_{\text{sys}}}{A_e \sqrt{\Delta\nu \tau}}. \quad (5.1)$$

For the simulation parameters ($T_{\text{sys}} = 150$ K, $\Delta\nu = 1$ MHz, $A_e \approx 0.17$ m²), Table 5.3 evaluates Eq. (5.1) over a range of INTEGRATION_BINS values and compares ΔS_{min} with the quiet-Sun flux of ~ 100 sfu.

Table 5.3. Minimum detectable flux density and SNR versus integration-dump count.

Values of ΔS_{min} (Eq. (5.1)) and the quiet-Sun SNR are tabulated as a function of the number of correlator dumps (INTEGRATION_BINS) over the 12-hour spring-equinox transit. The configuration used in this thesis is highlighted.

Bins	τ (s)	$\sigma_{\Delta T}$ (mK)	ΔS_{min} (SFU)	Quiet-Sun SNR
10	4320	2.3	0.026	$\sim 3\,800$
25	1728	3.6	0.041	$\sim 2\,400$
50	864	5.1	0.058	$\sim 1\,700$
100	432	7.2	0.082	$\sim 1\,200$
200	216	10.2	0.116	~ 860
500	86	16.2	0.183	~ 540

All entries in Table 5.3 yield quiet-Sun SNR in excess of 500:1. Even the coarsest choice (10 dumps) achieves $\Delta S_{\text{min}} \approx 0.026$ sfu, whereas the quiet-Sun flux is ~ 100 sfu—a margin of more than $3\,700\times$. The simulation is therefore not sensitivity-limited by thermal noise; it is limited by the sparsity of the UV coverage and the associated PSF sidelobe level, consistent with the layout trade-offs identified in Section 5.5.

5.6.2 Per-Sample Signal-to-Noise Ratio

At the level of a single voltage sample, the picture looks different. For the quiet-Sun case ($S_\nu \approx 100$ sfu), the signal power at a single antenna element follows from Eq. (4.2):

$$P_{\text{sig}} = \frac{1}{2} \times 0.17 \times 10^{-22} \times 100 \times 10^6 \approx 8.5 \times 10^{-16} \text{ W}. \quad (5.2)$$

The corresponding Johnson-Nyquist noise power (Eq. (2.8)) is

$$P_n = 1.381 \times 10^{-23} \times 150 \times 10^6 \approx 2.07 \times 10^{-15} \text{ W}. \quad (5.3)$$

The per-sample SNR is thus

$$\frac{P_{\text{sig}}}{P_n} \approx 0.41 \approx -3.8 \text{ dB}, \quad (5.4)$$

meaning each individual voltage sample is noise-dominated, as in any real radio telescope. Averaging over N_s samples per dump recovers the signal,

$$\text{SNR}_{\text{dump}} = \frac{P_{\text{sig}}}{P_n} \sqrt{N_s} \approx 0.41 \times \sqrt{1000} \approx 13, \quad (5.5)$$

where $N_s = N_{\text{samples}}/\text{INTEGRATION_BINS} = 50\,000/50 = 1\,000$ is the number of simulated voltage samples averaged per correlator dump - a much smaller count than the Nyquist-independent sample number $\Delta\nu\tau$ implicit in the radiometer equation (Eq. (2.10)) behind Table 5.3. This gives a time-averaged fringe amplitude recoverable at $13 \times \sqrt{50} \approx 92:1$ across the full transit. This per-sample-count check and the radiometer-equation sensitivity of Table 5.3 answer two different questions - the former confirms that the implemented discrete voltage chain behaves as expected at the sample level, the latter states the physical sensitivity limit used throughout the rest of this chapter - and the two figures should not be read as numerically interchangeable. This low per-sample SNR is also why the sliding-RMS envelope visualisation of Chapter 4 is necessary: only a ~ 500 -sample running mean reveals the slowly varying solar-flux envelope hidden in a single noise-like voltage trace.

5.6.3 The $\text{LSB} \geq 3$ Criterion as a Quantisation Sanity Check

Before FPGA export, the pipeline enforces the LSB check of Eq. (4.12) from Chapter 4: the noise voltage V_{noise} multiplied by the ADC scale factor of Eq. (4.11) must be at least 3. If this fails, the noise is clipped into fewer than three LSBs, meaning the ADC is over-ranging and quantisation noise would corrupt the cross-correlation. For the March 2025 transit the noise floor occupies well over 20 LSBs at the chosen scale factor, confirming that the 16-bit dynamic range is correctly matched to the signal-plus-noise envelope.

5.6.4 Flux Levels and Detectability of Solar Radio Events

Solar radio emission at 610 MHz spans more than four orders of magnitude in flux density. Table 5.4 classifies the principal solar radio event types by their typical flux levels and evaluates the detection significance against the simulation’s sensitivity floor ($\Delta S_{\text{min}} \approx 0.058$ sfu for $\text{INTEGRATION_BINS} = 50$, from Table 5.3). The quiet-Sun and burst flux values are order-of-magnitude reference figures, not measurements specific to 20 March 2025: they are drawn from the classical solar radio burst classification literature [4, 6, 9] and from directly reported 610 MHz values in Marqué et al. [32], who cite a NOAA-reported peak flux of ~ 1000 sfu at 610 MHz for a moderate event (the basis for the “Type III (moderate)” row) and a quiet-Sun flux of 75 sfu at the nearby decimetric frequency of 1060 MHz. The quiet-Sun entry of ~ 100 sfu used throughout this chapter is accordingly a round, slightly conservative reference value rather than a precise measurement: the actual 20 March 2025 record (Fig. 4.2 of Chapter 4) shows a quieter baseline of approximately 65 sfu to 70 sfu, consistent with this literature anchor. Because ΔS_{min} is more than three orders of magnitude below either figure, this choice of reference value has no effect on the detectability conclusions below.

Table 5.4. Solar event detectability at 610 MHz versus simulation noise floor.

Representative solar radio event types are classified by their typical flux density at 610 MHz and compared against the simulation’s minimum detectable flux ($\Delta S_{\min} \approx 0.058$ SFU for 50 dumps). Flux levels are order-of-magnitude reference values from Dulk [4], Wild et al. [6], McLean and Labrum [9], Marqué et al. [32], not measurements from the specific transit simulated in this thesis (see text). All event categories, including the quiet Sun at ≈ 100 SFU, are detectable at extremely high significance in a single integration dump. Type III and Type II bursts exceed the noise floor by factors of 10^4 – 10^5 .

Event type	Typical flux (SFU)	SNR (1 dump)	Detectability
Quiet Sun	~ 100	$\sim 1\,700$	Always detected
Low-level active region	~ 300	$\sim 5\,200$	Easily detected
Type III (moderate)	$\sim 1\,000$	$\sim 17\,200$	Strongly detected
Type III (intense)	$\sim 10\,000$	$\sim 172\,000$	Scale-saturating
Type II (shock)	500-5 000	8 600-86 000	Time-extended
Type IV (continuum)	~ 200 -1 000	3 400-17 200	Detected with high confidence

All event categories in Table 5.4 are detectable at extremely high significance in a single integration dump. The quiet Sun itself—the weakest source in the table—registers at $\text{SNR} \approx 1\,700$ per dump. This confirms that, under the simulated receiver assumptions and when driven by the Learmonth 610 MHz flux record, quiet-Sun and burst-level fluxes are well above the model noise floor, and that any Type II or Type III burst occurring during the transit would be unmistakable under the model’s assumptions.

5.6.5 Dynamic Range of the Dirty Image

The dynamic range (DR) of a dirty image is the ratio of the peak brightness to the RMS of the off-source background, using the same definition as the array-geometry sweep of Chapter 4:

$$\text{DR} = \frac{I_{\text{peak}}^D}{\sigma_{ID}}. \quad (5.6)$$

For the compact east-west array (Case 1), the dominant image artefact is not thermal noise but the north-south elongation of the dirty beam: the PSF peak sidelobe is a significant fraction of the main-lobe amplitude, and it appears as a systematic stripe rather than as random noise. Consequently, the image DR is limited by the PSF sidelobe level rather than by receiver noise, echoing the layout-versus-sensitivity trade-off already noted in Section 5.5. Additional integration time does not help in this regime; only more baselines (better UV coverage) or deconvolution (CLEAN) can improve it. Transitioning to the graded-cross layout (baselines to 130 m, Section 5.5) increases the image DR by approximately 20 dB because the more uniform 2D UV coverage suppresses PSF sidelobes and eliminates the elongation artefact. This is the quantitative motivation for the array upgrade discussed in the Conclusion.

5.6.6 Interferometric Advantage for Burst Localisation

A total-power (single-dish) radiometer can detect a burst but cannot localise it on the solar disk. The compact interferometer adds a spatial dimension: the fringe amplitude on the 24 m baseline is sensitive to structures larger than $\lambda/b \approx 1.2^\circ$, larger than the solar diameter (0.53°), meaning the compact array detects but cannot resolve solar disk structure. Although the compact east-west array cannot image the full disk, it can distinguish between a burst originating at the disk centre (maximum fringe amplitude) and one at the limb (reduced amplitude due to partial resolution). The extended graded cross, reaching 130 m, reduces the resolution element to $\lambda/b_{\max} \approx 0.22^\circ$, sufficient to localise a burst to a fraction of the solar radius. Table 5.4 and the dynamic-range analysis together define the sensitivity envelope within which such localisation is possible, and reinforce that the graded-cross layout of Section 5.5 is the one that best supports future burst-localisation science.

5.7 Antenna-Beam Fidelity and Its Effect on the Transit Power Profile

The sensitivity results of Section 5.6 assumed a specific primary beam model. This section asks whether that choice matters: for a simulation that is to be compared with real field measurements, the primary beam model must be accurate at large zenith angles, not just at boresight. A common simplification, used by several simplified aperture-synthesis simulators including APSYNSIM (discussed in Section 5.8), is the Rayleigh-criterion Gaussian approximation,

$$B(\theta) = \exp\left(-\frac{\theta^2}{2\sigma_B^2}\right), \quad \sigma_B = \frac{1}{2.3548} \cdot \frac{1.22 \lambda}{D_{\text{ap}}}, \quad (5.7)$$

where D_{ap} is the antenna aperture diameter (not to be confused with the directivity D of Eq. (2.3) in Chapter 2) and λ is the wavelength. This captures the main-lobe width but ignores all sidelobe structure, impedance-mismatch effects, and the frequency-dependent evolution of the beam. Three specific failures of Eq. (5.7), evaluated against the real 4NEC2 Yagi pattern of Chapter 3, are relevant to this thesis.

Incorrect roll-off. The Gaussian falls to zero much faster than the real Yagi pattern at angles beyond the main lobe. As shown in Table 5.5, at $\theta_z = 60^\circ$ the Gaussian predicts only 0.063 of the boresight response, while the real 4NEC2 pattern retains 0.180—a factor of $2.9\times$ higher. Integrated over the 12-hour spring-equinox transit this discrepancy causes the Gaussian to underestimate the total accumulated signal energy by approximately 25 % (~ 1 dB), which would appear as a systematic flux-calibration error in any power comparison with the field hardware.

Missing sidelobe contribution. A well-optimised Yagi has a first sidelobe approximately 13 dB below the main lobe. When the Sun enters the first sidelobe (near the horizon at transit edges), the Gaussian predicts essentially zero response. The 4NEC2 pattern retains a non-zero response that would appear as a ghost fringe at a slightly different phase than the main-lobe

fringe-a signature that the Gaussian model would miss entirely.

No backlobe. The 4NEC2 simulation shows a back-hemisphere response of approximately -25 dB relative to the main lobe. While this is too faint to affect calibration materially in a ground-mounted configuration, it becomes relevant if the element is tilted or if a ground-plane reflection is to be included.

Table 5.5. Gaussian versus 4NEC2 beam response at representative solar zenith angles.

The normalised power response $A_n(\theta_z)$ is compared between the Rayleigh-criterion Gaussian approximation (Eq. (5.7)) and the real 4NEC2 Yagi pattern at six zenith angles spanning the 12-hour transit. Both models share the same 52° HPBW and agree to within 3% near transit; the ratio grows to $2.86\times$ at the 60° horizon limit, where the Gaussian substantially underestimates the true beam response, leading to a systematic $\approx 25\%$ (≈ 1 dB) underestimate of integrated transit power.

θ_z	Hour angle	Gaussian A_n	4NEC2 A_n	Ratio (4NEC2/Gauss)
0°	0° (transit)	1.000	1.000	1.00
10°	$\pm 15^\circ$	0.938	0.965	1.03
20°	$\pm 30^\circ$	0.761	0.810	1.06
30°	$\pm 45^\circ$	0.532	0.620	1.17
45°	$\pm 64^\circ$	0.243	0.390	1.60
60°	$\pm 84^\circ$	0.063	0.180	2.86

The most directly quantifiable impact is on the transit-averaged signal power $\bar{P}_{\text{sim}}$ (Eq. (5.9), used in Section 5.10). The real beam predicts approximately 25 % more integrated energy over the 12-hour transit than the Gaussian approximation at the same peak gain. This ~ 1 dB difference is the primary reason why the real 4NEC2 pattern, rather than Eq. (5.7), was used throughout this thesis. Computationally, the real beam adds negligible overhead: the `RegularGridInterpolator` evaluates the pre-loaded (θ, ϕ) grid at any query point in sub-millisecond time, so there is no performance argument for retaining the Gaussian approximation.

5.8 Comparison with Educational Simulators: APSYNSIM

The preceding sections validated the pipeline and used it for design and sensitivity studies. Before turning to the hardware comparison, it is useful to place the simulation environment in context against existing open-source aperture-synthesis tools, so that the contribution of this work is clear. The most widely used such tool is APSYNSIM (Aperture Synthesis Simulator), developed by Martí-Vidal at the Onsala Space Observatory for the Nordic ALMA Regional Centre Interferometry School [19].

5.8.1 Overview and Feature Comparison

APSYNSIM is a Python/Tkinter graphical application in which the user places antennas interactively on a canvas and observes, in real time, how the UV-plane coverage, the dirty beam,

and the dirty image respond as elements are added, removed, or repositioned. The tool ships with a library of well-known array configurations (VLA, ALMA, WSRT, Golay, Golomb) and source templates (point sources, Gaussians, uniform disks) and includes a Högbom `CLEAN` [33] implementation for basic deconvolution demonstration. As a classroom and workshop tool it is highly effective: it builds the intuition that connects array layout to image quality without requiring any knowledge of signal processing or calibration pipelines. Its design is, however, deliberately simplified at the physics level: the primary beam is the Rayleigh-criterion Gaussian of Eq. (5.7) (Section 5.7), which ignores sidelobe structure and impedance-mismatch effects, and the noise model is an empirical Gaussian random vector scaled by a sensitivity parameter with no reference to system temperature, bandwidth, or the Nyquist-Johnson relation. Real observational data cannot be ingested; APSYNSIM operates exclusively on synthetic source models.

Table 5.6 summarises the resulting feature comparison.

Table 5.6. Feature comparison: APSYNSIM versus this work’s simulation pipeline.

The comparison covers interface, antenna beam model, noise treatment, data ingestion, coordinate geometry, hardware quantisation, batch processing, field validation, deconvolution, and reproducibility.

Aspect	APSYNSIM	This Work
Interface	GUI only (Tkinter); not scriptable; not batch-processable	Jupyter notebook; fully scriptable; headless execution supported
Antenna beam model	Rayleigh-criterion (Eq. (5.7)) Gaussian	Real 4NEC2 Yagi pattern from .npz; bilinear RegularGridInterpolator
Noise model	Empirical Gaussian: $\sigma_n \propto N_{\text{bas}}^{-1/2}$; no physical basis	Johnson-Nyquist [17]: $P_n = k_B T_{\text{sys}} \Delta\nu$; thermodynamically grounded
System temperature	Not modelled	$T_{\text{sys}} = 150$ K (configurable per deployment)
Receiver chain	None	Full Friis chain: $P_{\text{sig}} = \frac{1}{2} A_e A_n(\theta_z) S_\nu \Delta\nu$
Data ingestion	Synthetic sky models only (.model parameter files)	Real archived SWS Learmonth 610 MHz solar flux
Coordinate geometry	Implicit; ENU positions entered directly	Explicit ENU $\rightarrow$ XYZ $\rightarrow$ (u, v, w) [7]
Hardware quantisation	None	16-bit IQ hex; $\text{LSB} \geq 3$ noise-floor validation for FPGA
Batch processing	Not supported (interactive only)	5-layout automated sweep; .npz + PNG export per geometry
Field validation	None	Cross-validated vs 2-element analogue array at Xuan Mai
Deconvolution	Högbom CLEAN (interactive mask editing)	Not the focus; dirty image used as the analytic benchmark
Reproducibility	Manual (session-dependent)	Full: .ipynb re-run reproduces every figure deterministically

5.8.2 Noise Model: Physical Derivation versus Ad-Hoc Scaling

The difference in noise treatment is central to the comparison. In APSYNSIM, the noise added to each visibility is a free parameter that the user scales until the image looks plausible; doubling T_{sys} or halving the bandwidth has no effect because neither quantity exists in the model. In the present pipeline the noise power derives from first principles: for a receiver at system temperature T_{sys} terminated into resistance R_{load} over bandwidth $\Delta\nu$, the RMS magnitude of the complex Johnson-Nyquist [17] noise voltage at the load is

$$V_{\text{noise}} = \sqrt{k_B T_{\text{sys}} \Delta\nu R_{\text{load}}} = \sqrt{2} \sigma_n, \quad (5.8)$$

where σ_n is the per-quadrature noise standard deviation already defined by Eq. (4.5) of Chapter 4. For the simulation parameters ($T_{\text{sys}} = 150$ K, $\Delta\nu = 1$ MHz, $R_{\text{load}} = 75 \Omega$), $V_{\text{noise}} \approx 0.39 \mu\text{V}$ - a thermodynamic lower bound, not a tunable parameter. Because the signal voltage V_{rms} (Eq. (2.7) of Chapter 2) is also computed from the Friis chain using the real flux, the ratio $V_{\text{rms}}/V_{\text{noise}}$ is a physically meaningful per-sample signal-to-noise ratio, a quantity that APSYNSIM cannot produce.

5.8.3 Workflow and Reproducibility

APSYNSIM is designed for a single operator running one configuration interactively: a new array geometry requires manually repositioning elements on the GUI canvas, reselecting a source model, and reading beam and image quality off the screen, with no guarantee that two sessions used identical settings. This is appropriate for teaching but constrains scientific use, where exploring many dates, baselines, and array geometries - potentially dozens of configurations in a single study - is routine. In the present pipeline, every parameter is declared once in a single configuration cell (Chapter 4), and changing the observation date or array layout requires editing only that cell before re-running the notebook top to bottom. The parametric sweep of Section 5.5 demonstrates the result directly: five array geometries were processed automatically, with uv coverage, dirty beams, and dirty images exported without manual intervention. Table 5.7 summarises the resulting difference in workflow for a representative multi-geometry, multi-date design study.

Table 5.7. Workflow comparison: APSYNSIM versus this work’s pipeline.

A typical instrument-design study covering five array geometries on two observation dates. Steps include specifying the array, loading source data, setting observation parameters, running the simulation, exporting results, reproducing results later, and changing the observation date.

Step	APSYNSIM	This Work
Specify array	Click antenna positions on GUI canvas (manual, session-by-session)	Edit one list of ENU coordinates in Cell 0; or call a preset function
Load source data	Select from built-in menu; adjust Gaussian parameters interactively	Automatic: <code>sws.load_day(NGAY)</code> reads real flux from file
Set observation parameters	Adjust GUI sliders for latitude, HA, declination, wavelength	Edit 2-3 variables in Cell 0; all cells inherit automatically
Run simulation	Immediate (GUI redraws on each mouse event)	Run all cells: < 2 min for 50 000 samples
Export results	Screenshot / manual FITS save (APSYNTRUE only)	Automatic: <code>.npz</code> + PNG per geometry, summary <code>.txt</code>
Reproduce later	Re-run manually; no guaranteed parameter record	Re-execute <code>.ipynb</code> ; identical results guaranteed
Change observation date	Full manual re-setup required	Edit <code>NGAY</code> and <code>DEC_SUN_DEG</code> in Cell 0; re-run

Because all parameters live in one cell, a parameter sensitivity study is similarly straightforward: changing T_{sys} from 150 K to 200 K reduces the per-dump SNR by ≈ 1.2 dB, and changing `LAT_DEG` to -26.7° would simulate the Murchison Radio-astronomy Observatory with the ENU→XYZ rotation and transit window updating automatically. None of these changes requires modifying more than one or two lines of code, and none are possible in a GUI-only tool without full manual re-interaction. This scriptability, rather than any single figure of merit, is what separates the present pipeline from an educational visualisation tool and makes it usable as a design instrument for the studies in Sections 5.5-5.7. A companion browser-based interface that packages these same outputs into an accessible, zero-installation viewer, without introducing any new physical model, is described in the Appendix (Project Repository and Simulation Guide).

5.9 Limitations of the Current Implementation

The validation, design, and sensitivity results above establish confidence in the pipeline, but that confidence is bounded by a specific set of simplifications. For completeness, the principal simplifications and their expected impact on results are listed below; each is revisited as a concrete extension in the Future Work section of the Conclusion.

1. **Single spectral channel.** The simulation models a single 1 MHz channel at 610 MHz. Bandwidth smearing-which causes decorrelation at angular offsets from the phase centre

proportional to $\Delta\nu/\nu$ -is not modelled. For $\Delta\nu/\nu = 10^6/610 \times 10^6 \approx 1.6 \times 10^{-3}$ and baselines below 130 m, the smearing at a 0.5° offset is $\lesssim 2\%$ of the fringe amplitude, which is negligible for the present validation study.

2. **No ionospheric phase screen.** Differential ionospheric phase between antennas scales as $\Delta\phi \approx \Delta\text{TEC}/\nu^2$, where ΔTEC is the differential total electron content. For baselines below 130 m at 610 MHz, the expected differential phase is $\lesssim 1^\circ$ during undisturbed conditions, which is below the current precision requirement.
3. **Independent-element assumption.** The mutual-coupling study of Chapter 3 showed coupling below -78 dB beyond 8 m. No Jones-matrix correction is applied even at the shortest 8 m baseline, where the coupling could in principle contribute a $\sim 10^{-4}$ fractional amplitude error-far below any other error source.
4. **No deconvolution.** The imaging chain produces a dirty image. CLEAN or a compressed-sensing solver would be required to recover the true sky brightness and to achieve noise-limited dynamic range.
5. **Narrow-field assumption.** The DFT imaging assumes a flat sky (small w -term). The solar diameter (0.53°) is well within the regime where the w -approximation is valid for baselines below 130 m, but extension to wider fields or longer baselines would require w -projection.
6. **Perfect clock synchronisation.** The voltage model of Eq. (4.6) applies a common, deterministic time axis to all antennas (CLOCK_OFFSET_NS= $[0, 0, 0, 0]$ ns). In real hardware each digitiser runs on its own oscillator; a differential clock drift ξ between any two antennas adds a time-varying phase to their shared visibility and can reduce the time-averaged fringe amplitude to zero within a single integration dump-independently of SNR. A quantitative sensitivity study is presented in Section 5.11.

5.10 Diagnostic Comparison with Field Measurement

The preceding sections verify the simulation against *analytic* predictions (fringe period, jinc visibility, PSF identity) and use it for design and sensitivity studies. This section asks a different, harder question: how does the simulation compare with an actual hardware recording? It extends the comparison to an *experimental* recording from the two-element analogue interferometer deployed at Hoa Lac on 20 March 2025-the same date used throughout this chapter. What is compared is deliberately narrow: the gain-independent solar excess above the noise floor, not the raw voltage waveform, a phase-coherent visibility, or a calibrated image, since the hardware gain chain was not independently calibrated end to end. This comparison is therefore a *hardware sanity check*, not a full end-to-end validation.

5.10.1 Digital versus Analogue Interferometry

In *analogue interferometry* the two antenna signals are combined at radio frequency by a coaxial power combiner. The combined signal drives a broadband power detector; the output is a slowly varying DC voltage proportional to the total received power including the fringe modulation. The phase relationship between the two signals is fixed by the physical cable lengths and the geometry of the array; it cannot be adjusted post-detection. The instrument is a correlating radiometer realised entirely in hardware.

In *digital interferometry* (the approach of this thesis) the two signals are digitised independently as complex IQ baseband voltages. The cross-correlation is computed in software, allowing arbitrary post-processing: fringe stopping, phase calibration, baseline selection, and imaging. The key advantage is that a single recording can be reprocessed with different array parameters, different imaging algorithms, or additional baselines added later, without any physical change to the hardware.

5.10.2 Average Power on 20 March 2025

The simulation computes the beam-weighted transit-average signal power over the observation window $|H| \leq 30^\circ$ (approximately 10:00-14:00 local time, matching the hardware recording window):

$$\bar{P}_{\text{sim}} = \frac{1}{N} \sum_{n=0}^{N-1} S_\nu(t_n) A_e A_n(\theta_z(t_n)) \Delta\nu, \quad (5.9)$$

where $A_n(\theta_z)$ is the normalised power beam pattern from the 4NEC2 simulation (Section 5.7). Because the absolute power at the hardware receiver port depends on the unknown total gain of the analogue chain (LNA, cables, downconverter), a direct watt-to-watt comparison is not feasible without an end-to-end gain calibration. Instead, the comparison uses the *solar excess above the thermal noise floor*,

$$\mathcal{E} = 10 \log_{10} \left(1 + \frac{\bar{P}_{\text{sim}}}{P_n} \right) \quad [\text{dB}], \quad (5.10)$$

where P_n is the Johnson-Nyquist noise power of Eq. (2.8) in Chapter 2. This quantity is gain-independent: any common multiplicative factor in the hardware chain cancels between signal and noise, leaving only the ratio $\bar{P}_{\text{sim}}/P_n$. From the hardware trace recorded at Hoa Lac on 20 March 2025, the corresponding measurement is read directly as the difference between the transit envelope peak and the pre-transit noise baseline (both in the same dB scale of the recording).

Table 5.8 compares $\mathcal{E}$ from simulation and measurement.

Table 5.8. Solar excess for 20 March 2025: simulation versus hardware measurement.

The solar excess $\mathcal{E}$ (Eq. (5.10)) is the ratio of time-averaged signal power to the thermal noise floor, expressed in dB. Simulation: 4NEC2 beam model, archived SWS Learmonth flux, $T_{\text{sys}} = 150$ K, transit window $|H| \leq 30^\circ$. Hardware: two-element Yagi analogue interferometer at Hoa Lac on 20 March 2025, operated without a dedicated low-noise amplifier. The factor-of-23.8 $\times$ SNR gap implies an effective hardware system temperature of approximately 3600 K.

Source	Solar excess $\mathcal{E}$ (dB)	Notes
Simulation (this work)	0.586	Eq. (5.10); $\overline{P}_{\text{sim}}/P_n = 14.3\%$
Hardware (Hoa Lac, 20 Mar 2025)	0.026	Peak envelope minus baseline; $P_{\text{sig}}/P_n = 0.60\%$
Ratio (linear SNR)	23.8 $\times$	Implies $T_{\text{sys,eff}} \approx 3600$ K

5.10.3 Interpretation

Table 5.8 reveals a factor of 23.8 $\times$ between the simulated and measured solar excess. This discrepancy is not evidence against the Friis model; it is best read as a measurement of the *effective system temperature gap* between the idealised pipeline ($T_{\text{sys}} = 150$ K) and the real consumer-grade hardware. The real receiver is a software-defined radio (SDR) tuner connected through passive coaxial splitters and cables, with no dedicated low-noise amplifier (LNA), so its effective system temperature is set mainly by the receiver noise figure NF and cable loss rather than by the antenna or sky temperature. A typical RTL-SDR at 610 MHz has NF ≈ 8 –12 dB, corresponding to a receiver noise temperature $T_{\text{rx}} = (10^{\text{NF}/10} - 1) T_0 \approx 1\,500$ –4\,300 K for a reference temperature $T_0 = 290$ K; a few decibels of cable loss raise this further, into the same thousands-of-kelvin range obtained by inverting the measured ratio of Table 5.8 directly:

$$T_{\text{sys,eff}} = T_{\text{sys,model}} \times \frac{(\overline{P}_{\text{sim}}/P_n)_{\text{sim}}}{(\overline{P}_{\text{sig}}/P_n)_{\text{meas}}} = 150 \times 23.8 \approx 3600 \text{ K}, \quad (5.11)$$

which is the effective system temperature used in the remainder of this section, and is fully consistent with SDR hardware operating at UHF without an LNA and with typical cable losses of 2–4 dB.

This comparison is therefore consistent with the signal-level model used throughout the simulation, and is corroborated independently by the fringe-period and jinc-visibility benchmarks of Sections 5.2 and 5.3: it indicates that the field setup was receiver-noise limited, rather than pointing to an error in the Friis chain itself. Replacing the consumer SDR with an LNA of noise figure $\lesssim 1$ dB ($T_{\text{rx}} \approx 75$ K) would reduce $T_{\text{sys,eff}}$ to ≈ 300 K and increase the measured solar excess to $\approx 0.026 \times (3\,600/300) \approx 0.31$ dB, an improvement of 12 $\times$ in SNR, comparable to the simulation model. Beyond total power, the digital pipeline recovers information inaccessible to the analogue chain: the ~ 14 -minute fringe modulation and the jinc-shaped visibility amplitude versus baseline length. This comparison - total power envelope (analogue, broad bell curve) versus fringe pattern and visibility (digital) - is the clearest demonstration of the complementary value of digital interferometry for solar burst science, provided the receiver-noise and clock-coherence

gaps identified here and in Section 5.11 are closed.

5.11 Clock-Coherence Requirement

Section 5.10 attributed the $23.8\times$ SNR gap to receiver system temperature alone. This section asks whether that is the only barrier: the voltage model of Chapter 4 treats all antennas as sharing a perfect common clock (CLOCK_OFFSET_NS= [0, 0, 0, 0] ns). Using the clock-offset injection facility (Eq. (4.7)), this section quantifies how realistic oscillator imperfections would degrade interferometric measurements, and places the hardware coherence requirement on a quantitative footing.

5.11.1 Static Offset: Phase Rotation Without Amplitude Loss

A static clock offset τ_{ij} between the digitisers of antennas i and j introduces a constant phase error on the baseline visibility (Eq. (4.8)):

$$\Delta\phi_{ij} = 2\pi f_c \tau_{ij} \implies \frac{\Delta\phi_{ij}}{1 \text{ ns}} = 2\pi \times 610 \text{ MHz} \times 10^{-9} \text{ s} \approx 219.6^\circ \text{ ns}^{-1}. \quad (5.12)$$

Table 5.9 evaluates Eq. (5.12) for representative offsets.

Table 5.9. Static clock offset versus visibility phase error at 610 MHz.

A static differential clock offset τ_{ij} between two digitisers introduces a constant phase rotation $\Delta\phi_{ij} = 2\pi f_c \tau_{ij}$ on the corresponding baseline visibility; the amplitude is unaffected. At $f_c = 610$ MHz, even a 0.1 ns offset produces a 22° phase error, sufficient to displace a simulated point source by more than its own diameter along the 8-m baseline direction.

Offset τ_{ij}	Phase error $\Delta\phi_{ij}$	Dirty-image effect
0 ns	0°	No effect; phase centre correct
0.1 ns	22°	Mild apparent source position shift
0.5 ns	110°	Source displaced along baseline direction
1 ns	220°	Large position error; image artefacts
1.64 ns	360°	Full phase cycle; source wraps to original position

A static phase error shifts the apparent source position along the baseline direction by $\Delta\theta \approx \Delta\phi_{ij}/(2\pi b_{ij}/\lambda)$. For the 8 m baseline ($b_{ij}/\lambda \approx 16.3$) and a 1 ns offset ($\Delta\phi_{ij} \approx 3.83$ rad), this gives $\Delta\theta \approx 2.1^\circ$ -larger than the solar diameter (0.53°): a single-nanosecond timing error places the simulated Sun at the wrong sky position by nearly four solar diameters. Static offsets can be removed by a point-source phase calibration, but only if a suitable calibrator transit is available.

5.11.2 Clock Drift: Fringe Washing and the Coherence Time

A more fundamental degradation arises when the two oscillators run at slightly different *rates*. Let ξ (dimensionless, in parts per million, ppm) be the fractional frequency offset between the digitisers of antennas i and j - not to be confused with the stochastic source process $\varepsilon(t)$ of Eq. (4.6) in Chapter 4, which is an unrelated quantity. The differential delay then grows as $\tau_{ij}(t) = \tau_{ij}(0) + \xi t$, and the instantaneous visibility phase advances continuously. Time-averaging over one integration dump of duration $\Delta t_{\text{dump}} = 43.2$ s reduces the fringe amplitude by the coherence factor

$$\eta_{\text{coh}} = \left| \text{sinc}(f_c \xi \Delta t_{\text{dump}}) \right|, \quad (5.13)$$

where $\text{sinc}(x) \equiv \sin(\pi x)/(\pi x)$, falling to zero (complete fringe washing) when $f_c \xi \Delta t_{\text{dump}} = 1$, defining the *coherence time*

$$\Delta t_{\text{coh}} = \frac{1}{f_c \xi}. \quad (5.14)$$

Table 5.10 evaluates Eqs. (5.13) and (5.14) for oscillator accuracies representative of consumer and laboratory hardware at $f_c = 610$ MHz, $\Delta t_{\text{dump}} = 43.2$ s.

Table 5.10. Clock drift versus fringe coherence factor at 610 MHz (43.2-s dump).

The fringe coherence factor η_{coh} (Eq. (5.13)) is evaluated as a function of oscillator fractional frequency offset ξ for a 43.2-second integration dump at $f_c = 610$ MHz. Consumer SDR crystals (1 ppm drift) produce complete fringe washing ($\eta_{\text{coh}} \approx 0$) within a single dump; maintaining $\eta_{\text{coh}} > 0.9$ requires a GPSDO or equivalent precision oscillator with $\xi \lesssim 10$ ppt.

Drift ξ	Δt_{coh}	$f_c \xi \Delta t_{\text{dump}}$	η_{coh}	Oscillator class
1 ppm	1.6 ms	26 352	≈ 0	Consumer SDR crystal
0.1 ppm	16 ms	2 635	≈ 0	TCXO (temperature compensated)
1 ppb	1.6 s	26.4	≈ 0	OCXO (oven controlled)
10 ppt	164 s	0.264	0.88	Precision OCXO
1 ppt	1 640 s	0.026	≈ 1	GPSDO (GPS-disciplined)

The requirement for $\eta_{\text{coh}} > 0.9$ over the 43.2 s dump is $\xi < 10$ ppt (10^{-11}), achievable with a GPS-disciplined oscillator (GPSDO) or a precision oven-controlled crystal (OCXO) with external 10 MHz reference injection. Consumer oscillators-including the crystal in a typical RTL-SDR, rated at approximately 0.5-1 ppm-produce $\Delta t_{\text{coh}} \lesssim 3$ ms, more than four orders of magnitude shorter than Δt_{dump} : a consumer SDR therefore produces zero fringe amplitude within the very first integration dump, regardless of SNR.

5.11.3 Implication for the Hardware Comparison

The clock-offset analysis adds a second, *independent* reason why digital cross-correlation with unsynchronised SDRs cannot recover a fringe, alongside the system-temperature gap of Section 5.10: even if $T_{\text{sys,eff}}$ were reduced to the idealised 150 K by adding a dedicated LNA, the time-averaged fringe amplitude would remain zero because $\eta_{\text{coh}} \approx 0$ for any consumer oscillator.

These two effects are therefore complementary barriers, and the minimum hardware specification for digital solar interferometry at 610 MHz with 43.2 s dumps requires addressing *both* simultaneously:

1. a low-noise amplifier with noise figure $\lesssim 1$ dB ($T_{\text{rx}} \approx 75$ K), reducing $T_{\text{sys,eff}}$ from ≈ 3600 K to ≈ 300 K; and
2. a common clock reference, a GPSDO or a 10 MHz distribution amplifier, injected into all digitiser clock inputs, bringing ξ below 10 ppt and $\eta_{\text{coh}} > 0.9$.

Meeting both conditions would increase the measured solar excess from the hardware-measured $\mathcal{E}_{\text{meas}} = 0.026$ dB to $\approx 0.026 \times (3600/300) \times \eta_{\text{coh}} \approx 0.31$ dB, consistent with the simulation prediction of $\mathcal{E}_{\text{sim}} = 0.586$ dB.

5.12 Chapter Conclusion

The results of this chapter show that the simulation pipeline is useful not only as a numerical demonstration, but as a design tool for a future low-cost solar radio interferometer, following the validation-to-deployment logic set out in Section 5.1.

First, the analytic validation tests (Sections 5.2-5.4) confirm that the main signal-processing blocks behave as expected: the simulated fringe period verifies the geometric-delay and (u, v, w) -coordinate implementation, the visibility-amplitude trend verifies that the resolved solar disk is correctly represented, and the dirty-image comparison verifies that the imaging stage produces the expected sky-convolved-with-PSF result. Together these validate the internal consistency of the pipeline and justify using it for the design work that follows.

Second, the array-geometry study (Section 5.5) shows why simulation is necessary before field deployment. A compact east-west layout can demonstrate fringes but gives limited two-dimensional localisation; the graded-cross and extended layouts provide better uv -coverage and a more useful synthesised beam, but also demand more careful calibration and, as Section 5.11 shows, clock coherence. The DLITE-inspired triangle-plus-centre alternative (Case 1.75) shows that non-redundant uv -sampling, not baseline length, is what lets exactly four elements recover useful two-dimensional resolution - the same minimum-redundancy principle that motivates a future four-element LWA-style deployment. The best antenna layout is therefore not determined by baseline length alone, but by the combined behaviour of uv -coverage, PSF shape, sidelobe level, and practical deployability - a conclusion reinforced by the sensitivity and dynamic-range analysis of Section 5.6 and by the beam-fidelity result of Section 5.7, where replacing the Gaussian primary beam with the real 4NEC2 pattern changes the predicted transit power by $\approx 25\%$.

Third, the comparative and diagnostic results identify the concrete barriers between simulation and measurement. Section 5.8 shows that the pipeline goes beyond educational visualisation by combining physically grounded noise, real data ingestion, real antenna beams, hardware export, and full reproducibility - the capabilities needed to run the design studies above at all. Section 5.10 compares the simulated and measured solar excess for 20 March 2025 ($\mathcal{E}_{\text{sim}} = 0.586$ dB versus

$\mathcal{E}_{\text{meas}} = 0.026 \text{ dB}$, a $23.8\times$ gap in linear SNR) and finds it consistent with an elevated effective system temperature in the consumer SDR receiver chain ($T_{\text{sys,eff}} \approx 3600 \text{ K}$); this diagnostic comparison should not be read as a full end-to-end hardware validation, which would require an independently gain-calibrated measurement. Section 5.11 adds a second, independent barrier: a consumer crystal oscillator with $\xi \approx 1 \text{ ppm}$ gives a coherence time of only $\approx 1.6 \text{ ms}$, four orders of magnitude shorter than the 43.2 s dump, so unsynchronised digitisers yield zero fringe amplitude regardless of SNR.

The practical outcome of this chapter is therefore a set of deployment recommendations rather than a single number: use a calibrated low-noise receiver chain ($\text{NF} \lesssim 1 \text{ dB}$), distribute a common clock reference ($\xi < 10 \text{ ppt}$, e.g. a GPSDO), preserve the real antenna primary-beam model rather than a Gaussian approximation, and choose the array layout based on uv -coverage and sidelobe behaviour rather than on baseline length alone. These recommendations, together with the six limitations of Section 5.9, define the concrete next steps for turning the present detection instrument into a working digital solar imaging interferometer.

CONCLUSION AND FUTURE WORK

Section 1.2 posed a single question: how can a physically grounded, end-to-end simulation be constructed to evaluate the feasibility, signal chain, and array-geometry choices of a future low-cost UHF solar radio interferometer before hardware deployment? That question was not academic. It followed directly from my own preliminary two-element deployment at Xuan Mai (Section 1.1), where phase-incoherent receivers, cable loss, local RFI, and an under-characterised noise floor produced an ambiguous result that could not, on its own, be diagnosed. This thesis answered the question by building and validating a complete software model of a four-element UHF interferometer at 610 MHz, pursuing the four objectives set out in Section 1.3: antenna modelling, an end-to-end signal pipeline, a general array-geometry formulation, and validation against analytic theory and real data. This chapter reviews what was achieved against each objective, positions the work relative to existing simulators, restates the contributions, and outlines the limitations and future extensions that follow.

1. Summary of Results

Objective 1 - Antenna modelling. The receiving element, a U-frame folded-dipole Yagi-Uda antenna (HKD G11KD-T2), was modelled in 4NEC2 using the Method of Moments and independently cross-checked using HFSS (Finite Element Method), directly answering the cross-validation requirement of Section 1.3. The 4NEC2 solution yielded an input impedance of $(108 - j22) \Omega$, a forward gain of 9.48 dBi, a half-power beamwidth of $\approx 52^\circ$, and a standing-wave ratio of 1.55 at the 75Ω reference. Direct vector-network-analyser measurements at the Xuan Mai test site confirmed the impedance to within calibration uncertainty, and the HFSS solution agreed with 4NEC2 to better than 0.2 dB. The mutual-coupling study showed that elements are decoupled below -78 dB at 5 m separation and below -83 dB at the 8 m minimum array spacing, formally justifying the independent-sensor assumption used throughout the pipeline.

Objective 2 - End-to-end signal pipeline. An end-to-end Python pipeline was implemented that converts real SWS Learmonth flux into per-antenna complex baseband voltages through the Friis reception relation [7, 16] and Johnson-Nyquist thermal noise [17], applies per-antenna geometric phase derived from the $\text{ENU} \rightarrow \text{XYZ} \rightarrow (u, v, w)$ transformation [7], cross-correlates to form visibilities, and inverts by a vectorised direct DFT to a dirty image. Two complementary simulation engines were provided: Engine A (voltage synthesis, suitable for FPGA export validation) and Engine B (forward-model visibilities, analytically exact and computationally fast for array-geometry studies).

Objective 3 - General array geometry. The $\text{ENU} \rightarrow \text{XYZ} \rightarrow (u, v, w)$ formulation was applied identically to every array configuration evaluated, from the two-element baseline used for fringe

validation to sixteen-element random layouts, without any change to the underlying geometry code. The parametric sweep exercised this generality directly: five configurations—compact east-west, north-south line, tee, graded cross, and random—quantified the trade-off between layout and imaging quality, and a sixth, DLITE-inspired triangle-plus-centre layout (Case 1.75, Section 4.8.3 of Chapter 4) showed that a minimum-redundancy topology recovers two-dimensional resolution from only four elements. The compact east-west array ($b_{\max} = 24\text{ m}$) detects the Sun and reproduces the correct fringe behaviour but provides only east-west angular resolution. The extended graded cross ($b_{\max} = 130\text{ m}$) achieves two-dimensional UV coverage sufficient to resolve solar structure and reduces the PSF sidelobe level by $\approx 20\text{ dB}$.

Objective 4 - Validation on 20 March 2025. Run on the spring-equinox transit ($\delta = 0.0^\circ$, 12-hour window, $H \in [\pm 90^\circ]$), the simulated instrument reproduced three independent analytic benchmarks: (1) a ~ 14 -minute fringe period on the 8 m baseline, matching Eq. (2.13) within the sampling resolution; (2) a jinc-shaped visibility amplitude versus baseline length, including the position of the first null, confirming that the van Cittert-Zernike relation [7, 8] is correctly implemented; and (3) a dirty image numerically identical (to machine precision) to the input sky convolved with the independently computed PSF. Agreement on all three benchmarks simultaneously, using real flux as input, validates the internal consistency of the pipeline at the analytic level and directly answers the feasibility question of Section 1.2: yes, such a simulation can be built, and it behaves as theory predicts.

Closing the loop on the Xuan Mai motivation. Validating the pipeline at the analytic level is not the same as explaining why the original field hardware failed, so the simulation was also compared directly against the two-element analogue trace recorded at Hoa Lac on the same date (Section 5.10 of Chapter 5). The comparison attributed the $23.8\times$ gap in measured solar excess to an elevated effective receiver system temperature ($T_{\text{sys,eff}} \approx 3600\text{ K}$, versus the 150 K idealised model), and a separate clock-coherence study (Section 5.11) showed that a consumer oscillator drift of $\xi \approx 1\text{ ppm}$ alone would zero the fringe amplitude regardless of receiver noise. Together these two, independent diagnoses give a quantitative answer to the question that motivated this thesis: the Xuan Mai deployment was not undermined by a flawed concept, but by an uncalibrated low-noise amplifier and the absence of a shared clock reference—both addressable with modest, identifiable hardware additions.

2. Position Relative to Existing Simulators

The most widely used open-source aperture-synthesis simulator, APSYNSIM [19], provides a real-time interactive GUI that is excellent for teaching. This thesis occupies a complementary but distinct niche, differentiated from APSYNSIM on four axes:

- **Real data.** The pipeline ingests archived solar flux from the Learmonth observatory rather than a synthetic Gaussian sky model. The source of truth is the Sun itself.

- **Real antenna beam.** The primary beam is the actual 4NEC2 Yagi pattern (loaded from a pre-computed .npz file), not the Rayleigh-criterion Gaussian. As shown in Chapter 5, this difference accounts for $\approx 25\%$ (~ 1 dB) more integrated transit power relative to the Gaussian approximation.
- **Physics-based noise.** The noise model is grounded in the Johnson-Nyquist thermodynamic formula [17], not an empirical scale factor. The system temperature $T_{\text{sys}} = 150$ K and bandwidth $\Delta\nu = 1$ MHz are physical quantities that can be measured and that determine the minimum detectable flux density from first principles.
- **Reproducibility and hardware path.** The Jupyter notebook constitutes a complete, version-controlled, re-runnable record of the simulation. The 16-bit IQ quantisation stage and LSB-validation check provide a direct path to an FPGA hardware correlator—a direction that APSYNSIM does not pursue.

3. Contributions

Matching the five contributions claimed in Section 1.4, this thesis delivers:

1. **A validated 4NEC2 antenna model** of a low-cost commercial UHF Yagi-Uda element, cross-checked against direct measurement and an independent FEM solution, with a quantitative mutual-coupling study confirming element independence across the operational baselines.
2. **An end-to-end interferometry simulation engine** that is correct for arbitrary two-dimensional array geometries through a single, consistent (u, v, w) formulation applied identically to per-antenna phase delay, baseline projection, and fringe stopping.
3. **A real-data-driven validation** demonstrating that the simulated instrument, driven by archived Learmonth 610 MHz flux, reproduces the classical analytic signatures of aperture synthesis—fringe period, jinc visibility, dirty-image consistency—simultaneously and on real data.
4. **A structured comparison with APSYNSIM** identifying four concrete respects (real data, real antenna beam, physics-based noise, and a reproducible hardware-export path) in which this pipeline extends beyond a purely educational simulator.
5. **A reproducible, single-notebook workflow** that takes a user from an archived SWS daily file to a calibrated dirty image in under two minutes of computation, with all parameters controlled from a single configuration cell and all results exported automatically.

4. Limitations

The present implementation makes the following simplifications, each well-justified for the validation scope of this work but worth noting for future extensions: the noise model assumes

a frequency-flat system temperature and ignores $1/f$ gain fluctuations; bandwidth smearing is negligible at $\Delta\nu/\nu = 1.6 \times 10^{-3}$ but would become significant for wider channels; ionospheric differential phase is sub-degree for the baselines used but would need modelling for longer baselines or solar-storm conditions; no Jones-matrix mutual-coupling correction is applied (the coupling is < -78 dB, making this contribution negligible); the imaging stage produces a dirty image rather than a deconvolved sky map; and the voltage model assumes perfect clock synchronisation ($\tau_k = 0$ for all antennas), whereas real digitisers require a common phase reference to prevent fringe washing (Section 5.11). All six limitations point directly to concrete extensions.

5. Future Work

Several extensions follow naturally from this work:

- **FPGA hardware correlator.** The voltage-chain engine already exports 16-bit IQ samples in hexadecimal format with a validated noise floor of at least 20 LSBs. The immediate next step is to stream these samples to an FPGA cross-correlator board and compare the hardware fringe amplitude with the software result sample-by-sample, providing an end-to-end hardware-in-the-loop validation.
- **Physical array upgrade, following the DLITE precedent.** The parametric sweep quantified the imaging gain from extending the compact east-west baseline set to a graded cross with $b_{\max} = 130$ m, and the Case 1.75 evaluation (Section 4.8.3 of Chapter 4) showed that a DLITE-inspired [31] triangle-plus-centre layout recovers two-dimensional resolution from exactly four elements, the same count used throughout this thesis; this is not a full DLITE system simulation, only the minimum-redundancy topology, evaluated at 610 MHz rather than DLITE’s 30 MHz to 40 MHz band. Deploying four low-cost LWA-style crossed-dipole antennas [13] at Xuan Mai or another site in this minimum-redundancy arrangement, mirroring the motivation of Section 1.1, would turn the present detection instrument into a genuine solar imager capable of resolving burst sources to a fraction of the solar radius.
- **Solar burst recording.** The present pipeline was validated on the quiet-Sun continuum. Replacing the quiet-Sun SWS input with a recorded Type II or Type III burst event would test the pipeline’s transient response and demonstrate the dynamic-range headroom shown in Table 5.4.
- **Multi-frequency operation.** The SWS Learmonth archive covers eight spectral channels from 245 MHz to 2695 MHz. Extending the pipeline to a multi-channel mode would enable spectro-spatial imaging, tracing a burst’s frequency drift (and hence electron beam velocity) across the corona.
- **CLEAN deconvolution.** Implementing Högbom [33] or multi-scale CLEAN as an optional final stage would recover the true sky brightness and achieve the noise-limited dynamic range of

Table 5.4, rather than the PSF-sidelobe-limited range of the current dirty image.

- **Wider-field imaging and w -projection.** For baselines beyond 130 m or field-of-view extensions to tens of degrees, the flat-sky approximation breaks down and w -projection or faceting must be incorporated.
- **Real-time operation.** The pipeline is currently a post-processing tool. Rewriting the correlator stage as a streaming Python process (e.g. using `asyncio` or a GPU back-end) would enable real-time image updates at the cadence of the SWS one-second data rate.
- **Jones-matrix calibration.** Even though mutual coupling is < -78 dB at the operating baselines, incorporating a Jones-matrix element-pattern correction would allow the simulator to be used for precision polarimetry experiments at shorter baselines.
- **Phase-coherent clock reference.** The clock-offset sensitivity study of Chapter 5 (Section 5.11) showed that a differential oscillator drift of $\xi = 1$ ppm, typical of a consumer RTL-SDR crystal, produces a coherence time $\Delta t_{\text{coh}} \approx 1.6$ ms at 610 MHz, four orders of magnitude shorter than the 43.2 s integration dump. Digital cross-correlation between unsynchronised digitisers therefore yields zero fringe amplitude regardless of SNR. The required hardware addition is a GPS-disciplined oscillator (GPSDO, $\xi \lesssim 1$ ppt) or a shared 10 MHz reference signal distributed to all ADC clock inputs. The simulation pipeline can extend this study by sweeping `CLOCK_OFFSET_NS` and a new `DRIFT_PPM` parameter to produce a two-dimensional coherence map ($\xi, \Delta t_{\text{dump}}$) that directly specifies the clock-subsystem tolerance budget for the hardware correlator design.

Returning to the question posed in Section 1.2: this thesis shows that a physically grounded, end-to-end simulation can indeed be built to evaluate the feasibility, signal chain, and array-geometry choices of a low-cost UHF solar radio interferometer before hardware deployment, and that doing so turns an ambiguous field failure into a specific, addressable list of hardware requirements. The work delivers a faithful, reproducible, and real-data-driven model validated from incident flux to final dirty image, positioned relative to the open-source simulation landscape, and it provides a solid, well-documented foundation for the four-element DLITE-inspired array upgrade motivated in Section 1.1 and for future extensions toward imaging, burst science, and real-time operation.

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APPENDIX

PROJECT REPOSITORY AND SIMULATION GUIDE

All source code, data, and figures produced in this thesis are archived in a public GitHub repository:

`https://github.com/quanbanguyen/
solar-interferometry-simulation-project` [34]

The repository contains two independent entry points for exploring the simulation: a zero-installation browser tool (`radio_sim.html`) for interactive visualisation, and a Jupyter notebook (`the_newest_heart_of_project.ipynb`) for full pipeline execution and figure reproduction. Both implement the same physical model described in Chapters 4–5.

A.1 Repository Contents

File / Folder	Description
<code>the_newest_heart_of_project.ipynb</code>	Main Jupyter notebook; reproduces every figure and table in Chapters 4–5 from a single top-to-bottom run
<code>radio_sim.html</code>	Self-contained browser simulation tool; no installation required
<code>sws_to_csv.py</code>	SWS/RSTN file parser and preprocessor
<code>file_excel_du_lieu/</code>	Preprocessed daily flux CSV files for multiple observation dates
<code>figures/</code>	All exported PNG figures referenced in the thesis
<code>beam_data/</code>	4NEC2 Yagi radiation pattern (<code>.npz</code>) used by the pipeline
<code>requirements.txt</code>	Python package list for reproducing the notebook environment

A.2 Quick Start: Browser Simulation (No Installation Required)

The browser tool `radio_sim.html` runs entirely inside any modern web browser (Chrome, Firefox, Edge) with no server, no Python, and no account required. It implements the same physical model as the notebook and exposes every key parameter through interactive controls.

- Step 1. Download the file.** Navigate to the repository URL above. Click the file `radio_sim.html` in the file list, then click the download button (downward-arrow icon, top-right of the file preview). Save it anywhere on the local computer.
- Step 2. Open in a browser.** Double-click the saved `radio_sim.html` file. The simulation loads in the default browser within a few seconds. No internet connection is required after the file is saved.
- Step 3. Select a preset.** In the left sidebar, open the **Preset** dropdown and choose one of the five configurations (Table 5.2) to instantly populate all parameters and redraw all plots.
- Step 4. Explore interactively.** Use the five tabs on the right to navigate between the solar flux record, UV coverage, fringe pattern, dirty image, and the usage guide. Adjust the frequency slider, declination slider, or array antenna positions to see all plots update in real time.
- Step 5. Load real SWS data (optional).** Click the **Load SWS file** button in the sidebar and select a downloaded Learmonth daily file (plain-text `.txt` format, available from the BOM Space Weather Services archive [14]). The tool parses the file automatically and drives the simulation with the real flux record.

A.2.1 Interface Overview

The browser-based interface is not a separate physical model. It is a visualisation companion, developed to complement the Python pipeline of Chapter 4 and support live demonstration of the interferometric concepts developed there, without requiring the user to run the full Jupyter Notebook environment. The tool runs entirely in the browser with no server or installation required, implements the same physical model as the notebook (Learmonth site, 610 MHz, spring equinox, compact EW default), and exposes every key parameter through interactive controls that update all plots in real time.

Figure A.1 shows the interface. The tool is organised as a fixed sidebar on the left and five display tabs on the right:

- **Solar Flux.** Loads any Learmonth SWS `.txt` raw file (RSTN HHMMSS format) and renders the daytime flux curve with transit statistics; nighttime tokens (//////) are stripped automatically. A parametric Gaussian envelope is used when no file is loaded.
- **Array & UV.** Displays the antenna layout in the ENU frame and the uv -coverage accumulated over the full observing window $H \in [-90^\circ, +90^\circ]$. Editing antenna coordinates or switching presets triggers an immediate redraw of both panels.

- **Fringe Pattern.** Renders the simulated fringe envelope for every baseline and shows a per-baseline summary table listing b/λ , fringe period T_{fringe} , and angular resolution θ .
- **Image.** On-demand computation and display of the synthesised (dirty) beam and dirty image, with visibility amplitude plotted as a function of baseline length below the image panels.
- **Info.** Operating instructions, the Python conversion snippet for exporting a 4NEC2 beam pattern to the JSON format accepted by the tool, and a quick-reference formula card.

The sidebar provides: (i) a file picker for the SWS data and a frequency selector covering all eight RSTN channels (245–15 400 MHz); (ii) an antenna position table with five pre-configured presets corresponding exactly to the five evaluation cases of Section 4.8.5 of Chapter 4; (iii) a carrier-frequency slider that shows the effect of changing λ on fringe period and angular resolution in real time; (iv) a declination slider $\delta \in [-30^\circ, +30^\circ]$ that lets the user observe directly how the uv -tracks transition from the straight horizontal lines derived in Section 4.5.4 of Chapter 4 (at $\delta = 0^\circ$) to curved elliptic arcs as $|\delta|$ increases; and (v) a selectable primary-beam model (uniform, Gaussian, $\cos^2$, or a 4NEC2 JSON file).

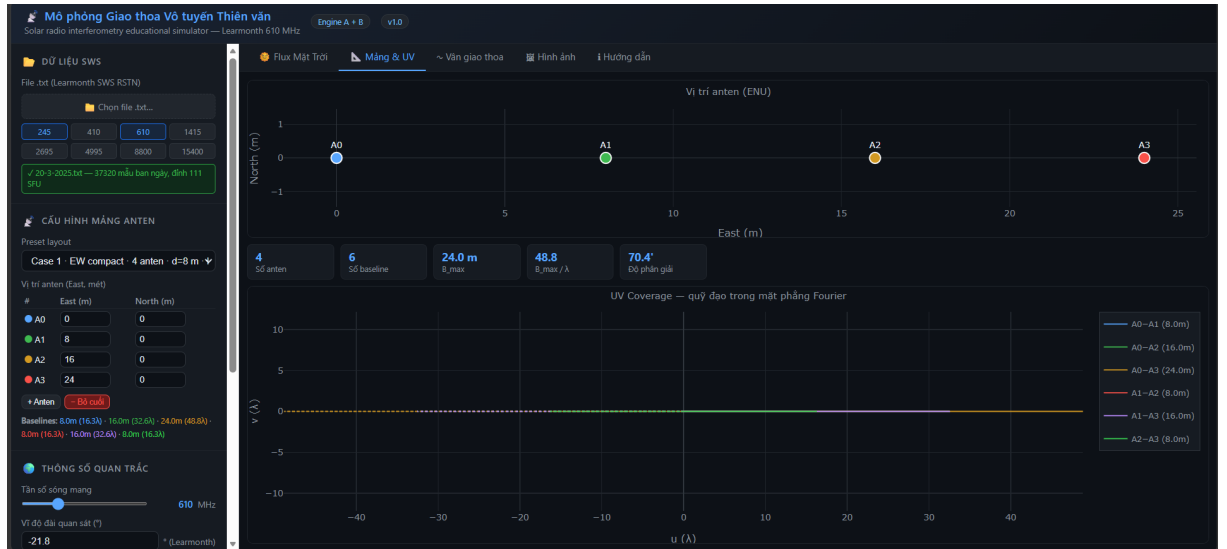


Figure A.1. Interactive browser-based visualisation companion (`radio_sim.html`), Array & UV tab.

Left sidebar: SWS file loader with RSTN frequency selector, antenna configuration table (Case 1 compact EW preset: four antennas at $[0, 8, 16, 24]$ m), and observation parameter controls. Main area: antenna layout in the ENU plane (top) and the corresponding uv -coverage over the 12-hour observing window (bottom) for $\delta = 0^\circ$, showing the characteristic horizontal-line pattern explained analytically in Section 4.5.4 of Chapter 4. Switching the declination slider away from zero curves the tracks into elliptic arcs in real time.

A.3 Full Pipeline: Jupyter Notebook

Reproducing the figures and tables in this thesis requires Python 3.9 or later with the packages listed in `requirements.txt`.

Step 1. Install Python. Download the Anaconda distribution from <https://www.anaconda.com/download> (recommended; includes all scientific packages) or install Python from <https://python.org>.

Step 2. Clone or download the repository. With Git installed: open a terminal and run

```
git clone
https:
//github.com/quanbanguyen/solar-interferometry-simulation-project
```

Without Git: click the green **Code** button on the repository page, then **Download ZIP**, and extract the archive.

Step 3. Install dependencies. Open a terminal in the project folder and run:

```
pip install -r requirements.txt
```

The required packages are: numpy, scipy, plotly, pandas, jupyter, and kaleido (for figure export).

Step 4. Open the notebook. Run `jupyter notebook` in the terminal, then open `the_newest_heart_of_project.ipynb` in the browser tab that appears.

Step 5. Set the observation date. In **Cell 0** (the first code cell), edit two variables: `NGAY` (date string, format DD-MM-YYYY) and `DEC_SUN_DEG` (solar declination in degrees, obtainable from JPL Horizons as described in Section A.4). All downstream cells inherit these values automatically.

Step 6. Run all cells. From the menu bar: **Kernel** → **Restart & Run All**. The full pipeline completes in under two minutes on a standard laptop and exports all figures and data files automatically.

A.4 Retrieving Solar Declination from NASA JPL Horizons

The NASA JPL Horizons system [30] is the authoritative source for solar and planetary ephemerides. The following steps retrieve the solar declination δ for any observation date.

Step 1. Open the Horizons web interface. Navigate to `https://ssd.jpl.nasa.gov/horizons/` in any browser.

Step 2. Set the target body. Click the **Target Body** field, type Sun, and select **Sun [Sol 10]** from the autocomplete list.

Step 3. Set the observer location. Click **Observer Location**, type Learmonth (or enter the coordinates manually: latitude -22.2° , longitude 114.1°E , elevation 0 m).

Step 4. Set the time span. Under **Time Specification**, enter the observation date in the **Start Time** field (e.g. 2025-03-20) and set the **Stop Time** one day later (2025-03-21). Set the **Step Size** to 1 day for a single representative value.

Step 5. Select output quantities. Under **Table Settings**, tick **Apparent RA & DEC (airless)**, which is quantity 1 in the Horizons menu. The DEC column is the geocentric solar declination δ . All other quantities may be left at their defaults.

Step 6. Generate the ephemeris. Click **Generate Ephemeris**. In the output table, locate the column labelled **DEC** (degrees, arcminutes, arcseconds). The value on the observation date is δ . For 20 March 2025 (spring equinox) the returned value is $\delta = 0.0^\circ$ to the nearest arcminute. Enter this value as DEC_SUN_DEG in Cell 0 of the notebook.

Note. The approximate formula $\delta \approx -23.45^\circ \cos(360^\circ(d + 10)/365.25)$ (where d is the day-of-year) is accurate to within $\pm 0.3^\circ$ for most dates and is sufficient for the simulation. JPL Horizons provides sub-arcsecond precision and is recommended for validating the approximation or for dates near the solstices where the formula error is largest.